

Program Analysis 2025-26

Lecture #12

Abstract Interpretation: maths



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Static analysis (recap)

All programs are interesting!

Proof (sketch).

Let $\mathcal{P}$ be a programming language.

Let us enumerate all syntactically valid programs in $\mathcal{P}$ (e.g., in lexicographic order).

Assume, for contradiction, that some programs are not interesting.

Then, there exists a "smallest non-interesting program".

Being the smallest such program makes it interesting, hence the contradiction.

Corollary: No program is interesting

Proof (sketch). If all programs are likely interesting, none of them really is such!

What went wrong?

The previous argument is obviously just a joke and our proof cheats

"Interesting" is neither a formal nor a semantic property, it depends on the enumeration, it is self-referential, it is not a property of program behaviour

So what kind of properties do we actually care about?

Semantic properties depends on the input-output behaviour of the program

A property is *non-trivial* if: it holds for some program and not for others

Rice's Theorem: Every non-trivial semantic property of programs is undecidable.

Yes, No, Don't know

Due to Rice's Theorem, all semantic properties $P(c)$ of programs are undecidable, thus we cannot design fully precise static analysers able to automatically infer $P(c)$

Nevertheless, we aim to ensure the **correctness** (**soundness**) of static analysis:

- If the analysis **outputs YES**, then the $P(c)$ is **certainly verified**
- If the analysis **does not output YES** then **it may happen** that $P(c)$ is **not verified**

"not YES" means "Don't know"

Example: Halting problem

Let $H_{\sigma_0} \triangleq \{c \mid \llbracket c \rrbracket \sigma_0 \neq \emptyset\}$

i.e., it is the set of programs that **terminate** on the initial memory $\sigma_0 \triangleq \lambda x.0$

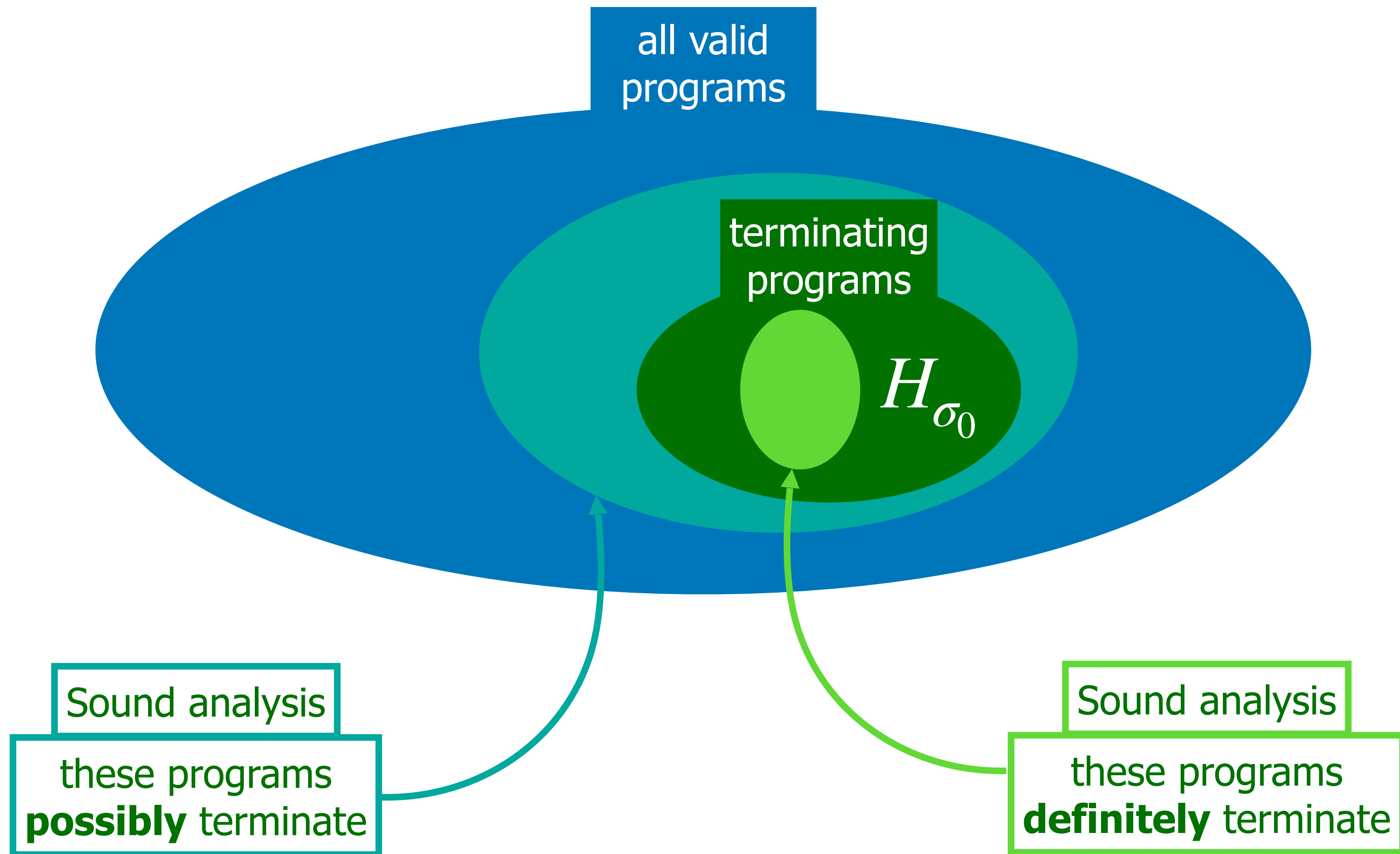
Halting problem (HP): Given c , does $c \in H_{\sigma_0}$?

For any (Turing equivalent) programming language, HP is **undecidable**:

No analyser can always correctly answer in finite time if $c \in H_{\sigma_0}$ or not

it means "for any program c "

Example: Halting problem

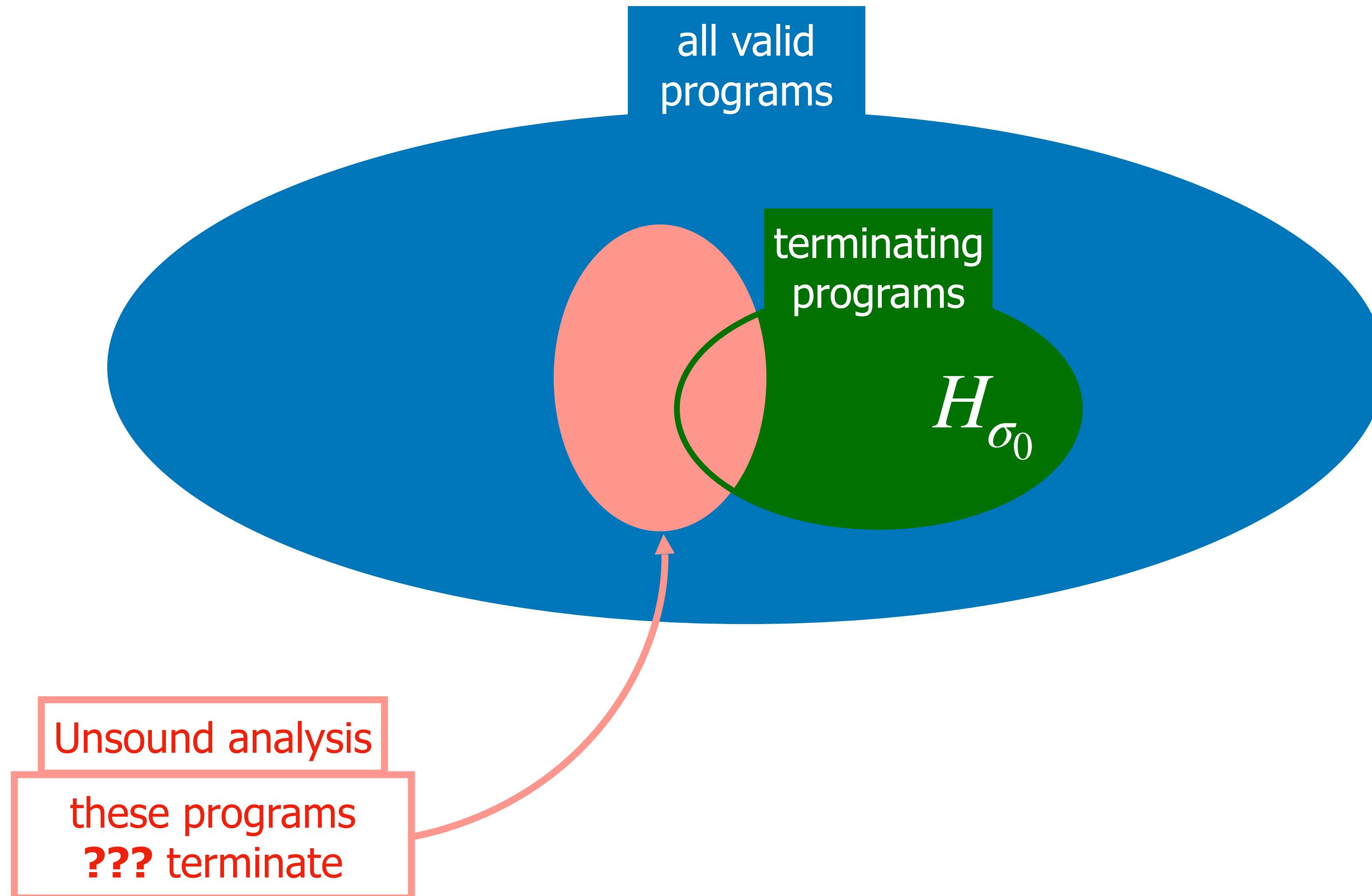


Example: Halting problem

We thus have two types of termination analysis:

- “**Definite** termination analysis”
 - **YES** the input program **definitely** terminates
 - **NO** the input program **could** not terminate
- “**Possible** termination analysis”
 - **YES** the input program **could** terminate
 - **NO** the input program **definitely** does not terminate

Example: Halting problem



Unsound analyses

Imagine a static program analysis that, in order to **detect divisions by zero**, syntactically finds the strings `/0`

```
x = 100/0;
```

← **true alarm**

```
print("25/09/2016");
```

← **false alarm**

```
y=100/02;
```

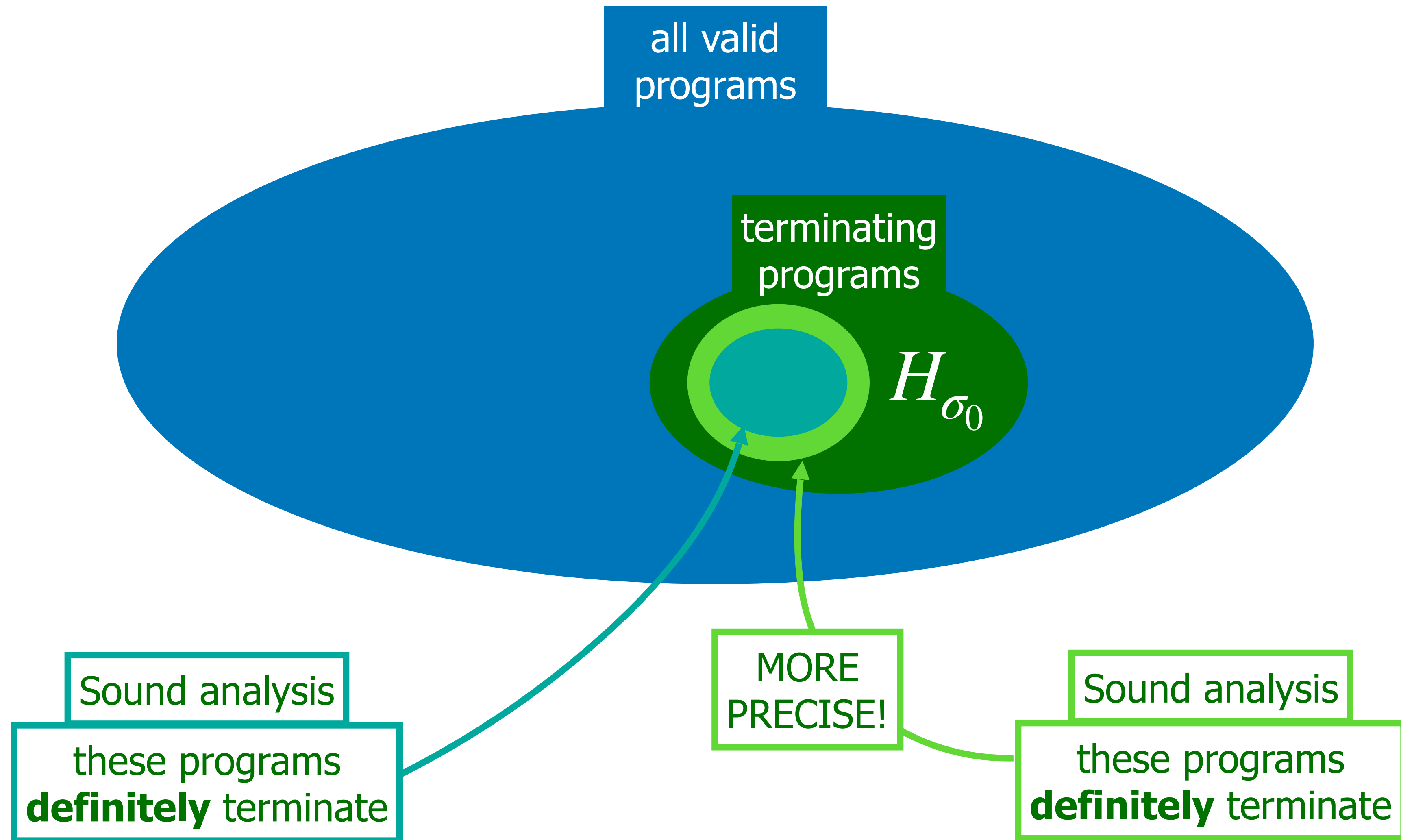
← **false alarm**

```
z=0;  
print(5/z);
```

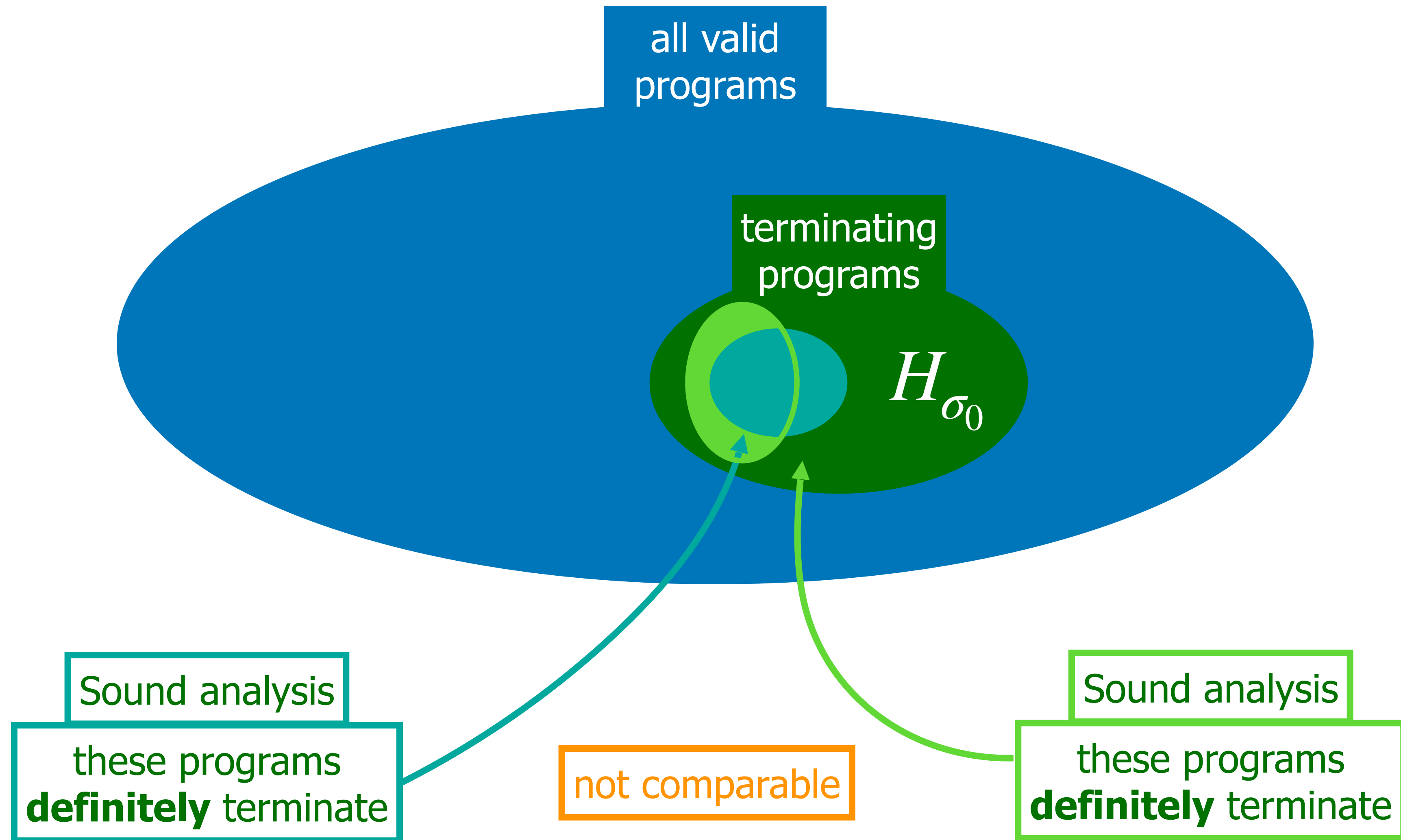
← **false negative**

Static program analyses with false negatives are called **unsound**

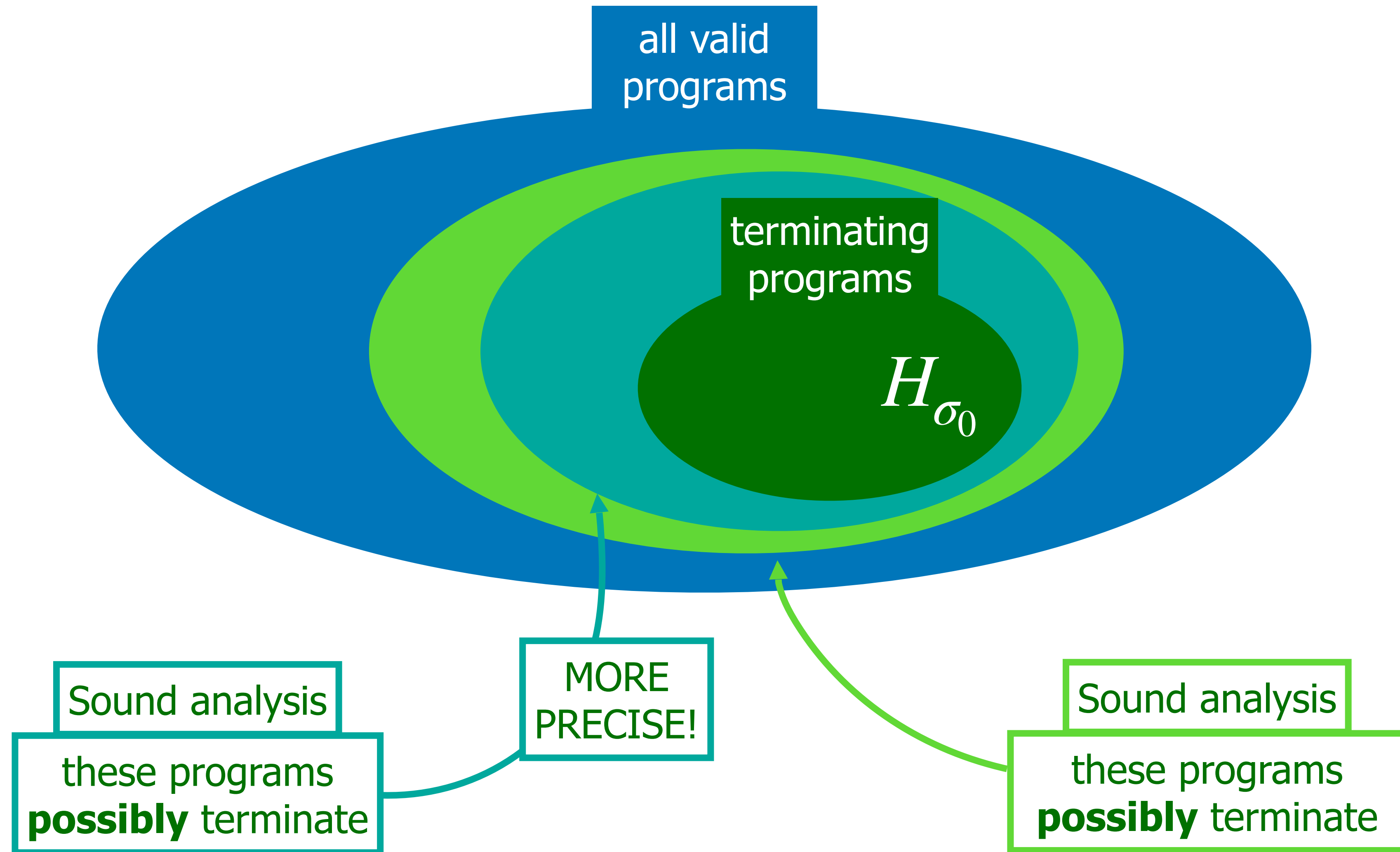
Example: Precision



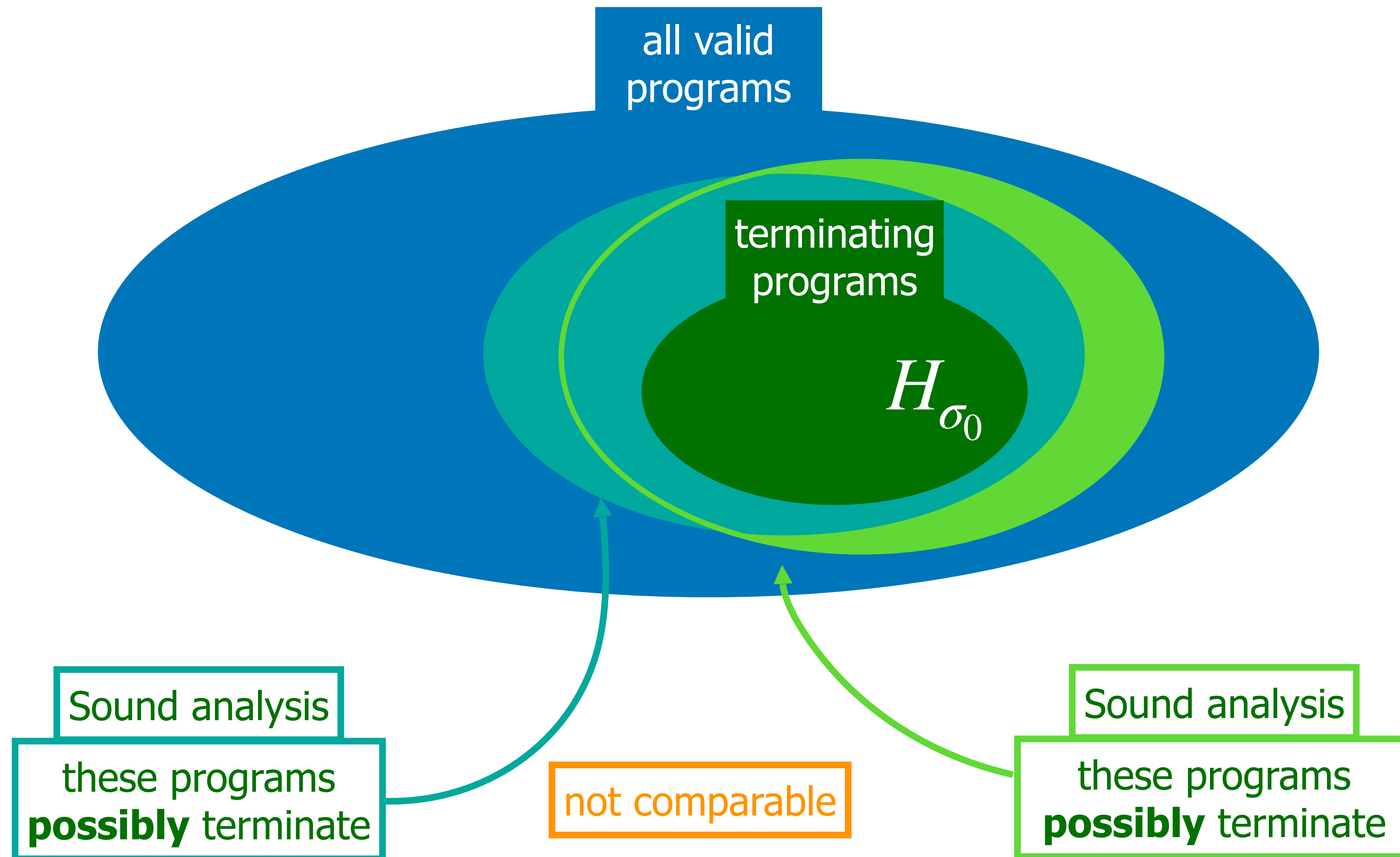
Example: Precision



Example: Precision



Example: Precision



Static analysis design

To design a **sound static analysis** we need to:

1. have a reference program semantics $\llbracket \cdot \rrbracket$
2. specify what and how the static analysis $\llbracket \cdot \rrbracket^\#$ computes
3. prove the correctness of $\llbracket \cdot \rrbracket^\#$ w.r.t. $\llbracket \cdot \rrbracket$
4. (efficiently) implement a static analyzer for $\llbracket \cdot \rrbracket^\#$

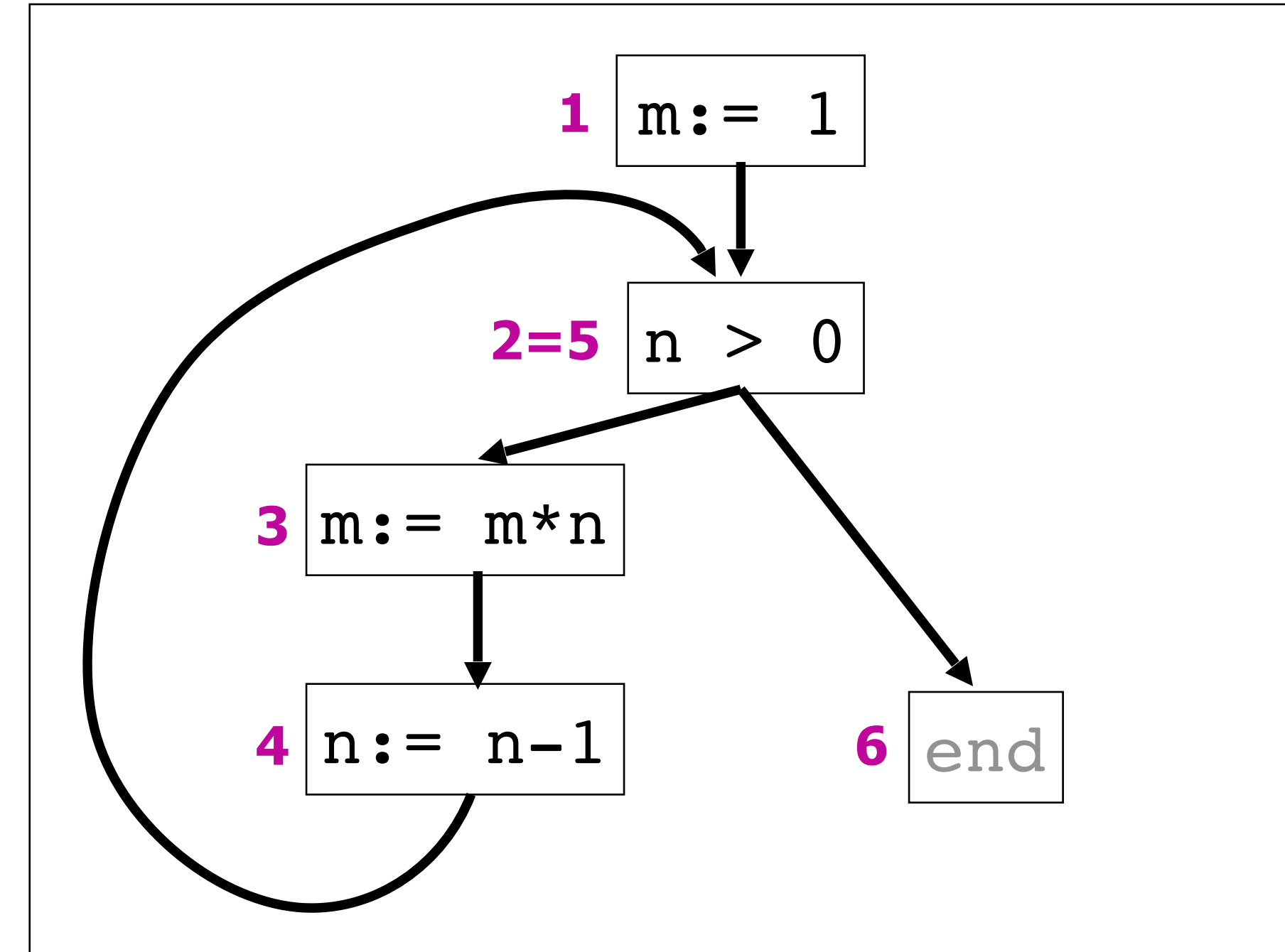
Sound abstractions

Program points & Control flow graph

```
{true}  
m := 1;  
while n > 0 do {  
    m := m * n;  
    n := n - 1  
}  
{m > 0}?
```

```
(1) m := 1;  
(2) while n > 0 do {  
(3)     m := m * n;  
(4)     n := n - 1  
(5=2) }  
(6) // end
```

With program points



Control flow graph

Abstract values

We infer by static analysis that at program point (6) the value of m will be **positive** for any input value n .

In order to achieve this, the analysis must propagate “**sign information**”, disregarding the actual values of variables.

(Non-relational abstraction)

Each program variable can take one of these “abstract values”:

$\mathbb{Z}_{>0}$

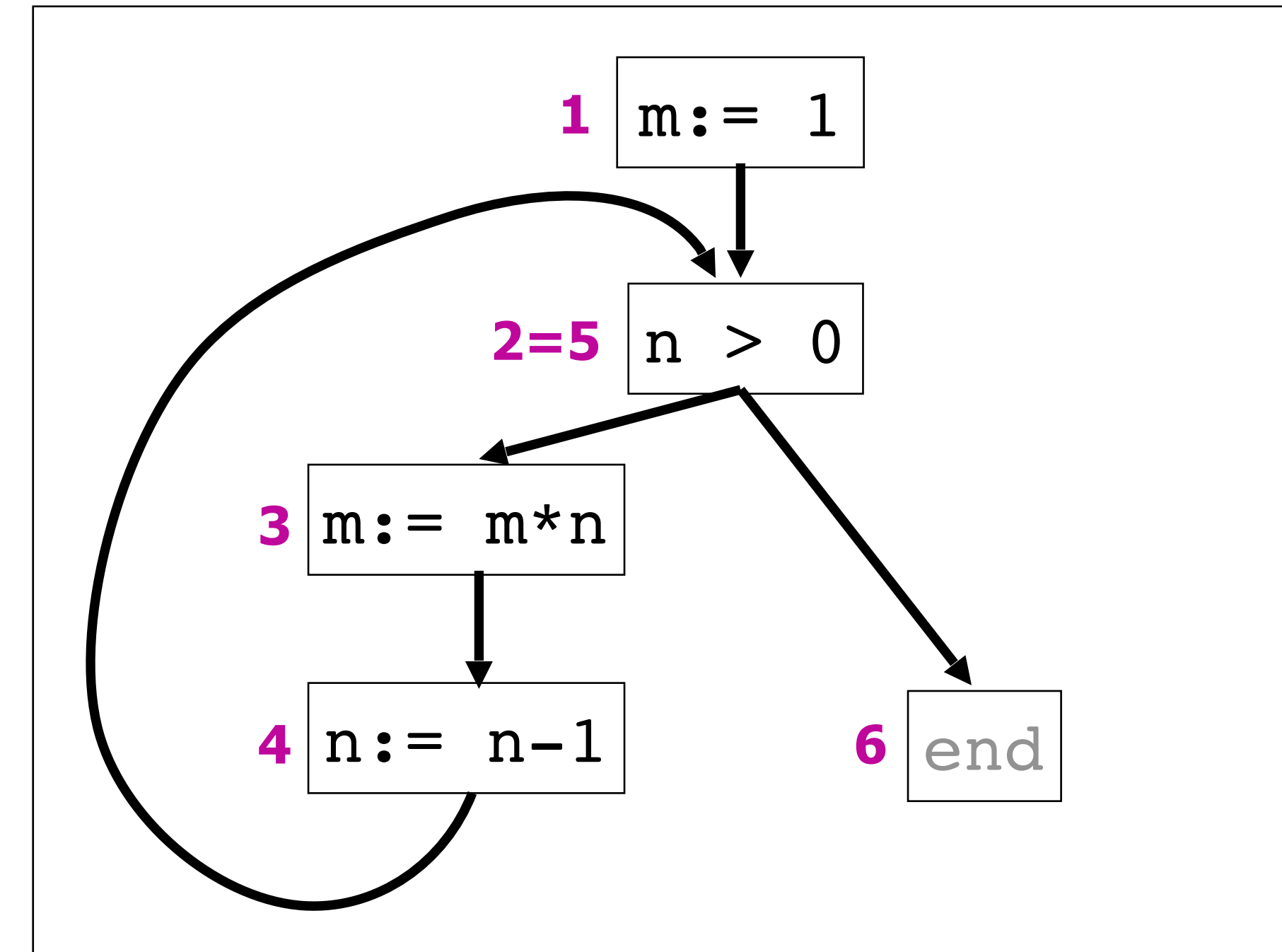
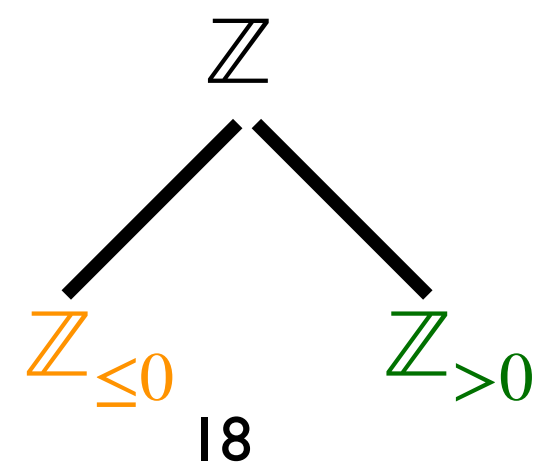
its actual value is **positive**

$\mathbb{Z}_{\leq 0}$

its actual value is **not positive**

$\mathbb{Z}$

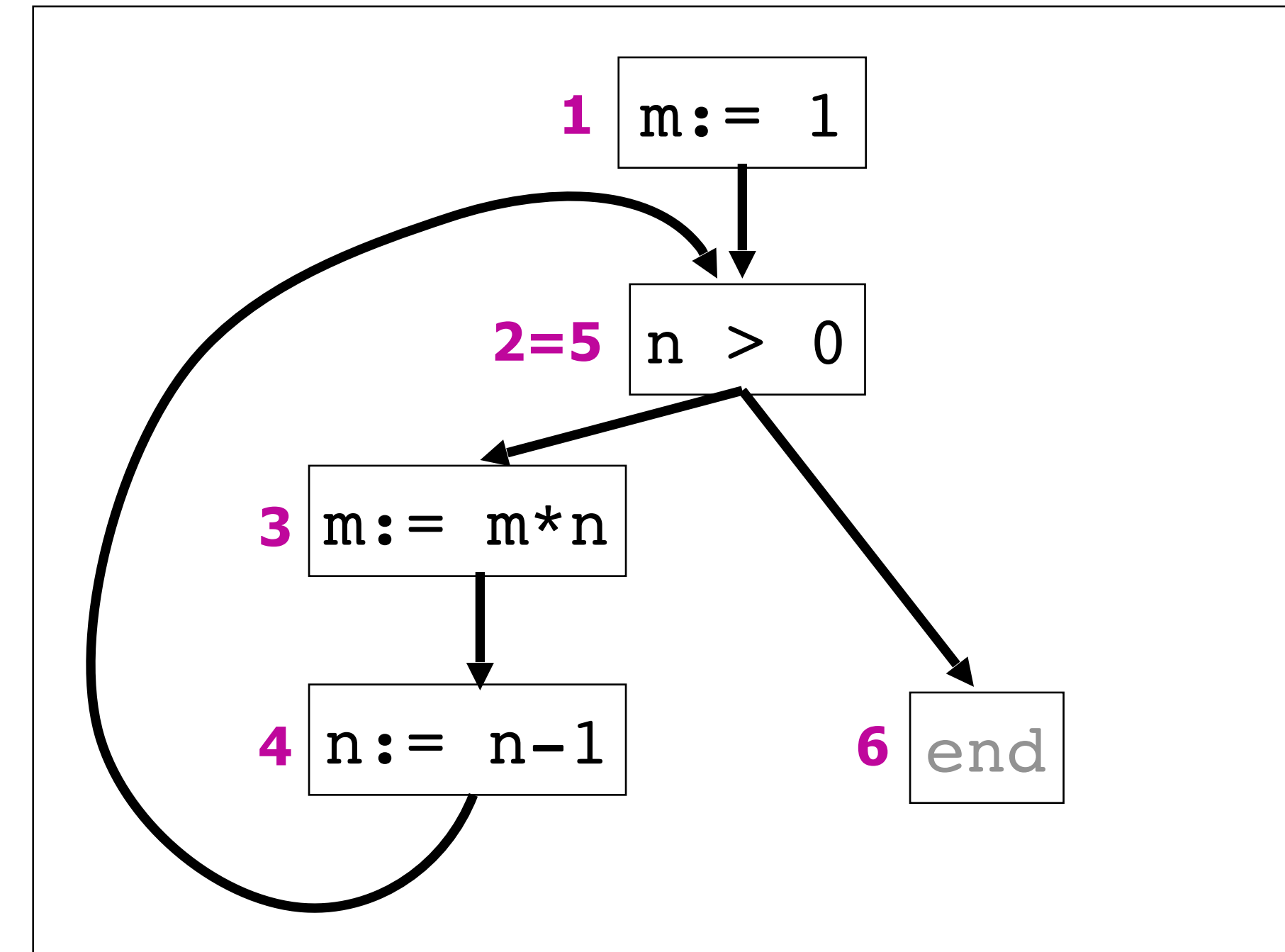
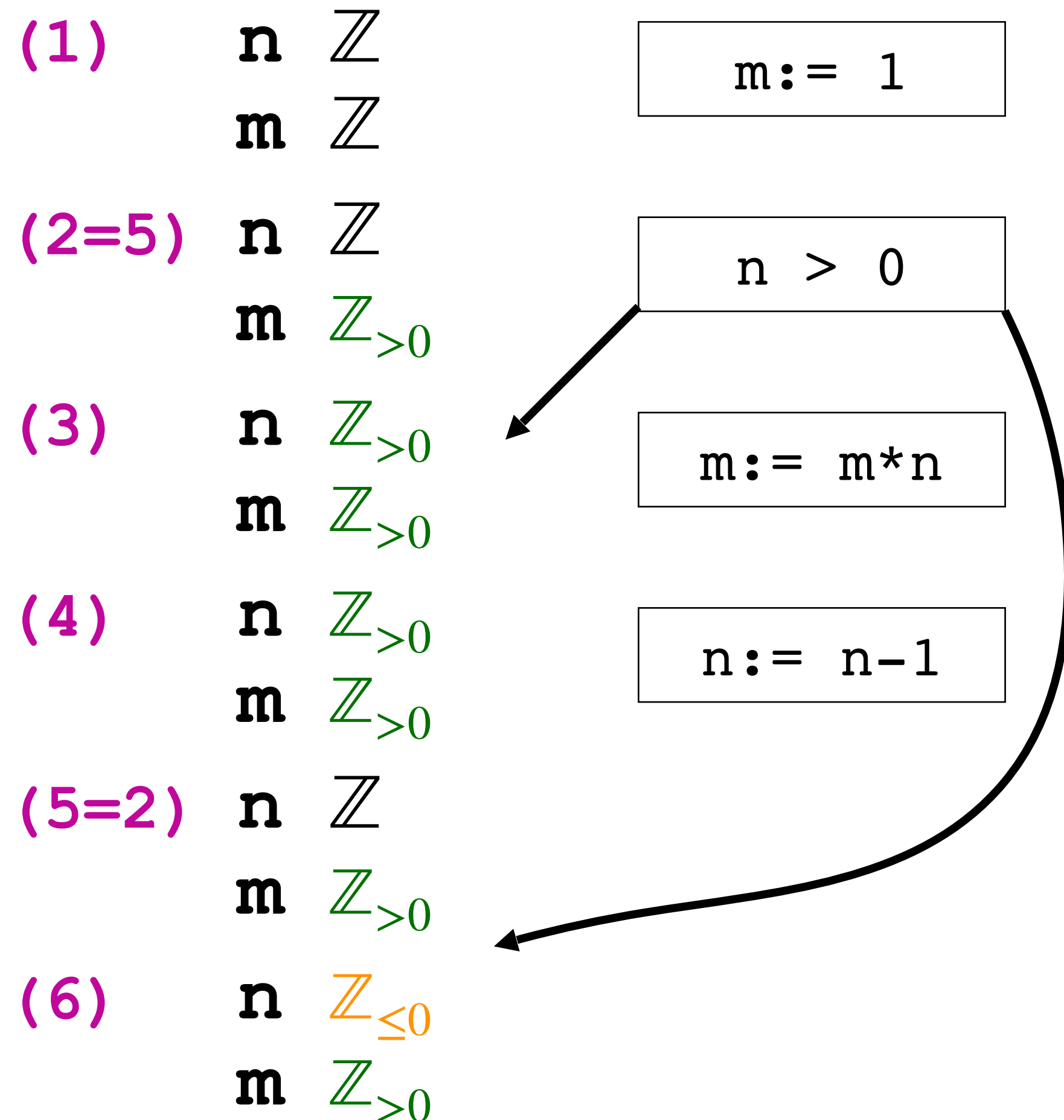
any value (**don't know**)



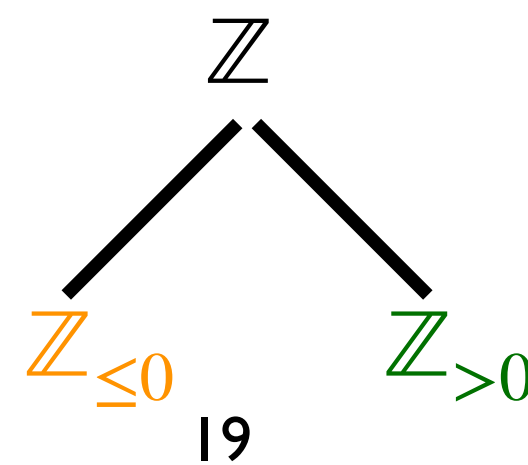
Control flow graph

Example: Sign analysis

Sign Invariants

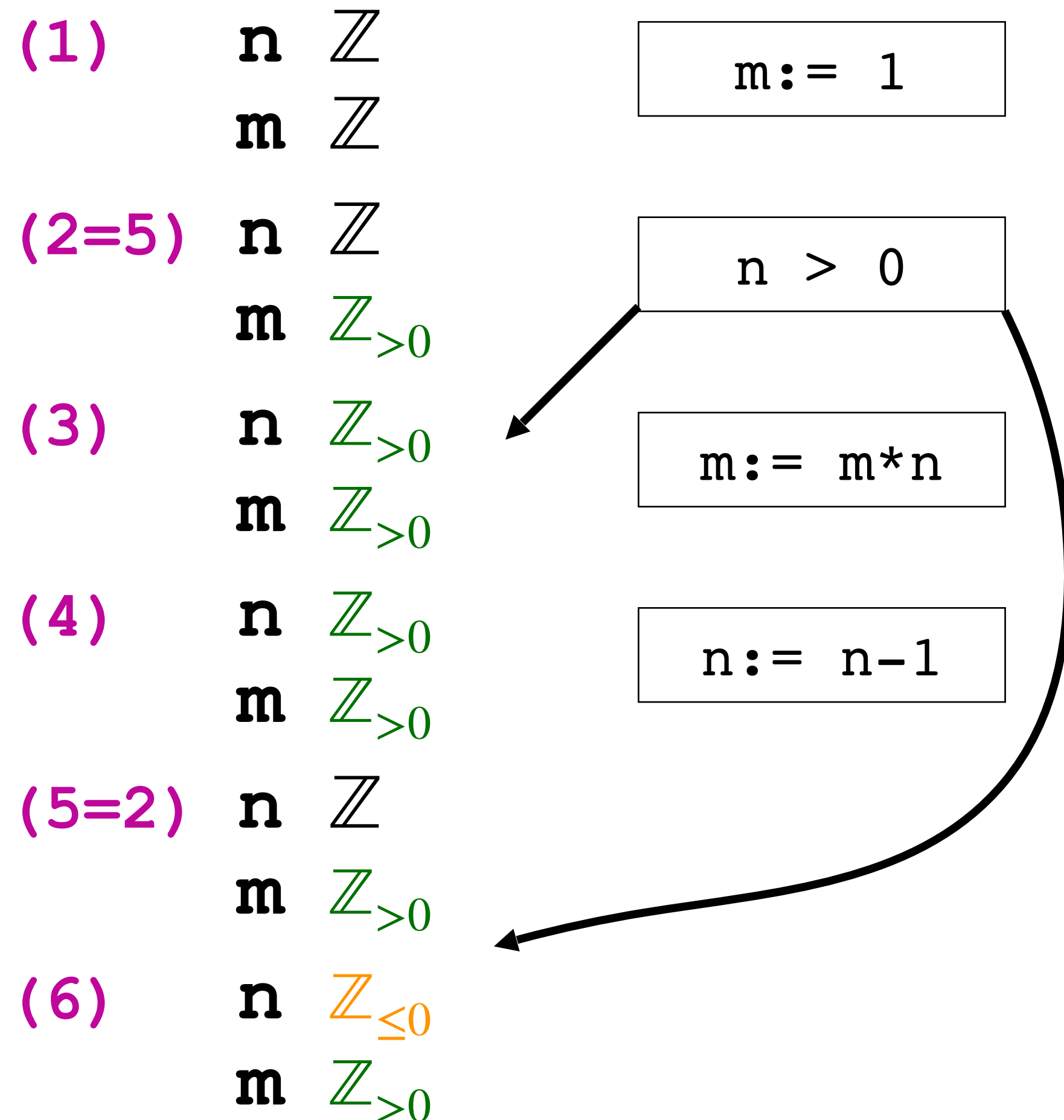


Control flow graph



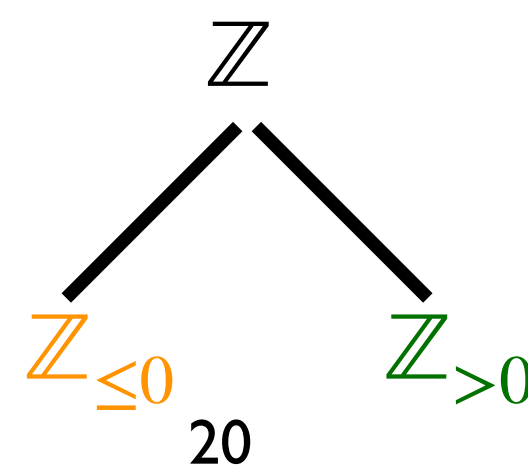
Example: Sign analysis

Sign Invariants



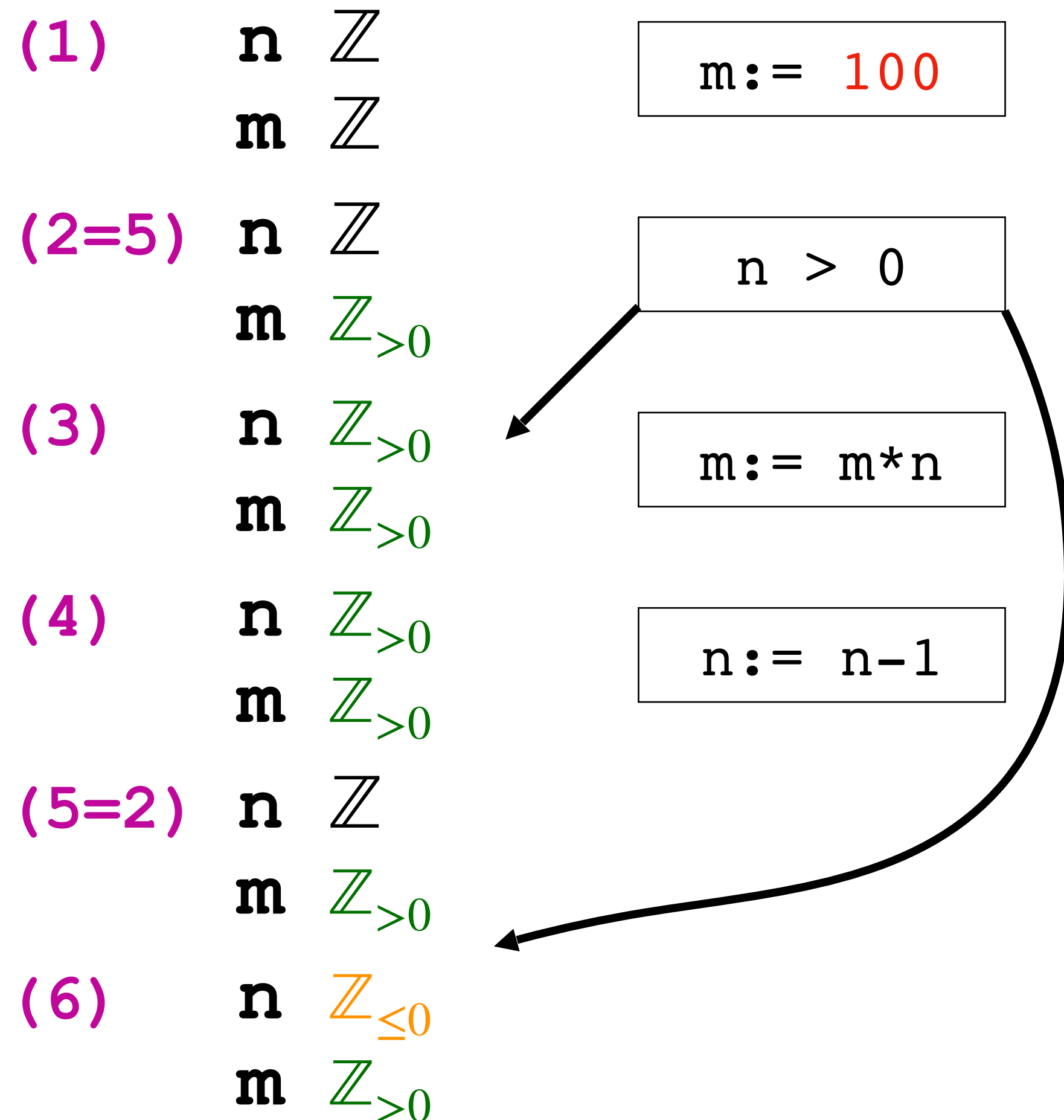
```
(1) m := 1;  
(2) while n > 0 do {  
(3)     m := m * n;  
(4)     n := n - 1  
(5=2) }  
(6) // end
```

Computes $k!$
(for k the initial value of n)



Example: Sign analysis

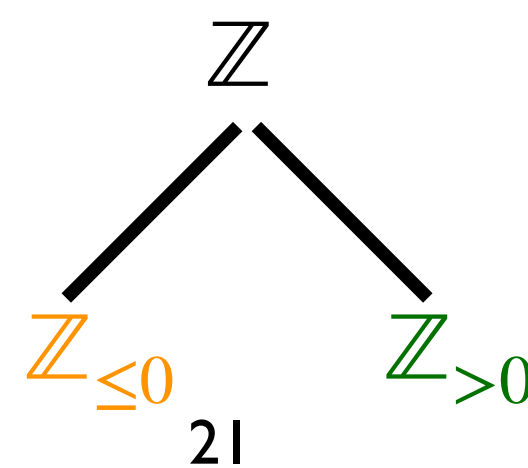
Sign Invariants



Same analysis

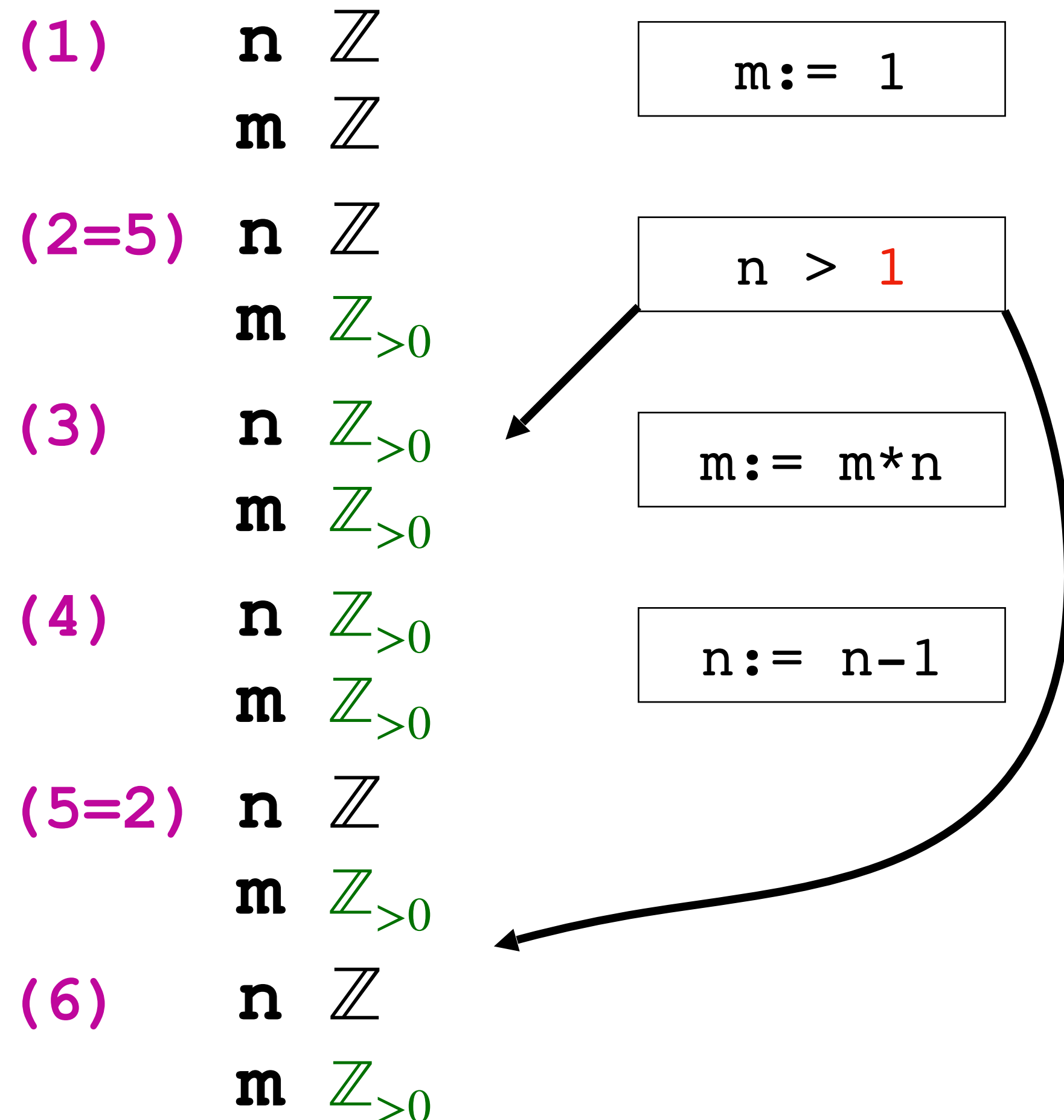
```
(1) m := 100;  
(2) while n > 0 do {  
(3)     m := m * n;  
(4)     n := n - 1  
(5=2) }  
(6) // end
```

Computes $100(k!)$
(for k the initial value of n)



Example: Sign analysis

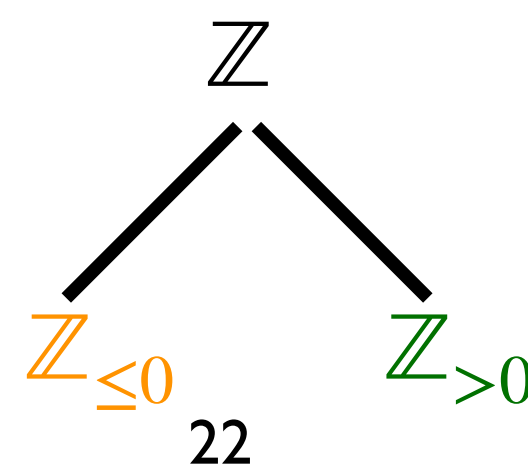
Sign Invariants



Best we can say
($n \leq 1$ at the end)

```
(1) m := 1;  
(2) while n > 1 do {  
(3)     m := m * n;  
(4)     n := n - 1  
(5=2) }  
(6) // end
```

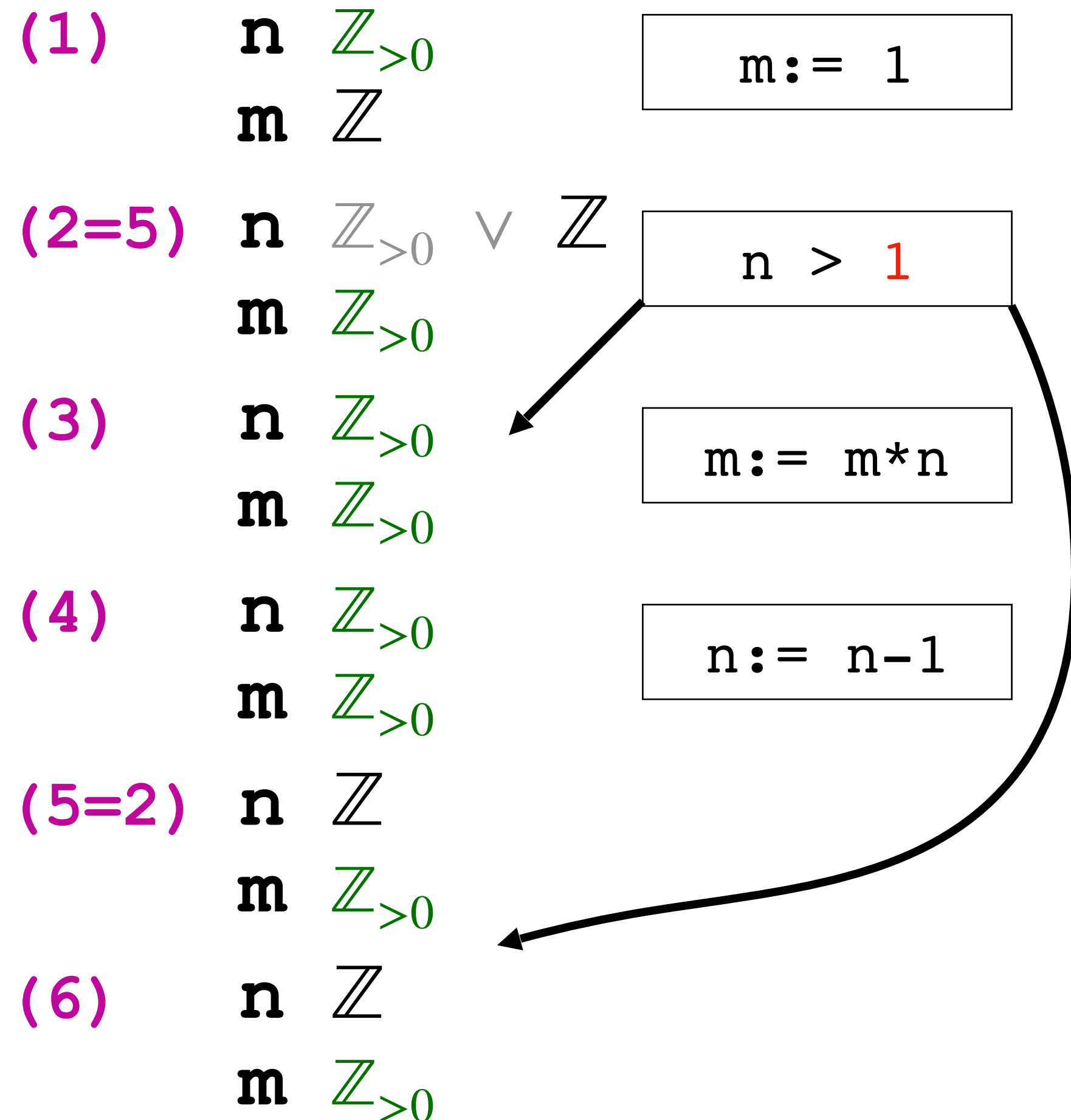
Computes $k!$
(for k the initial value of n)



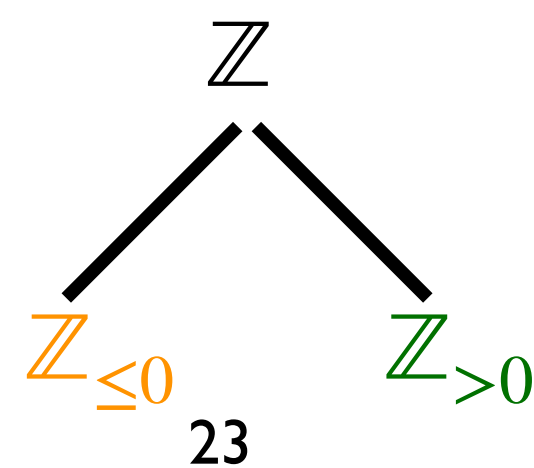
Example: Sign analysis

Sign Invariants

$\{n > 0\}$



Precision loss:
 $(n = 1 \text{ at the end})$
 $\mathbb{Z}$ instead of $\mathbb{Z}_{>0}$



```
(1) m := 1;
(2) while n > 1 do {
(3)     m := m * n;
(4)     n := n - 1
(5=2) }
(6) // end
```

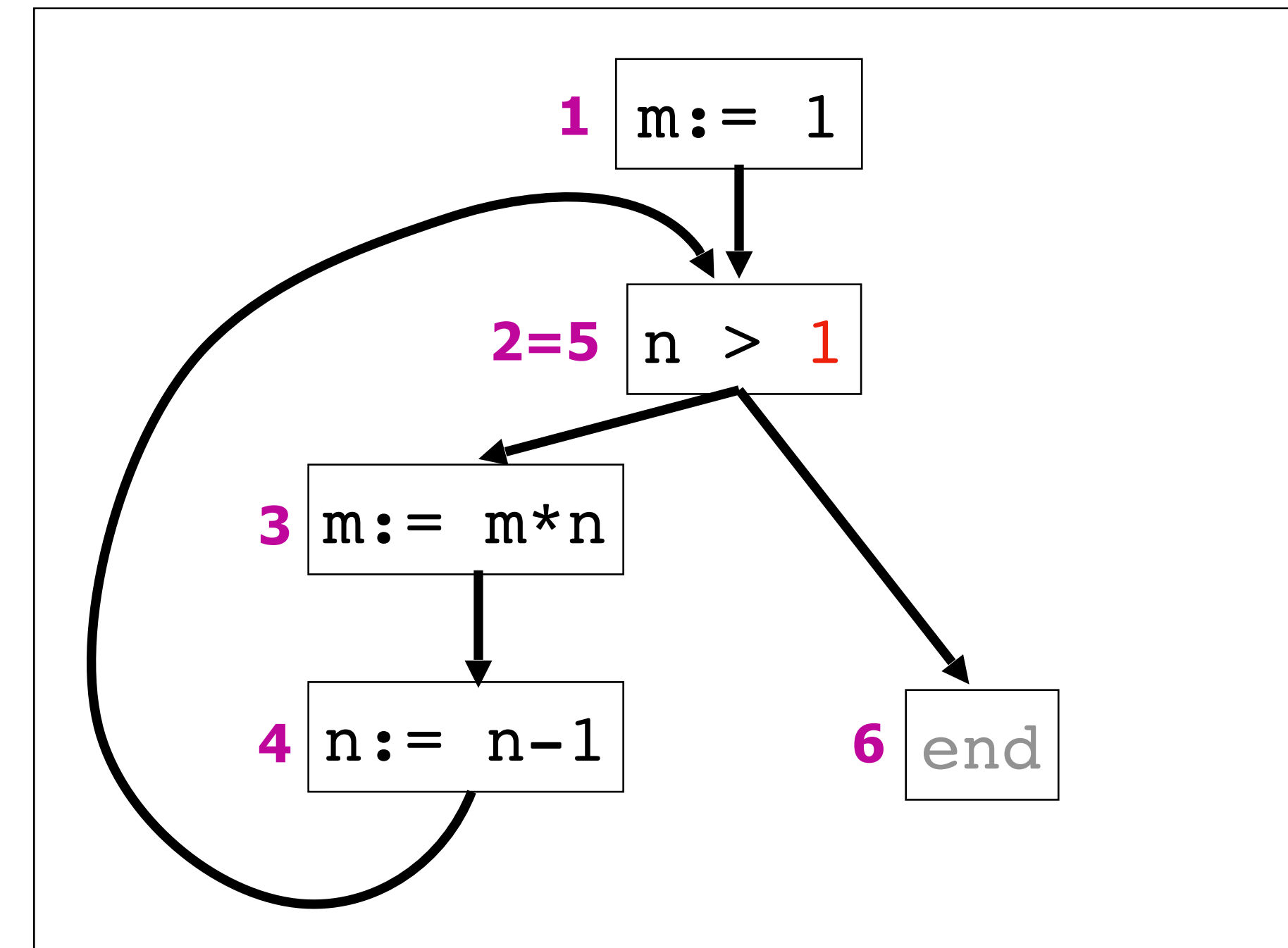
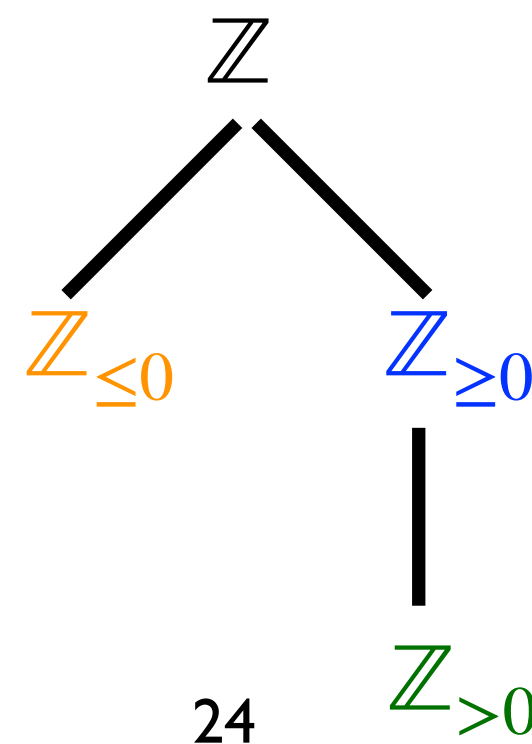
Computes $k!$
 (for k the initial value of n)

Domain refinement

To increase precision about the final result, we can refine the “**sign information**”.

Each program variable can now take one of these “abstract values”:

$\mathbb{Z}_{\geq 0}$	its actual value is not negative
$\mathbb{Z}_{> 0}$	its actual value is positive
$\mathbb{Z}_{\leq 0}$	its actual value is not positive
$\mathbb{Z}$	any value (don't know)

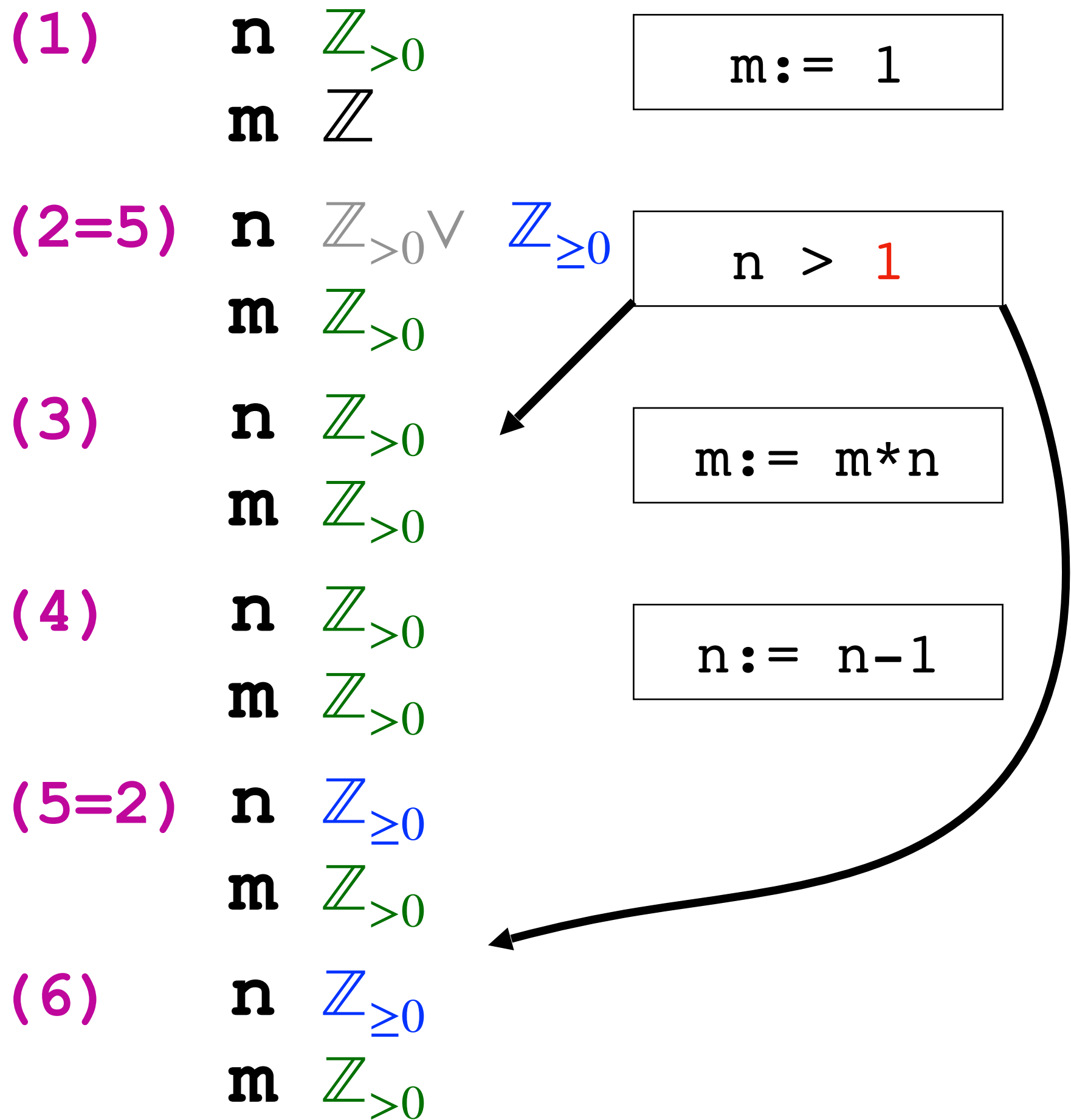


Control flow graph

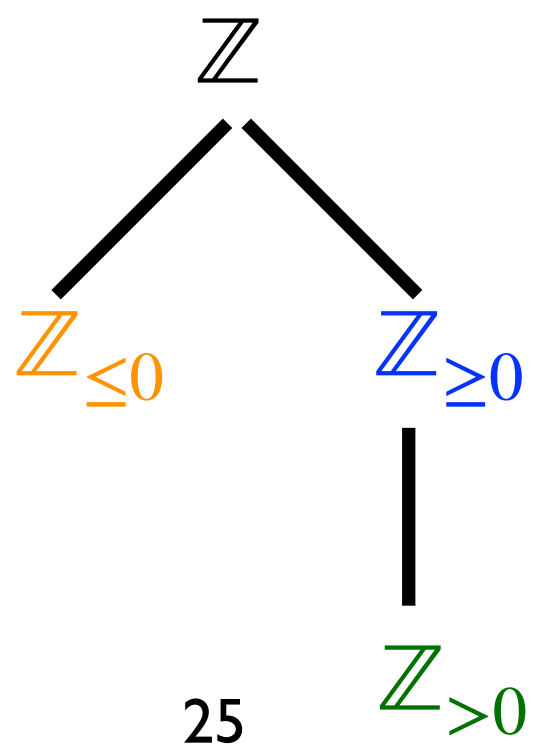
Example: Refined sign analysis

Sign Invariants

$\{n > 0\}$



Precision loss:
($n = 1$ at the end)
 $\mathbb{Z}_{\geq 0}$ instead of $\mathbb{Z}_{>0}$



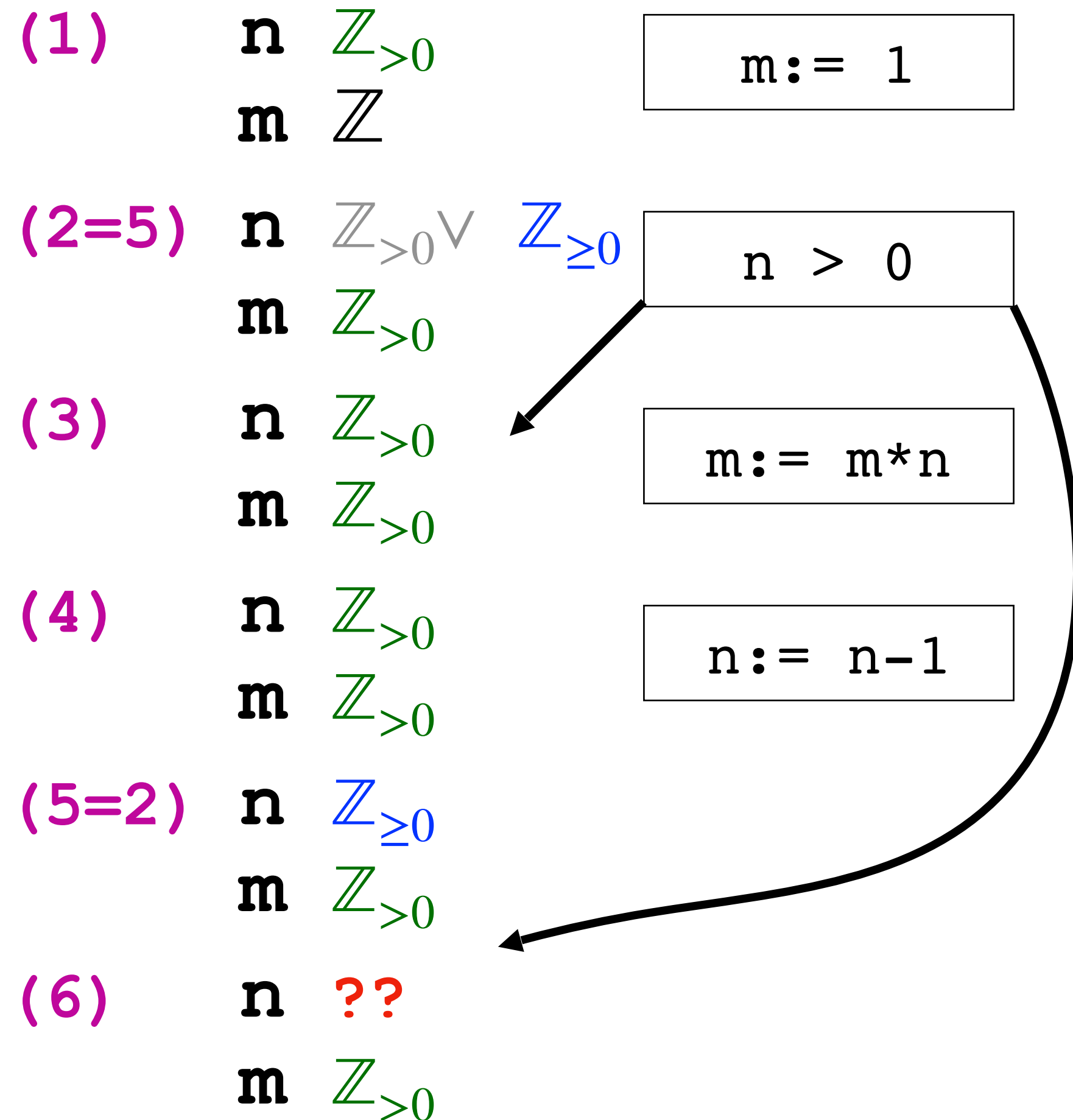
```
(1) m := 1;  
(2) while n > 1 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes $k!$
(for k the initial value of n)

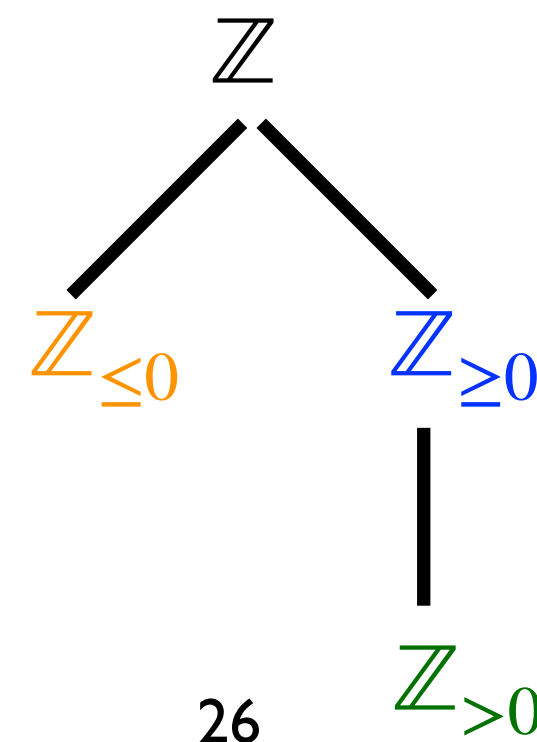
Example: Refined sign analysis

Sign Invariants

$\{n > 0\}$



Which result?
($n = 0$ at the end)
 $\mathbb{Z}_{\leq 0}$ or $\mathbb{Z}_{\geq 0}$?



```
(1) m := 1;  
(2) while n > 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

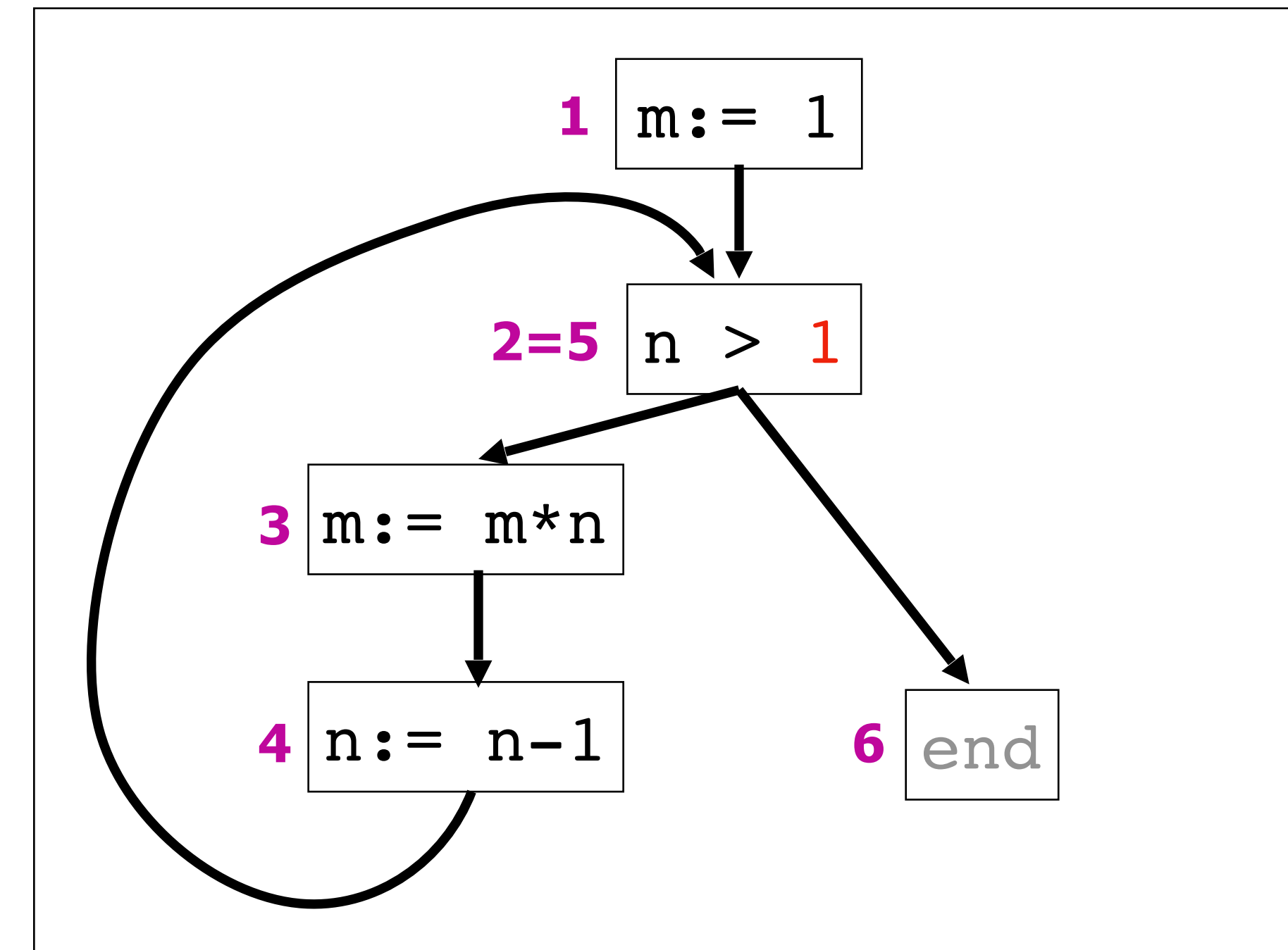
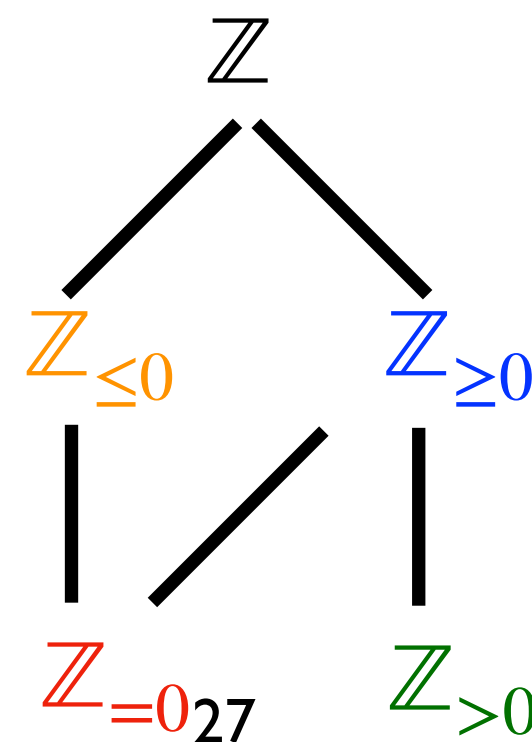
Computes $k!$
(for k the initial value of n)

Domain refinement 2

To increase precision about the final result, we can refine the “**sign information**”.

Each program variable can now take one of these “abstract values”:

$\mathbb{Z}_{=0}$	its actual value is zero
$\mathbb{Z}_{\geq 0}$	its actual value is not negative
$\mathbb{Z}_{> 0}$	its actual value is positive
$\mathbb{Z}_{\leq 0}$	its actual value is not positive
$\mathbb{Z}$	any value (don't know)

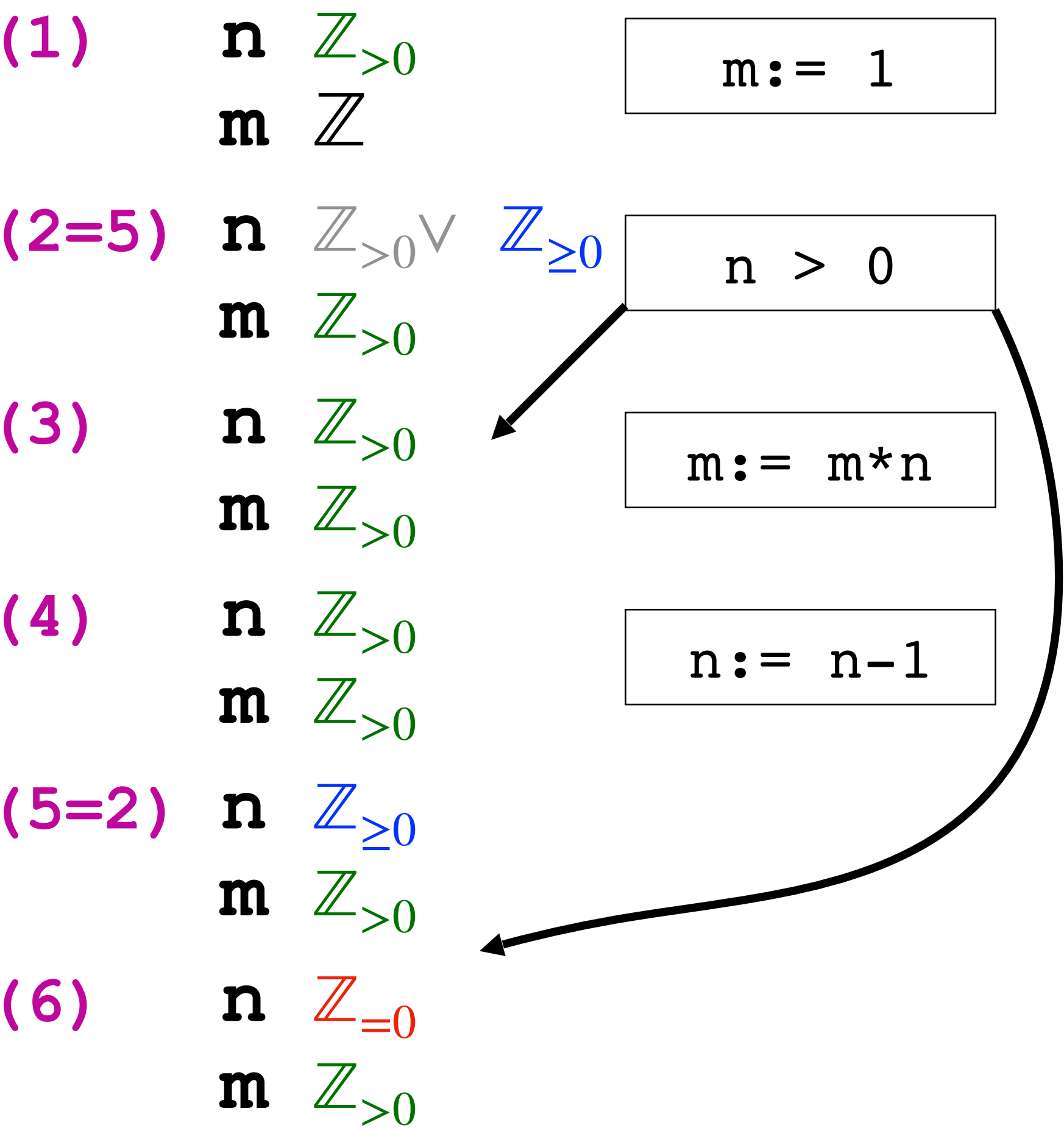


Control flow graph

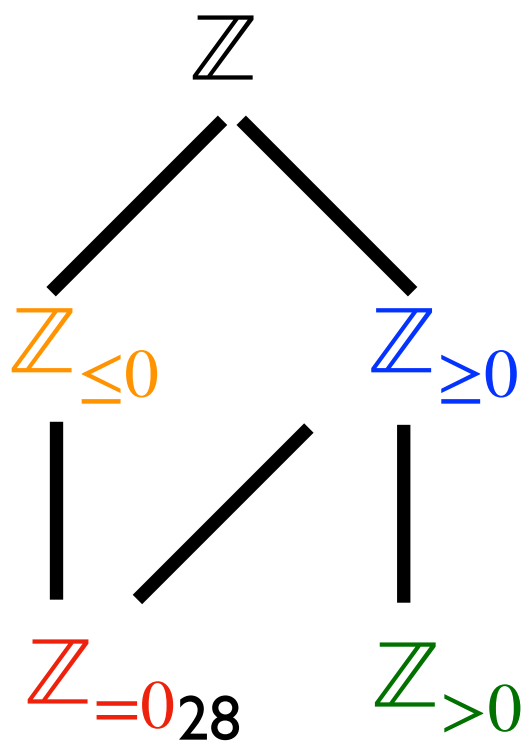
Example: Refined sign analysis 2

Sign Invariants

$\{n > 0\}$



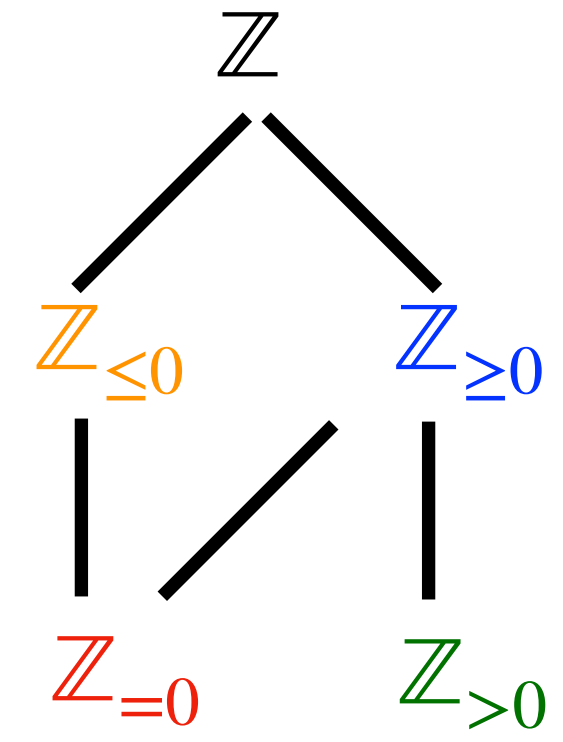
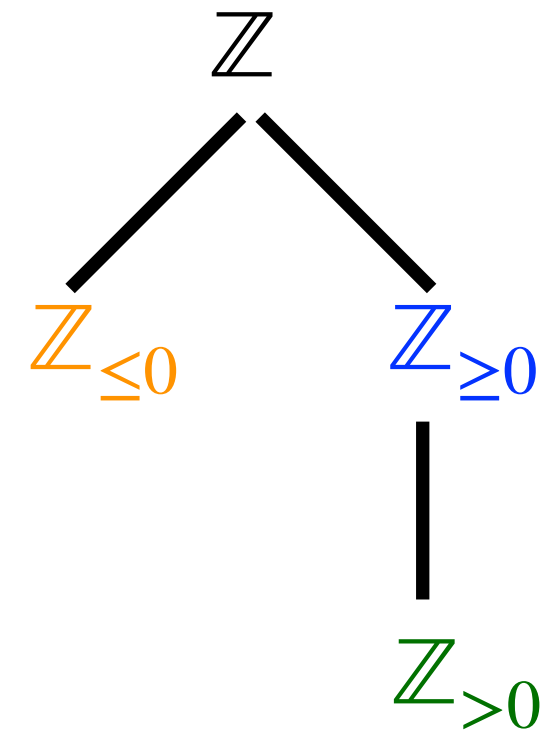
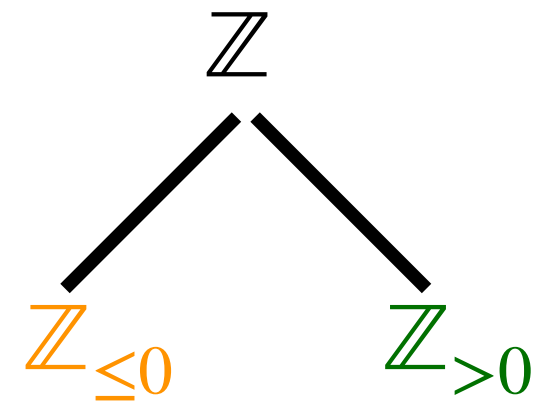
Best precision



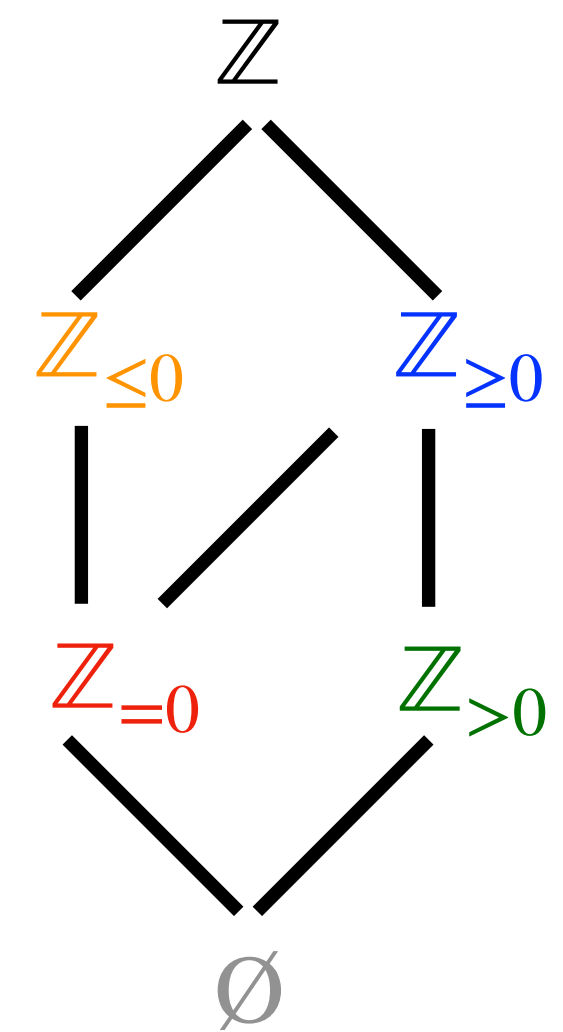
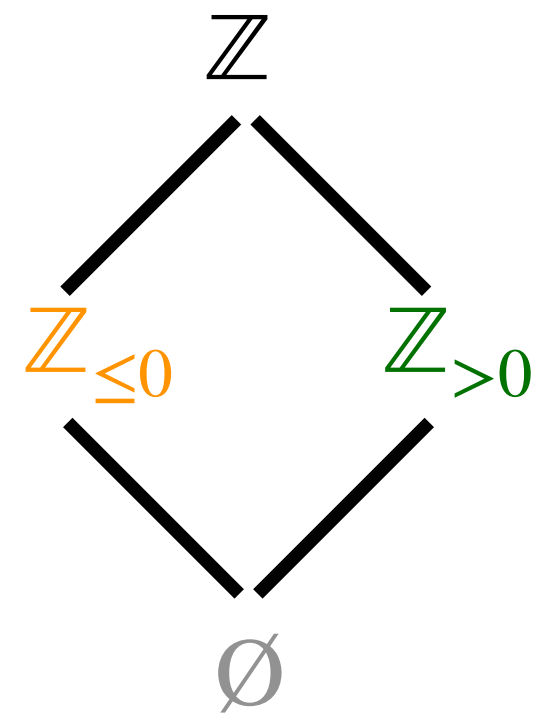
```
(1) m := 1;  
(2) while n > 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes $k!$
(for k the initial value of n)

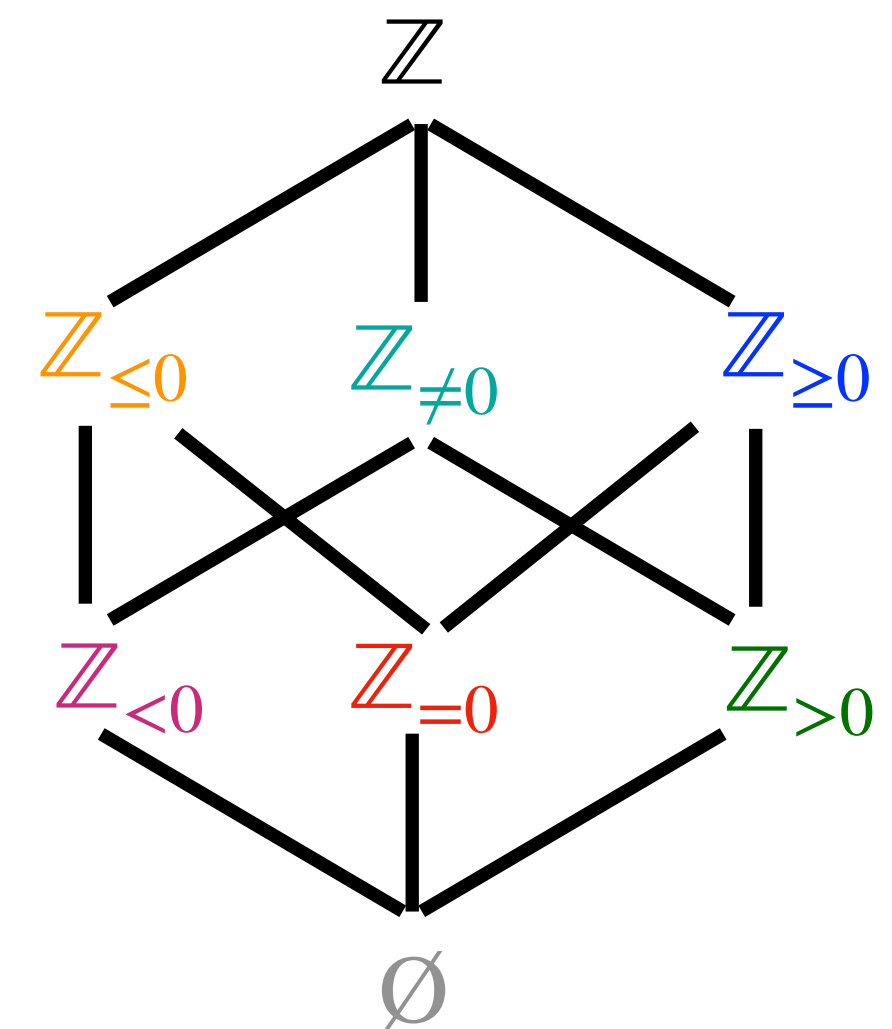
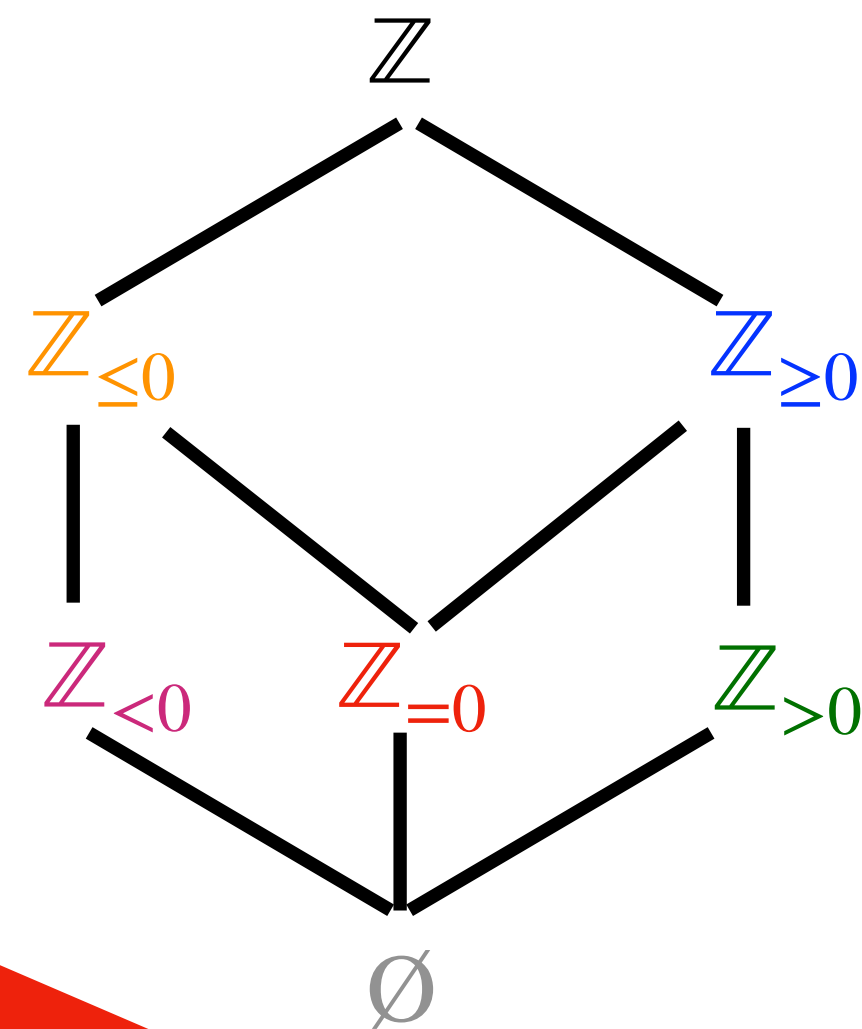
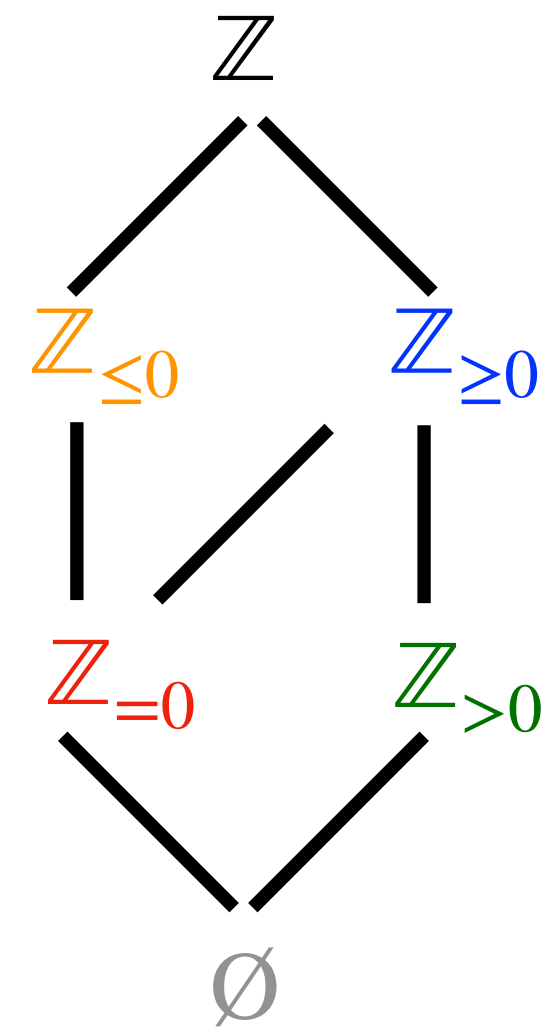
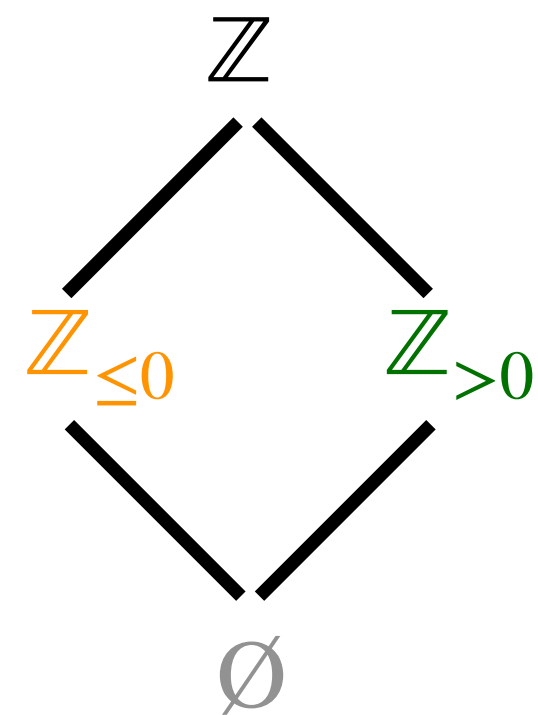
Which domain?



Must be closed by meet!



Which domain?



moving to coarser domains
(not necessarily less precise analysis!)

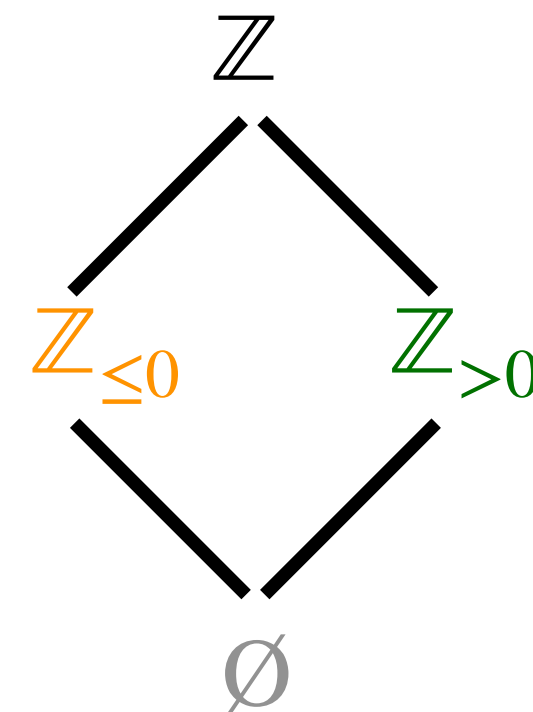
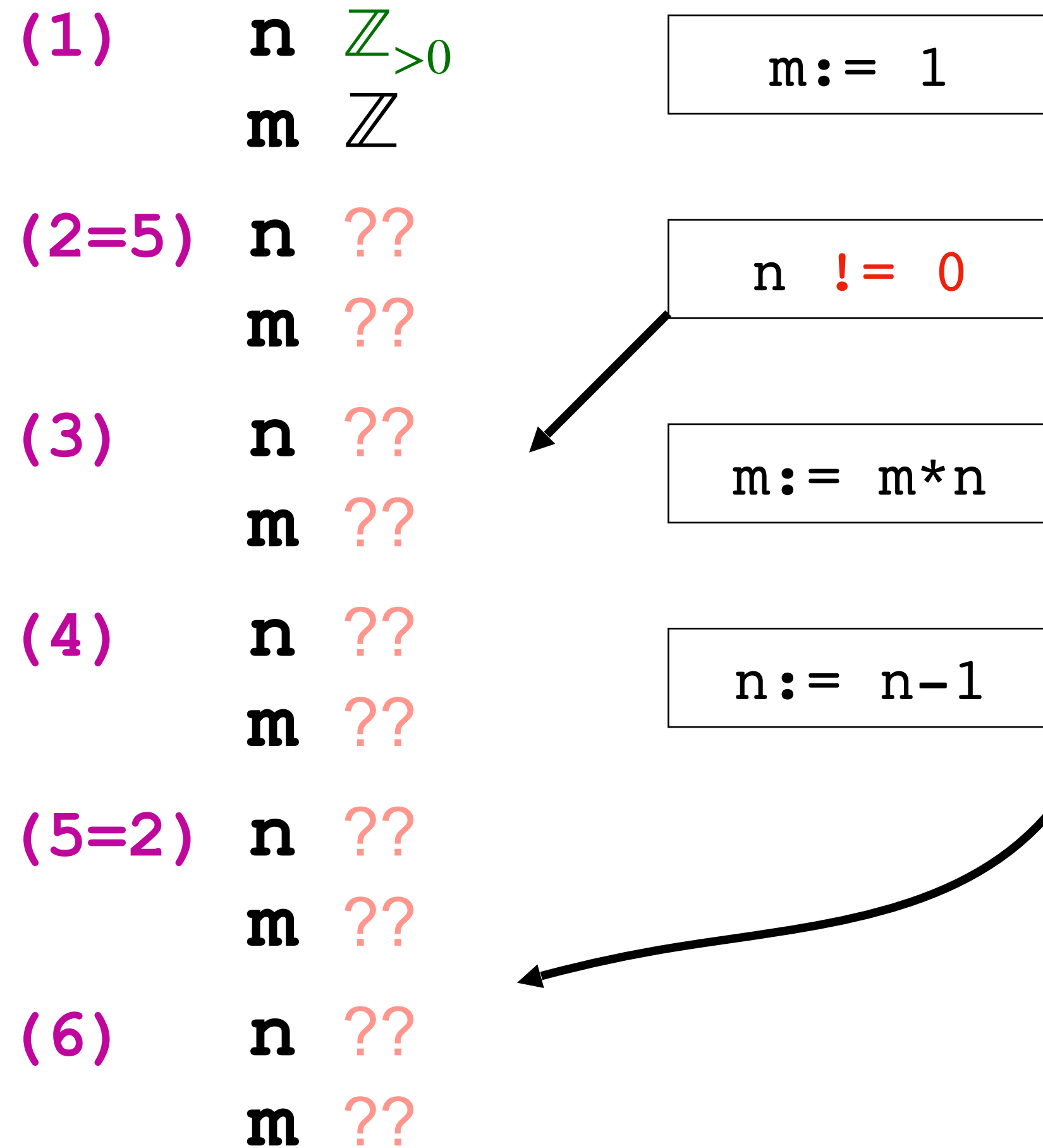
moving to finer domains
(not necessarily more precise analysis!)

Exercises

Which result? 1

Sign Invariants

$\{n > 0\}$



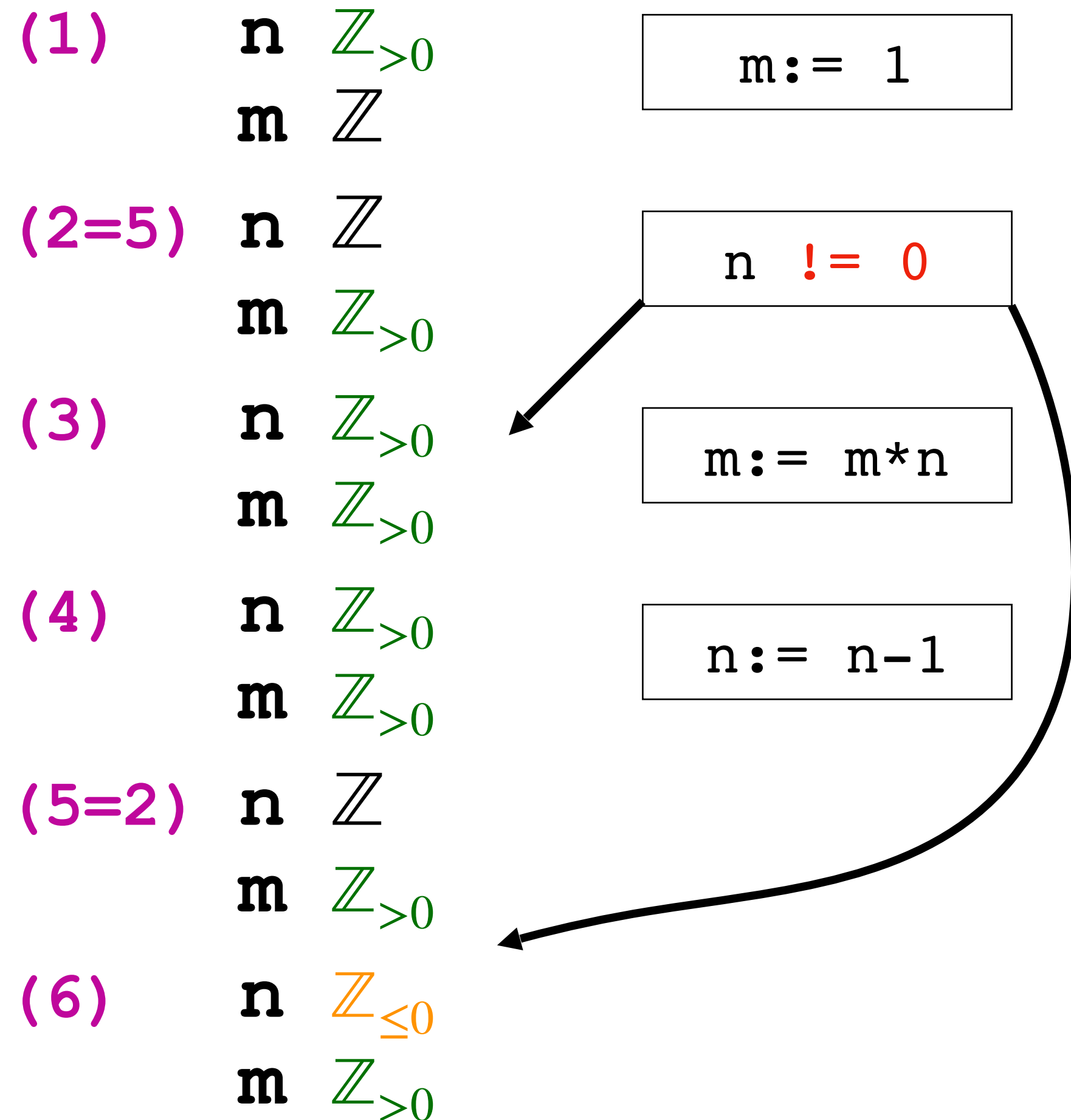
```
(1) m := 1;  
(2) while n != 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes $k!$
(for k the initial value of n)

Which result? 1

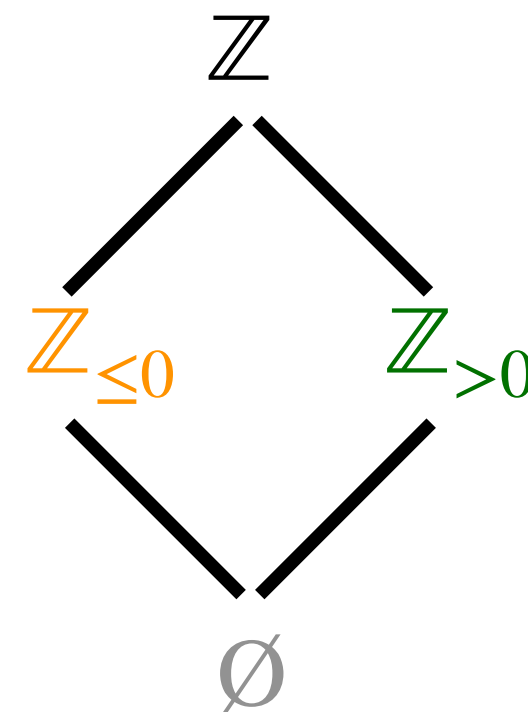
Sign Invariants

$\{n > 0\}$



```
(1) m := 1;  
(2) while n != 0 do {  
(3)     m := m * n;  
(4)     n := n - 1  
(5=2) }  
(6) // end
```

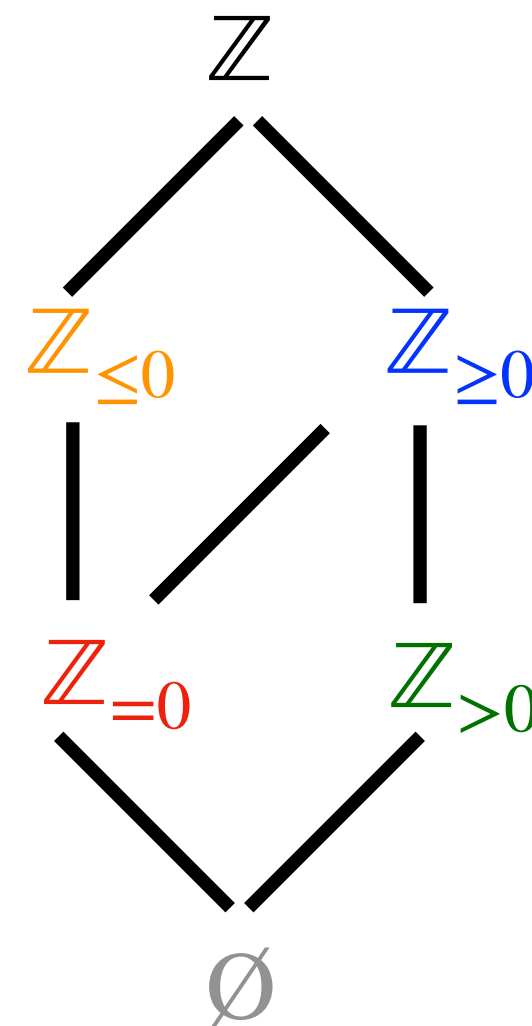
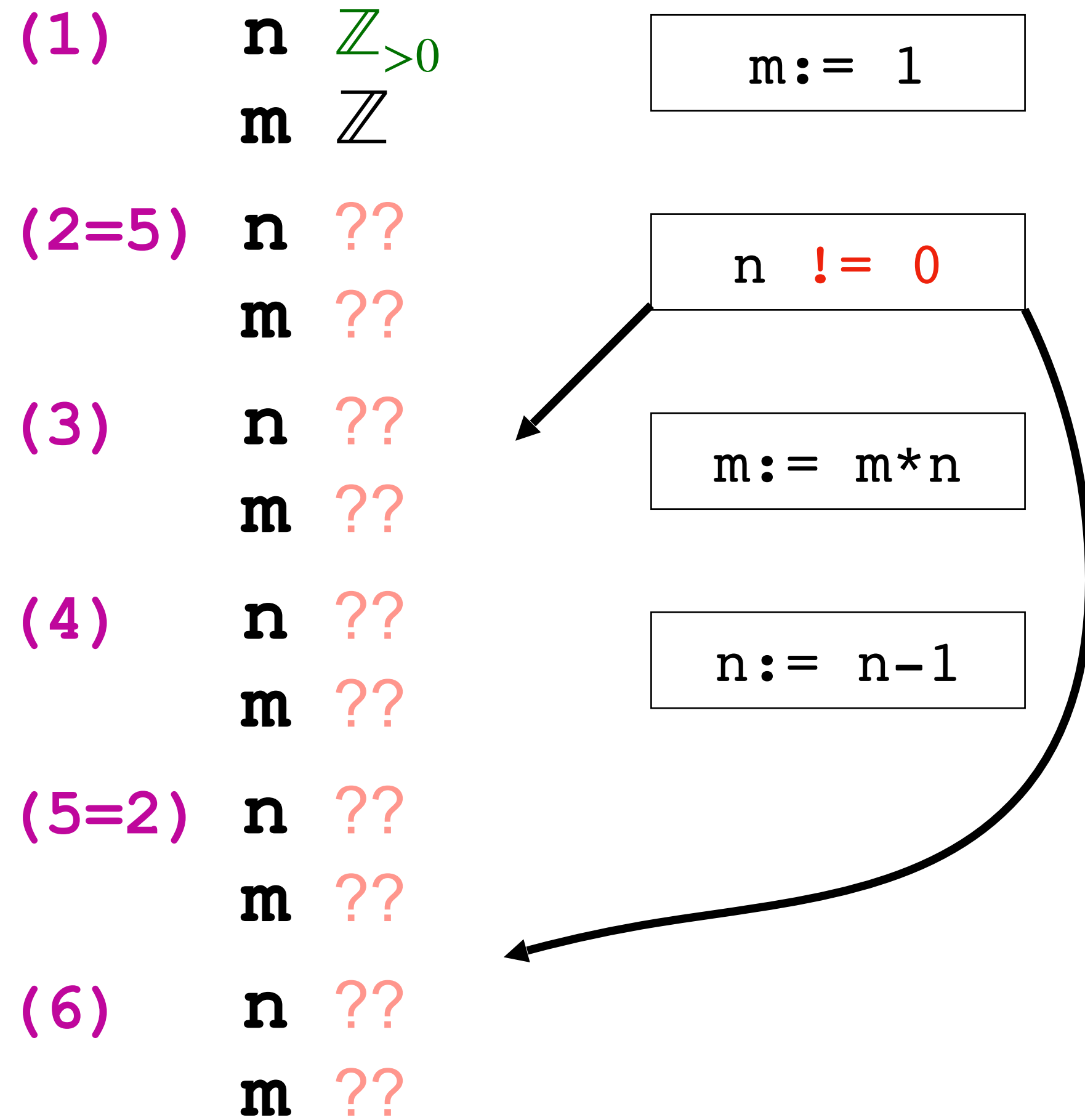
Computes $k!$
(for k the initial value of n)



Which result? 2

Sign Invariants

$\{n > 0\}$



```

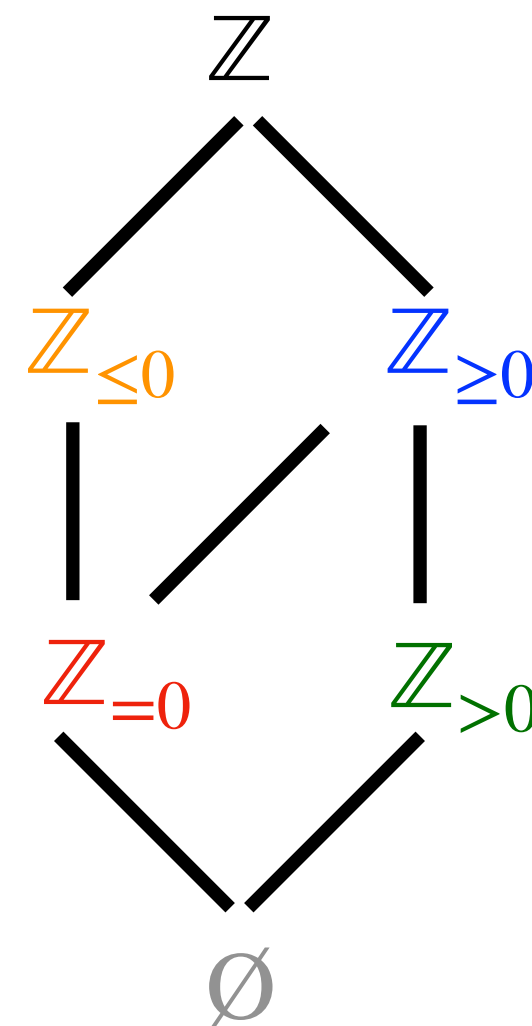
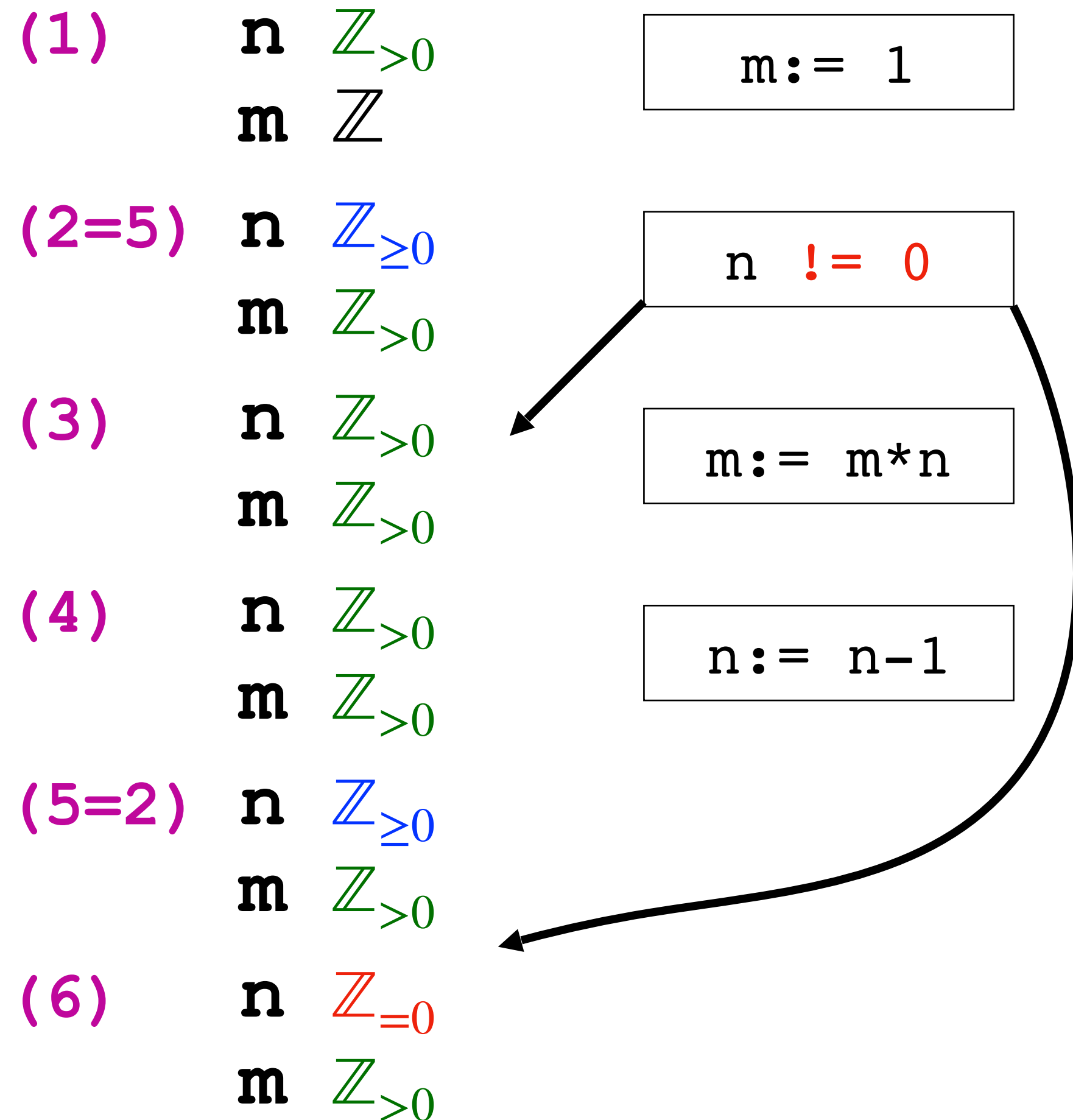
(1) m := 1;
(2) while n != 0 do {
(3)     m := m * n;
(4)     n := n - 1
(5=2) }
(6) // end
  
```

Computes $k!$
 (for k the initial value of n)

Which result? 2

Sign Invariants

$\{n > 0\}$



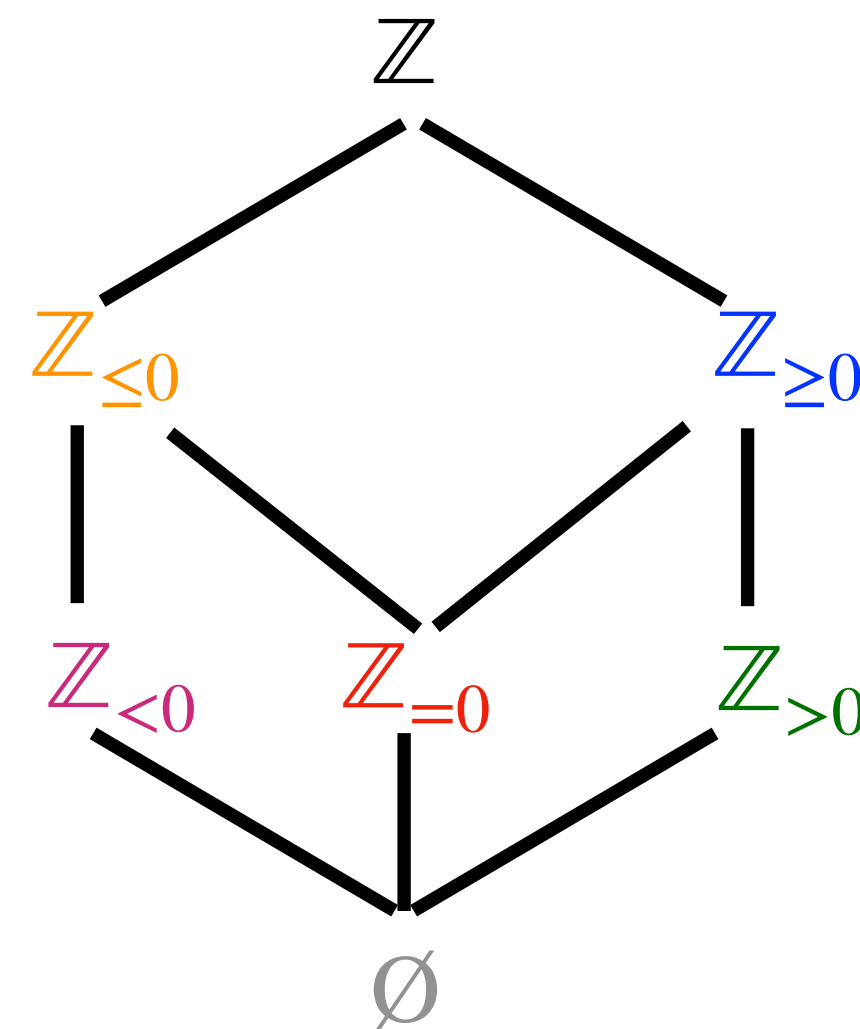
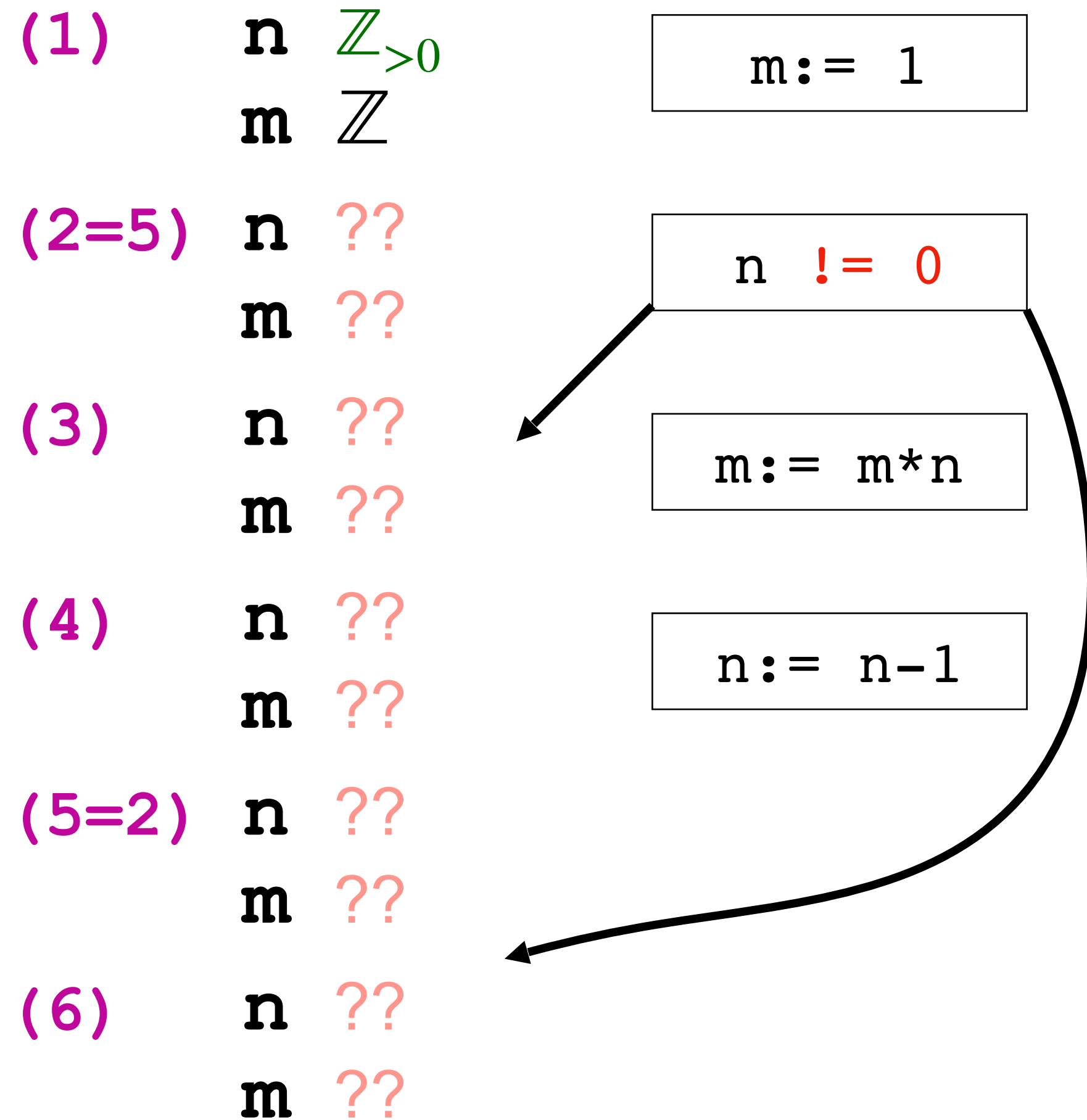
```
(1) m := 1;  
(2) while n != 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes $k!$
(for k the initial value of n)

Which result? 3

Sign Invariants

$\{n > 0\}$



```

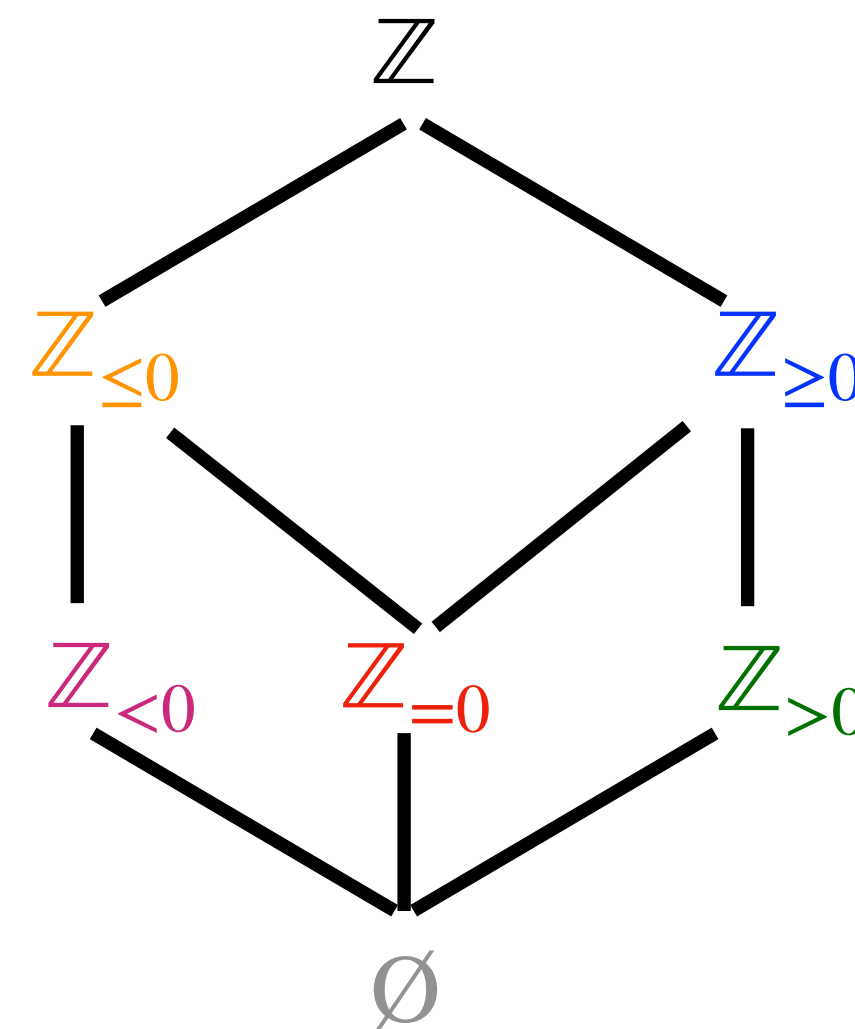
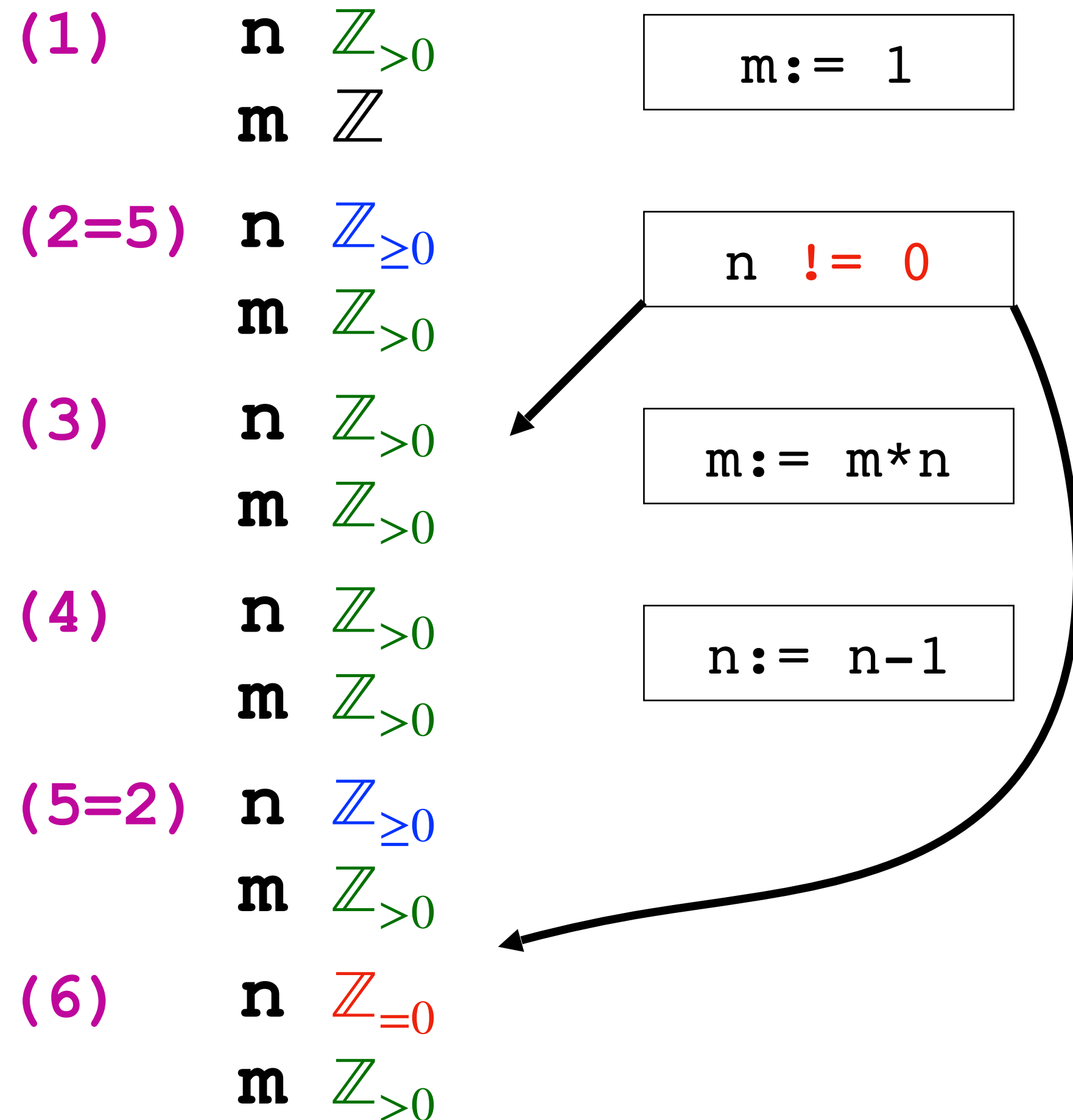
(1) m := 1;
(2) while n != 0 do {
(3)     m := m * n;
(4)     n := n - 1
(5=2) }
(6) // end
    
```

Computes $k!$
 (for k the initial value of n)

Which result? 3

Sign Invariants

$\{n > 0\}$



```
(1) m := 1;
(2) while n != 0 do {
(3)     m := m * n;
(4)     n := n - 1
(5=2) }
(6) // end
```

Computes $k!$
 (for k the initial value of n)

Which result? 4

Sign Invariants

{true}

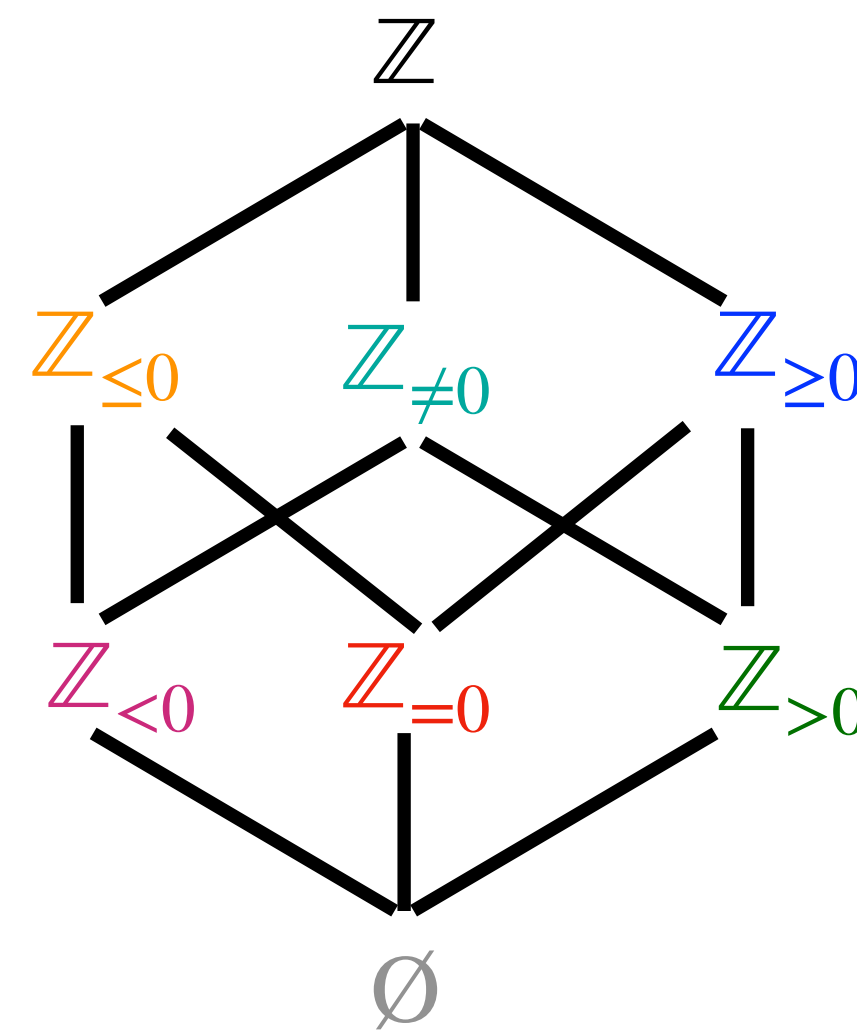
m := 1

n != 0

m := m * n

n := n - 1

(1) n $\mathbb{Z}$
m $\mathbb{Z}$
(2=5) n ??
m ??
(3) n ??
m ??
(4) n ??
m ??
(5=2) n ??
m ??
(6) n ??
m ??



```
(1) m := 1;  
(2) while n != 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes k!
(for k the initial value of n)

Which result? 4

Sign Invariants

{true}

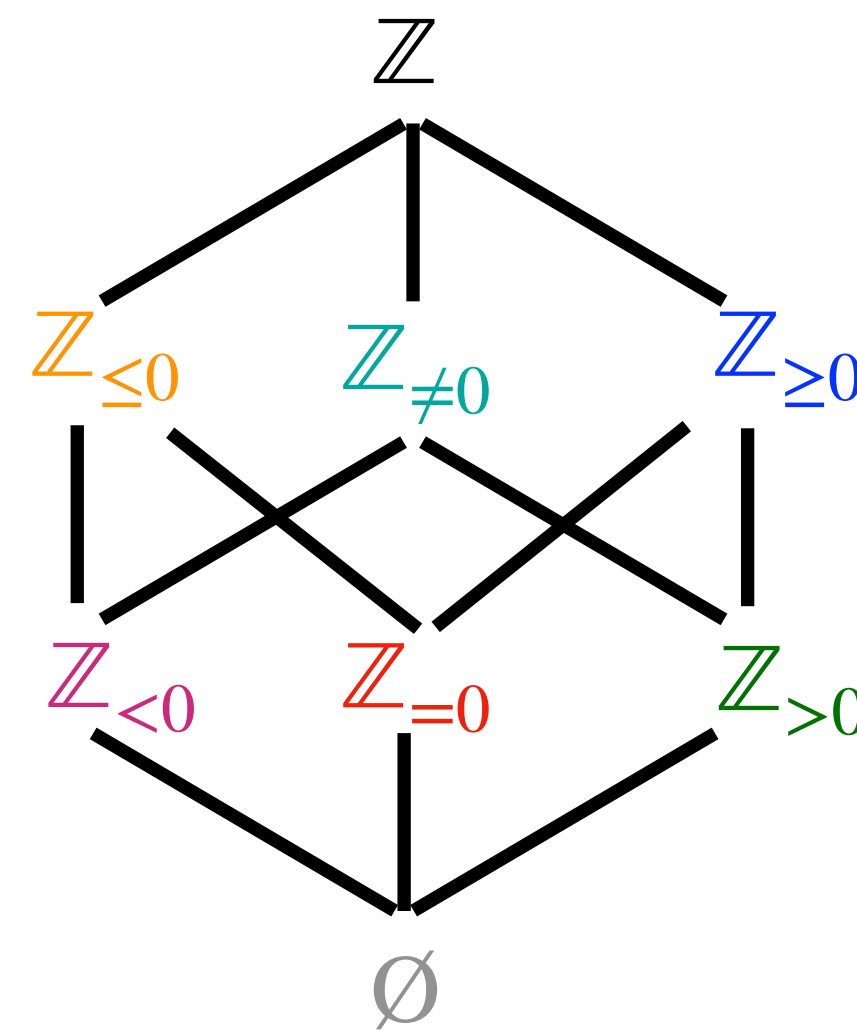
m := 1

n != 0

m := m * n

n := n - 1

(1) n $\mathbb{Z}$
m $\mathbb{Z}$
(2=5) n $\mathbb{Z}$
m $\mathbb{Z}_{\neq 0}$
(3) n $\mathbb{Z}_{\neq 0}$
m $\mathbb{Z}_{> 0}$
(4) n $\mathbb{Z}_{\neq 0}$
m $\mathbb{Z}_{\neq 0}$
(5=2) n $\mathbb{Z}$
m $\mathbb{Z}_{\neq 0}$
(6) n $\mathbb{Z}_{= 0}$
m $\mathbb{Z}_{\neq 0}$



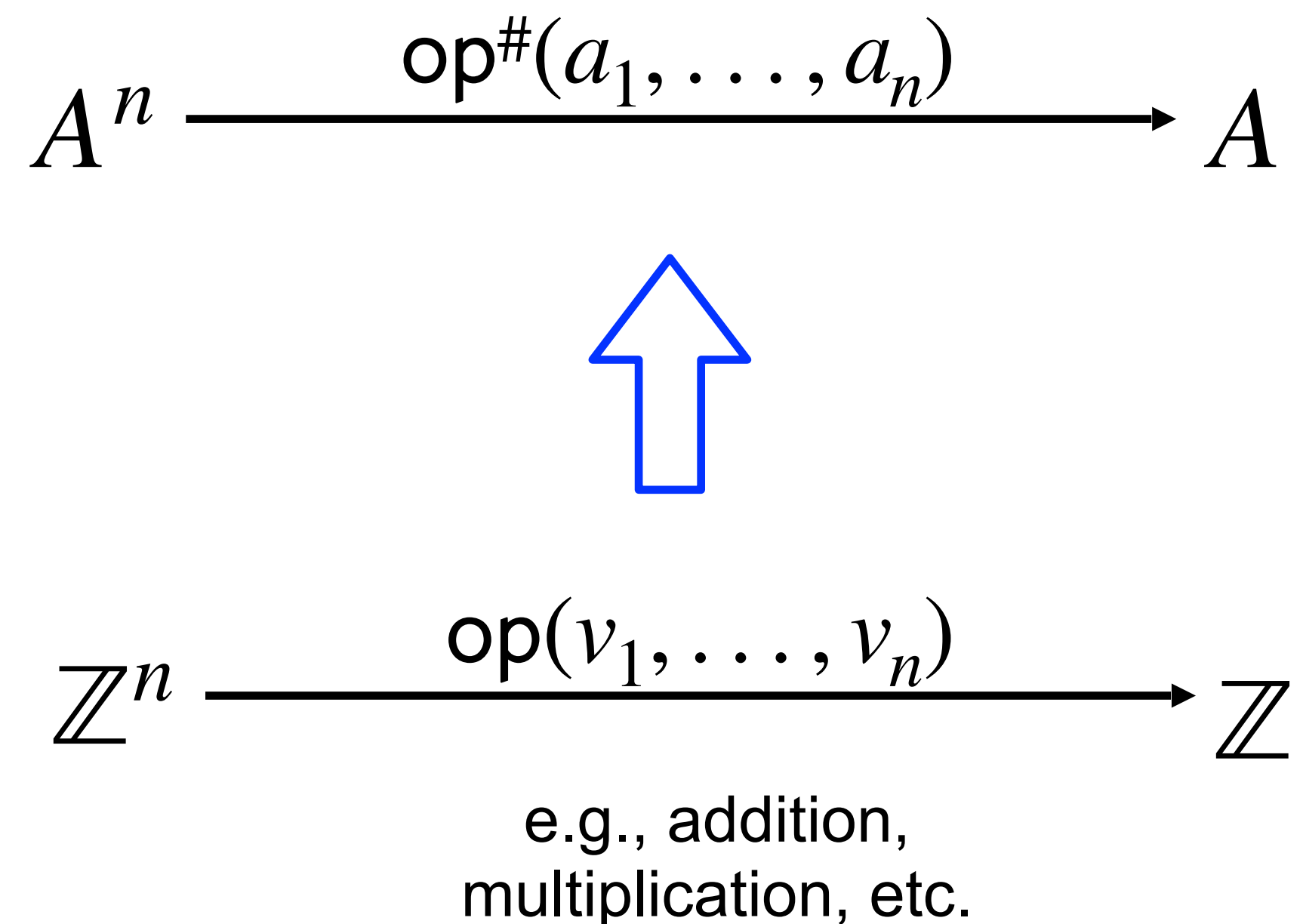
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(1) m := 1;  
(2) while n != 0 do {  
(3)   m := m * n;  
(4)   n := n - 1  
(5=2) }  
(6) // end
```

Computes k! or diverges
(for k the initial value of n)

Abstract operations

Transfer functions

A **transfer function** defines how the abstract state of a program changes after the execution of an instruction

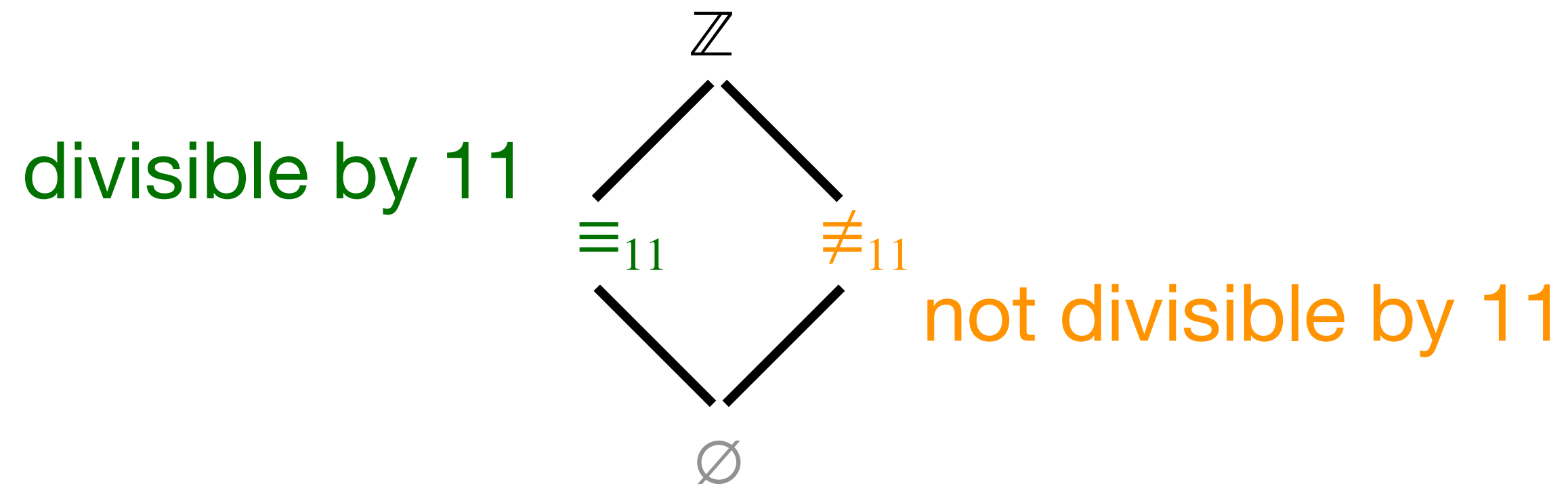


Soundness requirement:

if a_i is a sound abstraction of v_i for any i ,

then $a = \text{op}^\#(a_1, \dots, a_n)$ must be a **sound** abstraction of $v = \text{op}(v_1, \dots, v_n)$

Examples



$$\neq_{11} \times^{\#} \neq_{11} = \neq_{11} \quad \text{sound}$$

$$\neq_{11} \times^{\#} \neq_{11} = \mathbb{Z} \quad \text{sound}$$

$$\neq_{11} \times^{\#} \mathbb{Z} = \neq_{11} \quad \text{unsound}$$

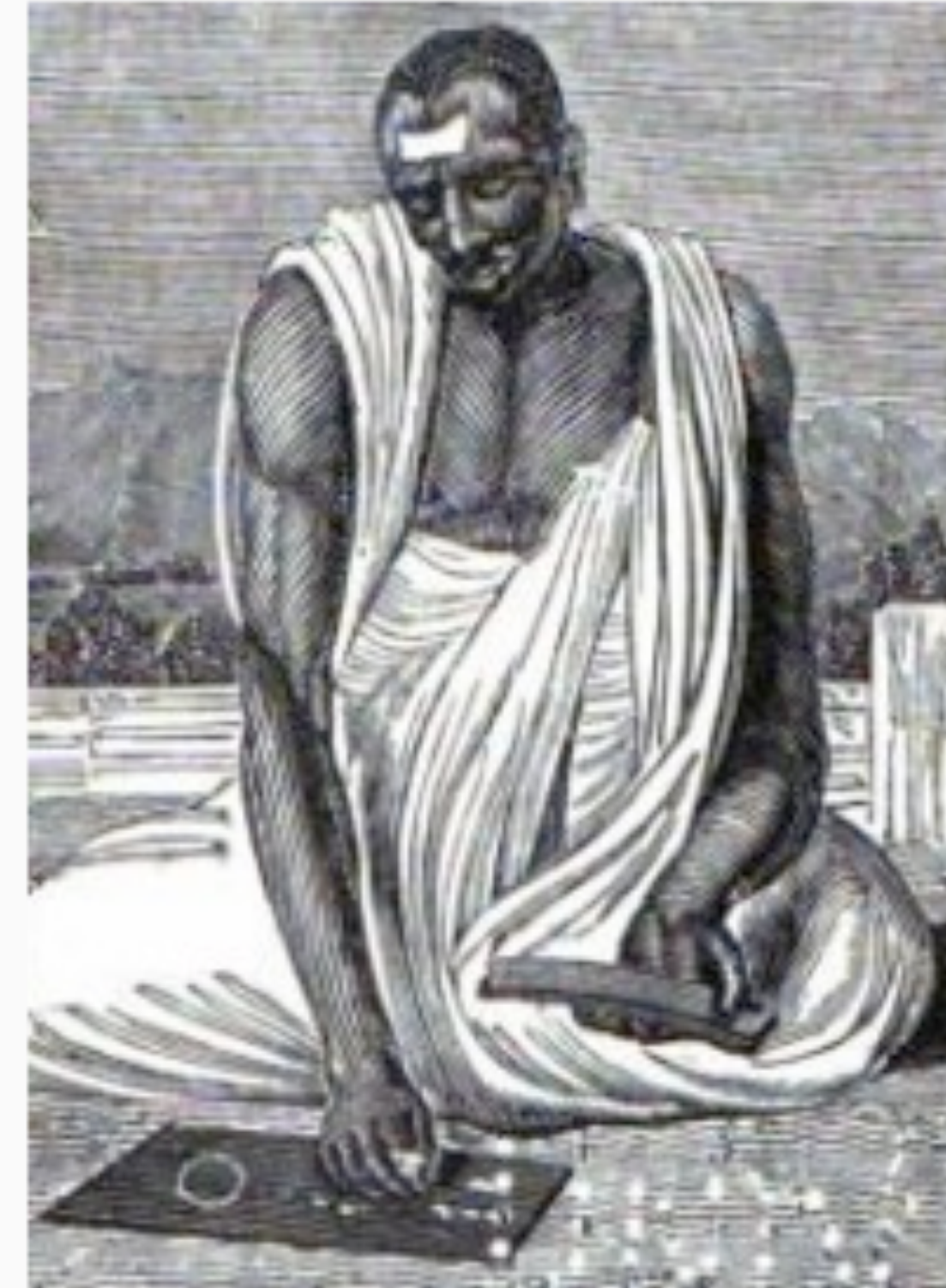
$$\equiv_{11} \times^{\#} \mathbb{Z} = \neq_{11} \quad \text{unsound}$$

$$\equiv_{11} \times^{\#} \mathbb{Z} = \equiv_{11} \quad \text{sound}$$

Brahmagupta

Brahmagupta (Sanskrit: ब्रह्मगुप्त; (598–c.670 CE) was an Indian mathematician and astronomer who wrote two important works on Mathematics and Astronomy: the *Brāhmasphuṭasiddhānta* (Extensive Treatise of Brahma) (628), a theoretical treatise, and the *Khaṇḍakhadyaka*, a more practical text.

Brahmagupta



Born	598 CE
Died	c.670 CE
Fields	Mathematics, Astronomy
Known for	Zero, modern Number system

The rule of signs by Brahmagupta (628)

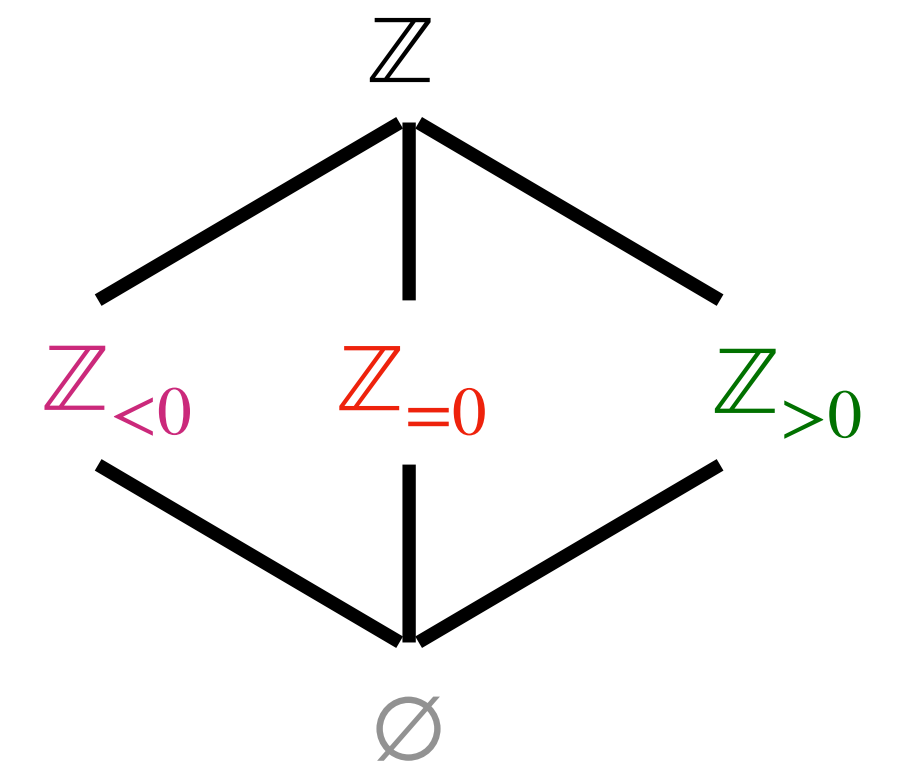
18.30. [The sum] of two positives is positives, of two negatives negative;

[...] $\mathbb{Z}_{>0} +^{\#} \mathbb{Z}_{>0} = \mathbb{Z}_{>0}$ $\mathbb{Z}_{<0} +^{\#} \mathbb{Z}_{<0} = \mathbb{Z}_{<0}$

18.32. A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.

18.33. The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros is zero.

18.34. A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.



Here Brahmagupta states that $\frac{0}{0} = 0$ and as for the question of $\frac{a}{0}$ where $a \neq 0$ he did not commit himself.^[26] His rules for arithmetic on negative numbers and zero are quite close to the modern understanding, except that in modern mathematics division by zero is left undefined.

wrong $\mathbb{Z}_{=0} /^{\#} \mathbb{Z}_{=0} = \mathbb{Z}_{=0}$

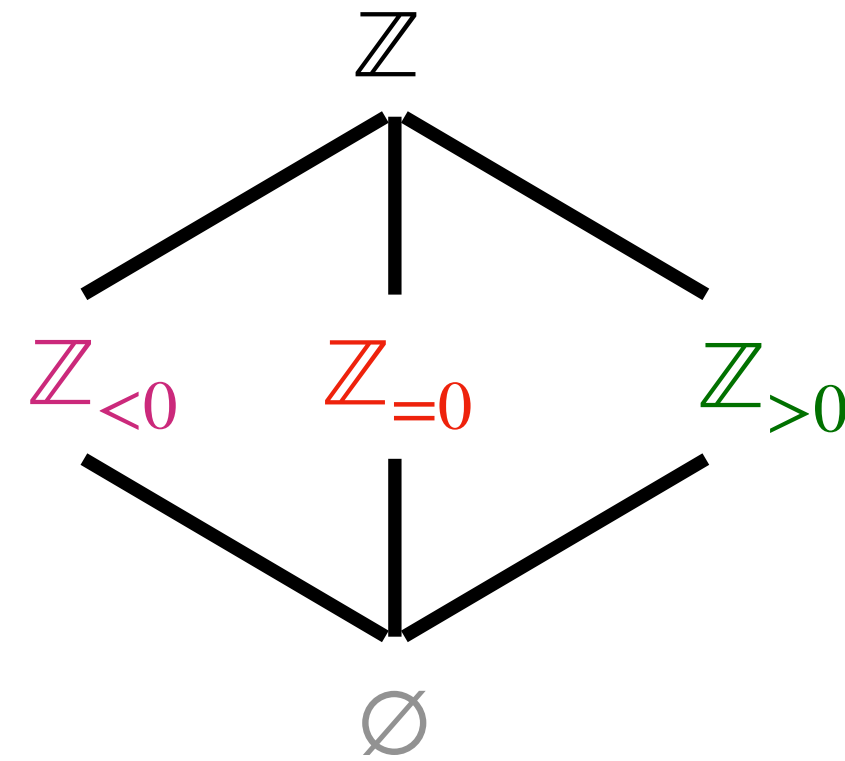
Expressions

$$e ::= n \mid e_1 \times e_2$$

$$\llbracket n \rrbracket = n \in \mathbb{Z}$$

$$\llbracket e_1 \times e_2 \rrbracket = \llbracket e_1 \rrbracket \times \llbracket e_2 \rrbracket$$

$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$		(commutative)
$\mathbb{Z}_{=0}$			
$\mathbb{Z}_{>0}$			



$$A = \{\mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}\}$$

$$\sigma(n) = \mathbb{Z}_{<0} \quad \text{if } n < 0$$

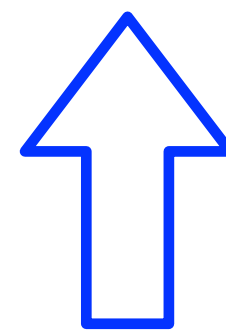
$$\sigma(0) = \mathbb{Z}_{=0}$$

$$\sigma(n) = \mathbb{Z}_{>0} \quad \text{if } n > 0$$

$$\sigma(e_1 \times e_2) = \sigma(e_1) \times^\# \sigma(e_2)$$

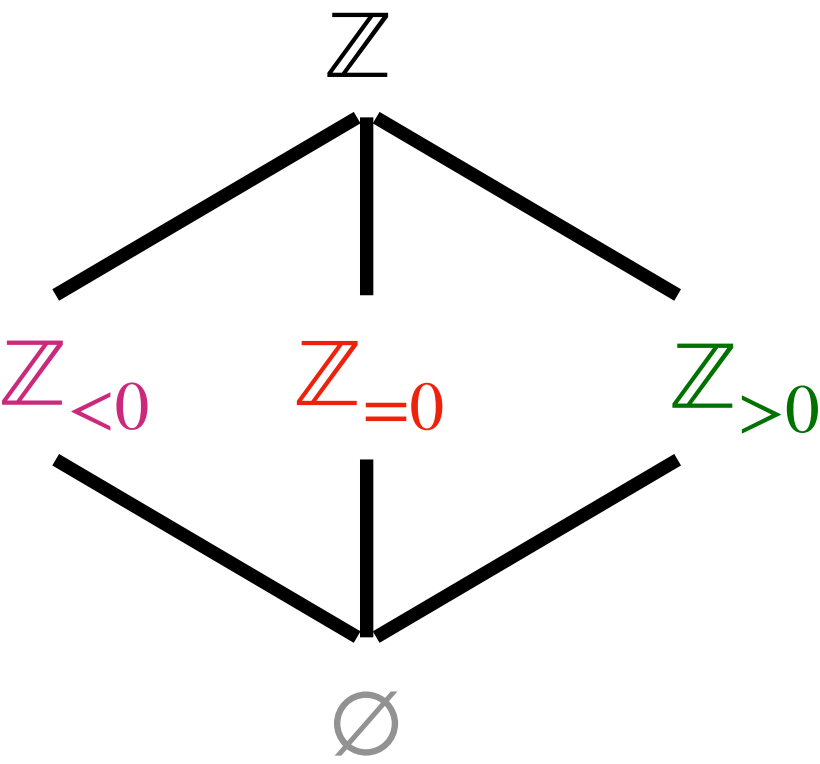
$$a_1 \times^\# a_2 = ?$$

$$A^2 \xrightarrow{a_1 \times^\# a_2} A$$



$$\mathbb{Z}^2 \xrightarrow{v_1 \times v_2} \mathbb{Z}$$

A remark about soundness



$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$	(commutative)	
$\mathbb{Z}_{=0}$			
$\mathbb{Z}_{>0}$			

$\mathbb{Z}_{<0} \times^\# \mathbb{Z}_{<0}$ $\mathbb{Z}_{>0}$

-1

-2

-3

-4

⋮

$\times$

-1

-2

-3

-4

⋮

1, 2, 3, 4, ...

2, 4, 6, 8,

3, 6, 9, 12, ...

4, 8, 12, 16,

⋮⋮⋮

46

Expressions

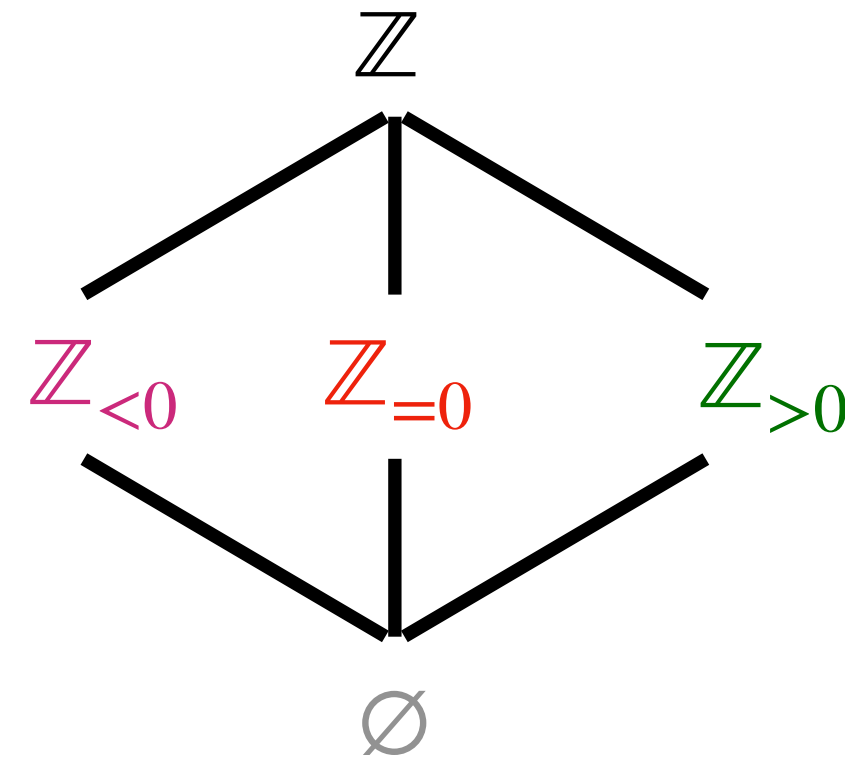
$$e ::= n \mid e_1 \times e_2$$

$$\llbracket n \rrbracket = n \in \mathbb{Z}$$

$$\llbracket e_1 \times e_2 \rrbracket = \llbracket e_1 \rrbracket \times \llbracket e_2 \rrbracket$$

$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$		
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	
$\mathbb{Z}_{>0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$

(commutative)



$$A = \{\mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}\}$$

$$\sigma(n) = \mathbb{Z}_{<0} \quad \text{if } n < 0$$

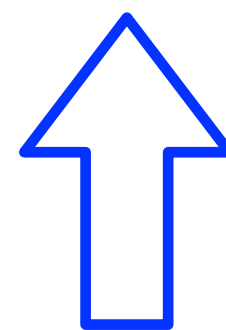
$$\sigma(0) = \mathbb{Z}_{=0}$$

$$\sigma(n) = \mathbb{Z}_{>0} \quad \text{if } n > 0$$

$$\sigma(e_1 \times e_2) = \sigma(e_1) \times^\# \sigma(e_2)$$

$$a_1 \times^\# a_2 = ?$$

$$A^2 \xrightarrow{a_1 \times^\# a_2} A$$



$$\mathbb{Z}^2 \xrightarrow{v_1 \times v_2} \mathbb{Z}$$

Correctness

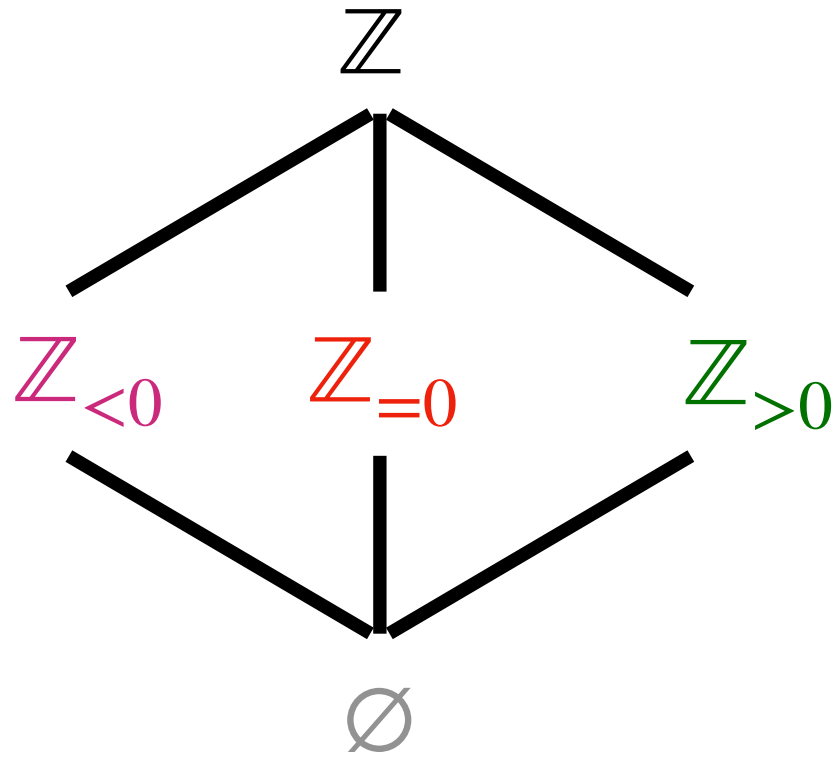
$e ::= n \mid e_1 \times e_2$

$\llbracket n \rrbracket = n \in \mathbb{Z}$

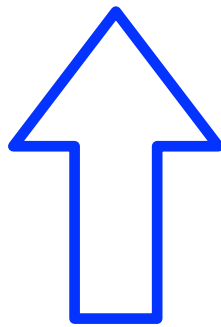
$\llbracket e_1 \times e_2 \rrbracket = \llbracket e_1 \rrbracket \times \llbracket e_2 \rrbracket$

$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$		
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	
$\mathbb{Z}_{>0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$

(commutative)



$A^2 \xrightarrow{a_1 \times^\# a_2} A$



$\mathbb{Z}^2 \xrightarrow{v_1 \times v_2} \mathbb{Z}$

$A = \{\mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}\}$

for any expression e :

$\sigma(e) = \mathbb{Z}_{<0} \Rightarrow^? \llbracket e \rrbracket < 0$

$\sigma(e) = \mathbb{Z}_{=0} \Rightarrow^? \llbracket e \rrbracket = 0$

$\sigma(e) = \mathbb{Z}_{>0} \Rightarrow^? \llbracket e \rrbracket > 0$

(proof by structural induction)

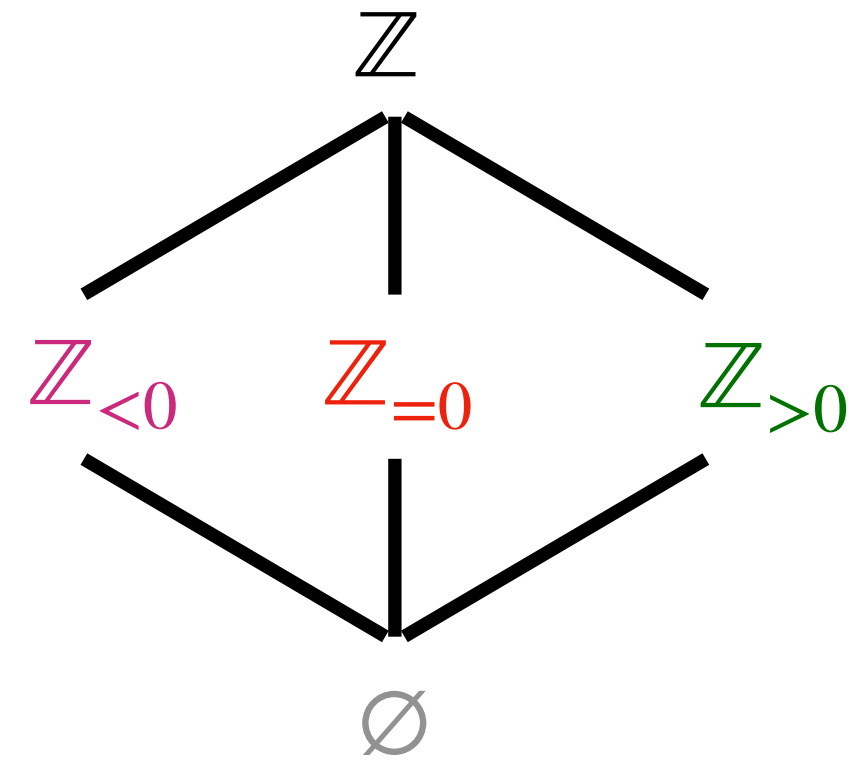
Completeness

$$e ::= n \mid e_1 \times e_2$$

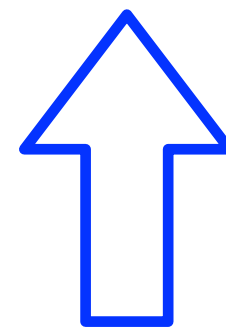
$$\llbracket n \rrbracket = n \in \mathbb{Z}$$

$$\llbracket e_1 \times e_2 \rrbracket = \llbracket e_1 \rrbracket \times \llbracket e_2 \rrbracket$$

$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$		(commutative)
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	
$\mathbb{Z}_{>0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$



$$A^2 \xrightarrow{a_1 \times^\# a_2} A$$



$$\mathbb{Z}^2 \xrightarrow{v_1 \times v_2} \mathbb{Z}$$

$$A = \{\mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}\}$$

for any expression e :

$$\sigma(e) = \mathbb{Z}_{<0} \iff \llbracket e \rrbracket < 0$$

$$\sigma(e) = \mathbb{Z}_{=0} \iff \llbracket e \rrbracket = 0$$

$$\sigma(e) = \mathbb{Z}_{>0} \iff \llbracket e \rrbracket > 0$$

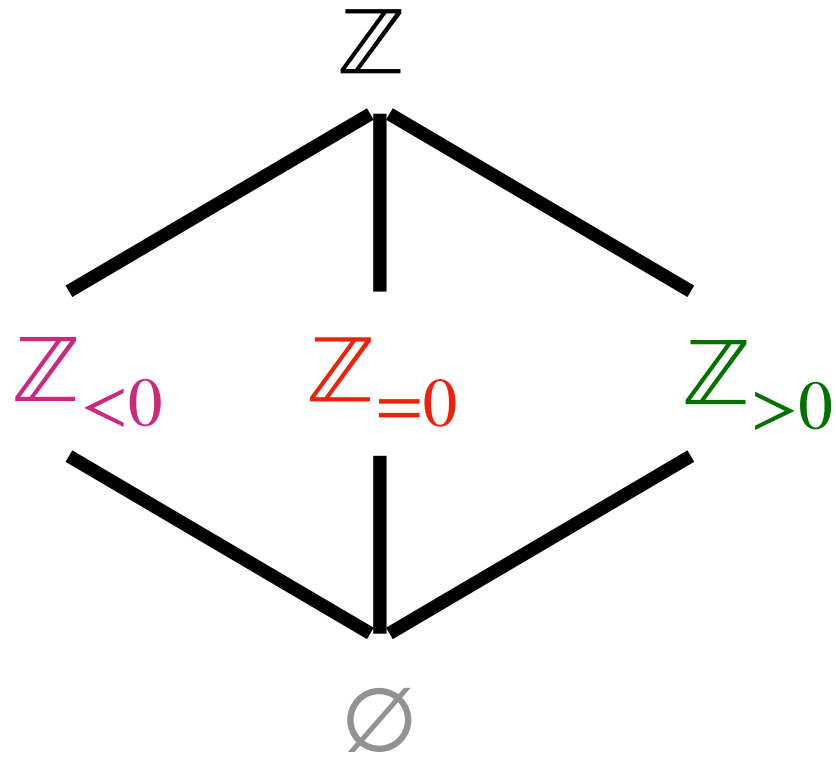
(proof by structural induction)

Adding addition

$e ::= n \mid e_1 \times e_2 \mid e_1 + e_2$

$\llbracket e_1 + e_2 \rrbracket = \llbracket e_1 \rrbracket + \llbracket e_2 \rrbracket$

$+^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{<0}$	(commutative)	
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	
$\mathbb{Z}_{>0}$	??	$\mathbb{Z}_{>0}$	$\mathbb{Z}_{>0}$

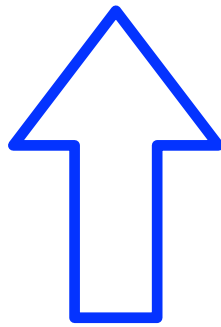


$A = \{ \mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0} \}$

$\sigma(e_1 + e_2) = \sigma(e_1) +^\# \sigma(e_2)$

$a_1 +^\# a_2 = ?$

$A^2 \xrightarrow{a_1 +^\# a_2} A$



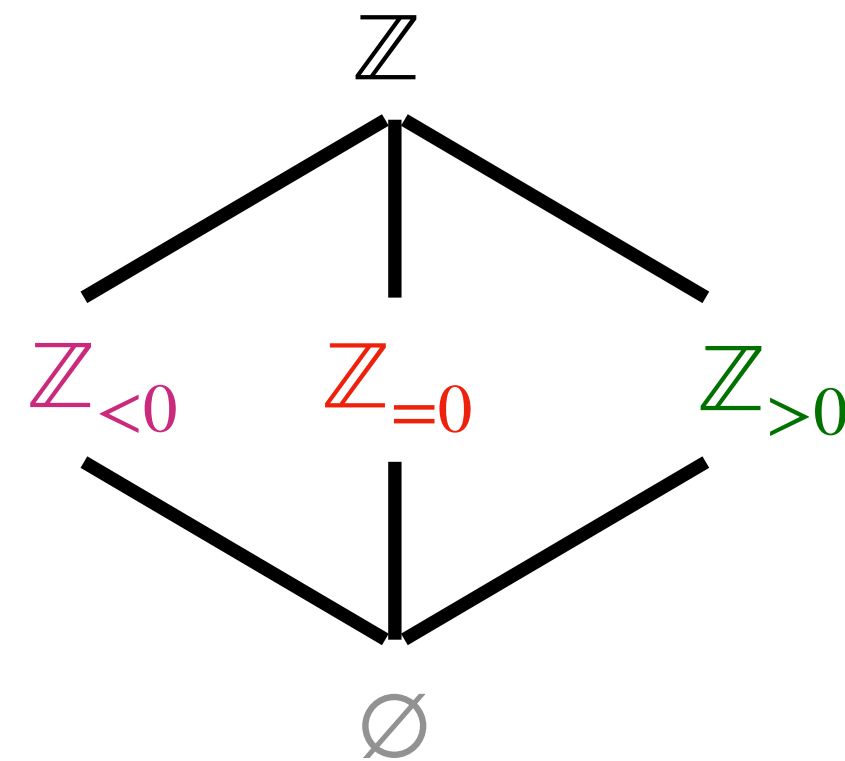
$\mathbb{Z}^2 \xrightarrow{v_1 + v_2} \mathbb{Z}$

Adding addition

$$e ::= n \mid e_1 \times e_2 \mid e_1 + e_2$$

$$\llbracket e_1 + e_2 \rrbracket = \llbracket e_1 \rrbracket + \llbracket e_2 \rrbracket$$

$+^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$	$\mathbb{Z}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{<0}$	(commutative)		
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$		
$\mathbb{Z}_{>0}$	$\mathbb{Z}$	$\mathbb{Z}_{>0}$	$\mathbb{Z}_{>0}$	
$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$



(exploit another abstract value)

$$A = \{\mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}, \mathbb{Z}\}$$

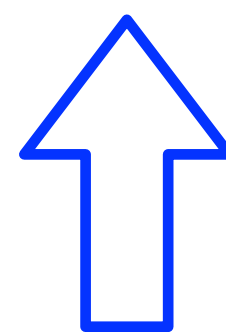
$$\sigma(e_1 + e_2) = \sigma(e_1) +^\# \sigma(e_2)$$

$$a_1 +^\# a_2 = ?$$

$\times^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$	$\mathbb{Z}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}_{>0}$	(commutative)		
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$		
$\mathbb{Z}_{>0}$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$	
$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}$	$\mathbb{Z}$

need to extend previous operations

$$A^2 \xrightarrow{a_1 +^\# a_2} A$$

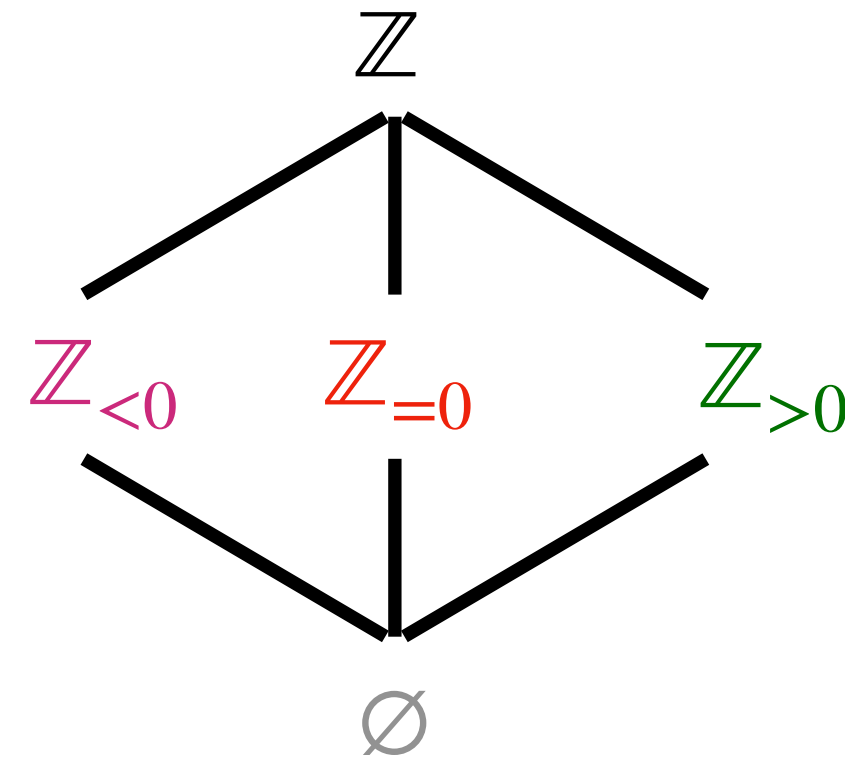


$$\mathbb{Z}^2 \xrightarrow{v_1 + v_2} \mathbb{Z}$$

Adding division

$e ::= \dots \mid e_1 / e_2$

$\llbracket e_1 / e_2 \rrbracket = \llbracket e_1 \rrbracket / \llbracket e_2 \rrbracket$ (if $\llbracket e_2 \rrbracket \neq 0$)



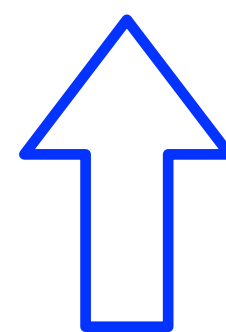
$A = \{ \mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}, \mathbb{Z} \}$

$\sigma(e_1 / e_2) = \sigma(e_1) /^\# \sigma(e_2)$

$a_1 /^\# a_2 = ?$

$/^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$	$\mathbb{Z}$
$\mathbb{Z}_{<0}$	$\mathbb{Z}$	??	$\mathbb{Z}$	$\mathbb{Z}$
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	??	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$
$\mathbb{Z}_{>0}$	$\mathbb{Z}$	??	$\mathbb{Z}$	$\mathbb{Z}$
$\mathbb{Z}$	$\mathbb{Z}$	??	$\mathbb{Z}$	$\mathbb{Z}$

$$A^2 \xrightarrow{a_1 /^\# a_2} A$$

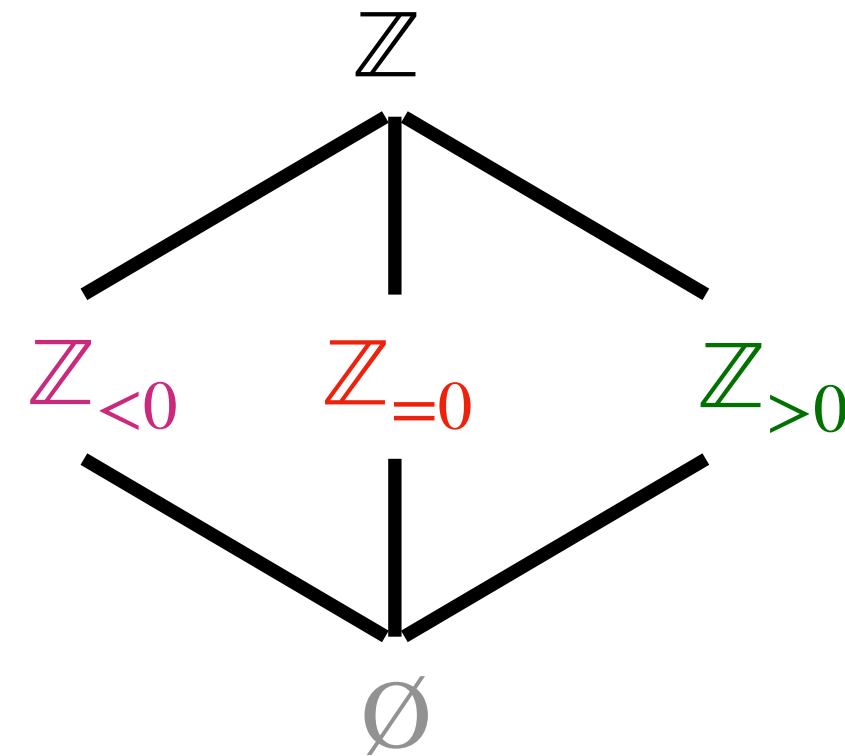


$$\mathbb{Z}^2 \xrightarrow{v_1 / v_2} \mathbb{Z}$$

Adding division

$e ::= \dots \mid e_1/e_2$

$\llbracket e_1/e_2 \rrbracket = \llbracket e_1 \rrbracket / \llbracket e_2 \rrbracket$ (if $\llbracket e_2 \rrbracket \neq 0$)



(exploit another abstract value)

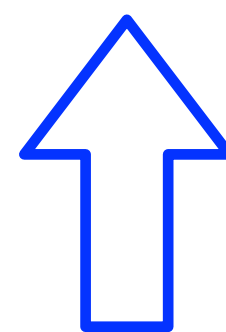
$$A = \{\emptyset, \mathbb{Z}_{<0}, \mathbb{Z}_{=0}, \mathbb{Z}_{>0}, \mathbb{Z}\}$$

$$\sigma(e_1/e_2) = \sigma(e_1) /^\# \sigma(e_2)$$

$$a_1 /^\# a_2 = ?$$

$/^\#$	$\mathbb{Z}_{<0}$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{>0}$	$\mathbb{Z}$	$\emptyset$
$\mathbb{Z}_{<0}$	$\mathbb{Z}$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\emptyset$
$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\emptyset$	$\mathbb{Z}_{=0}$	$\mathbb{Z}_{=0}$	$\emptyset$
$\mathbb{Z}_{>0}$	$\mathbb{Z}$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\emptyset$
$\mathbb{Z}$	$\mathbb{Z}$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

$$A^2 \xrightarrow{a_1 /^\# a_2} A$$



$$\mathbb{Z}^2 \xrightarrow{v_1/v_2} \mathbb{Z}$$

Properties as abstractions

Concrete values are mapped to abstract ones

Different concrete values can be mapped to the same abstract one

To some extent, an abstract value represents all its underlying concrete values

$\gamma : A \rightarrow \mathcal{P}(\mathbb{Z})$ called **concretization map**

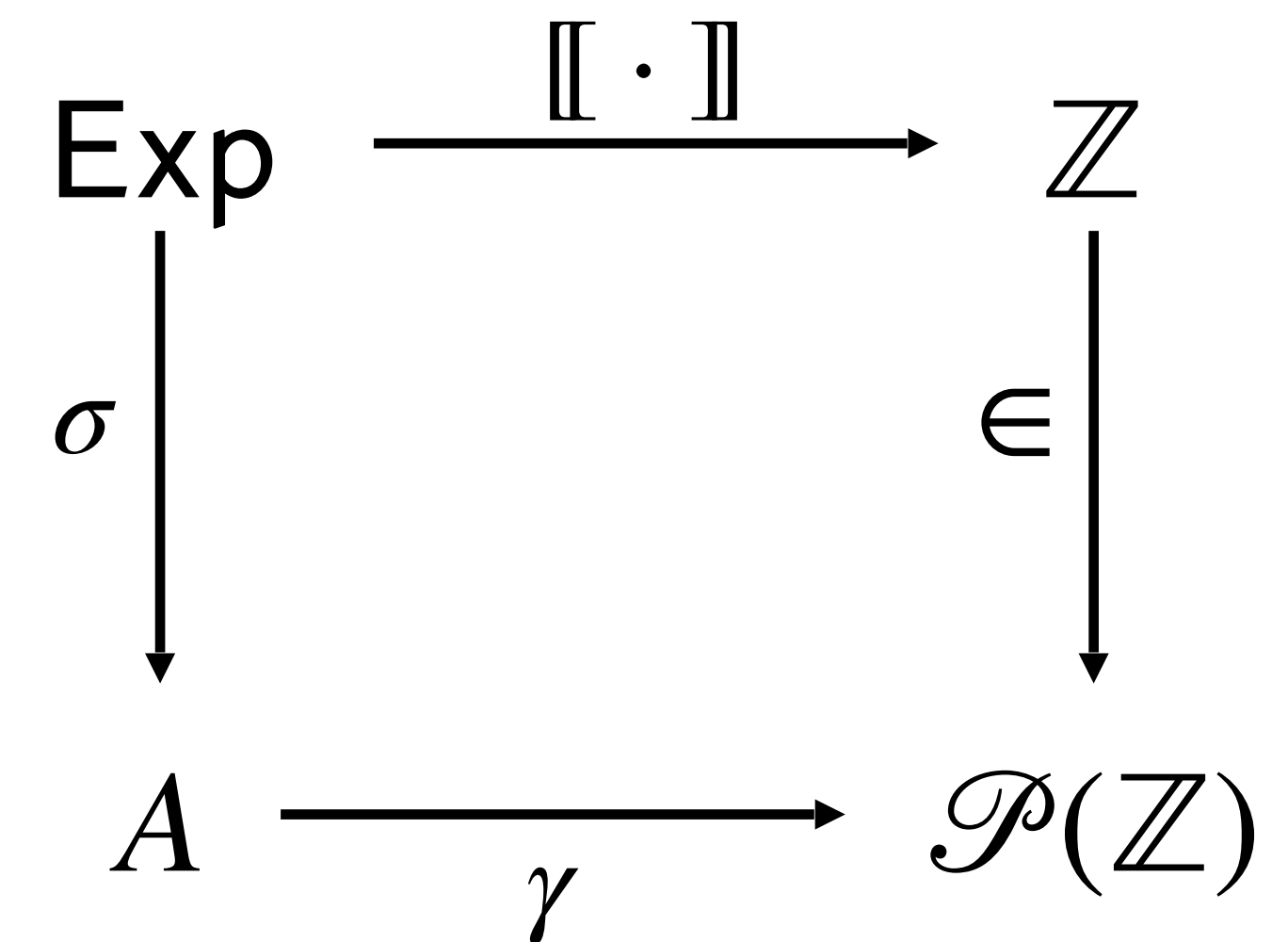
$$\gamma(\mathbb{Z}_{<0}) = \{n \in \mathbb{Z} \mid n < 0\}$$

$$\gamma(\mathbb{Z}_{=0}) = \{0\}$$

$$\gamma(\mathbb{Z}_{>0}) = \{n \in \mathbb{Z} \mid n > 0\}$$

$\gamma(\mathbb{Z}) = \{n \in \mathbb{Z}\}$ note that it contains each of $\gamma(\mathbb{Z}_{<0})$, $\gamma(\mathbb{Z}_{=0})$ and $\gamma(\mathbb{Z}_{>0})$

$\gamma(\emptyset) = \emptyset$ note that it is contained in all the previous ones



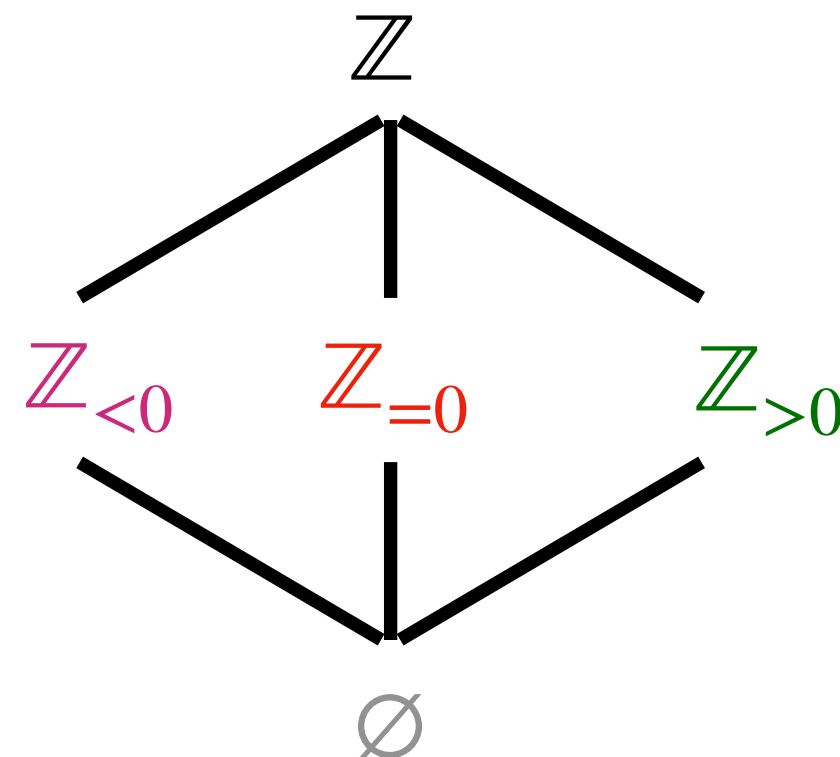
The abstract domain

The **abstract domain** is a lattice whose partial order $\sqsubseteq$ defined the notion of approximation/precision

Reminder: lattice means that each subset of the abstract domain has lub and glb

The partial order must be coherent with the concretization map: $\gamma : A \rightarrow \mathcal{P}(\mathbb{Z})$

$$a_1 \sqsubseteq a_2 \implies \gamma(a_1) \subseteq \gamma(a_2)$$



The abstraction map

The **abstraction map** $\alpha : \mathcal{P}(\mathbb{Z}) \rightarrow A$ is the counterpart of the concretization map γ

The function $\alpha : \mathcal{P}(\mathbb{Z}) \rightarrow A$ maps a set of concrete values S into the **most precise** abstract value that is a (sound) approximation of S

In our example:

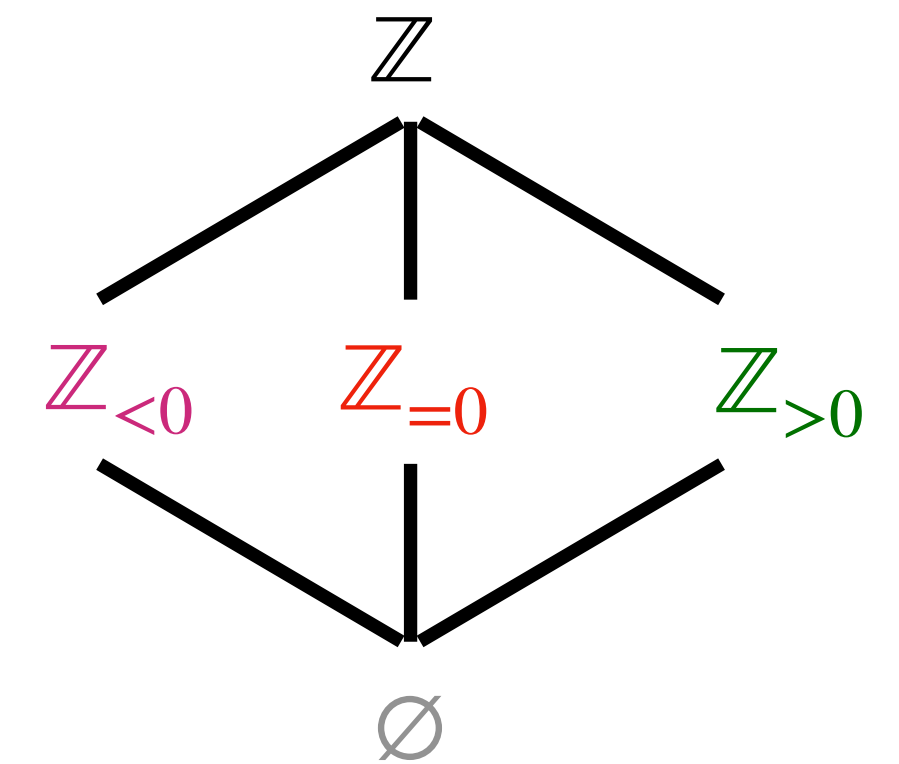
$$\alpha(\emptyset) = \emptyset$$

$$\alpha(\{0\}) = \mathbb{Z}_{=0}$$

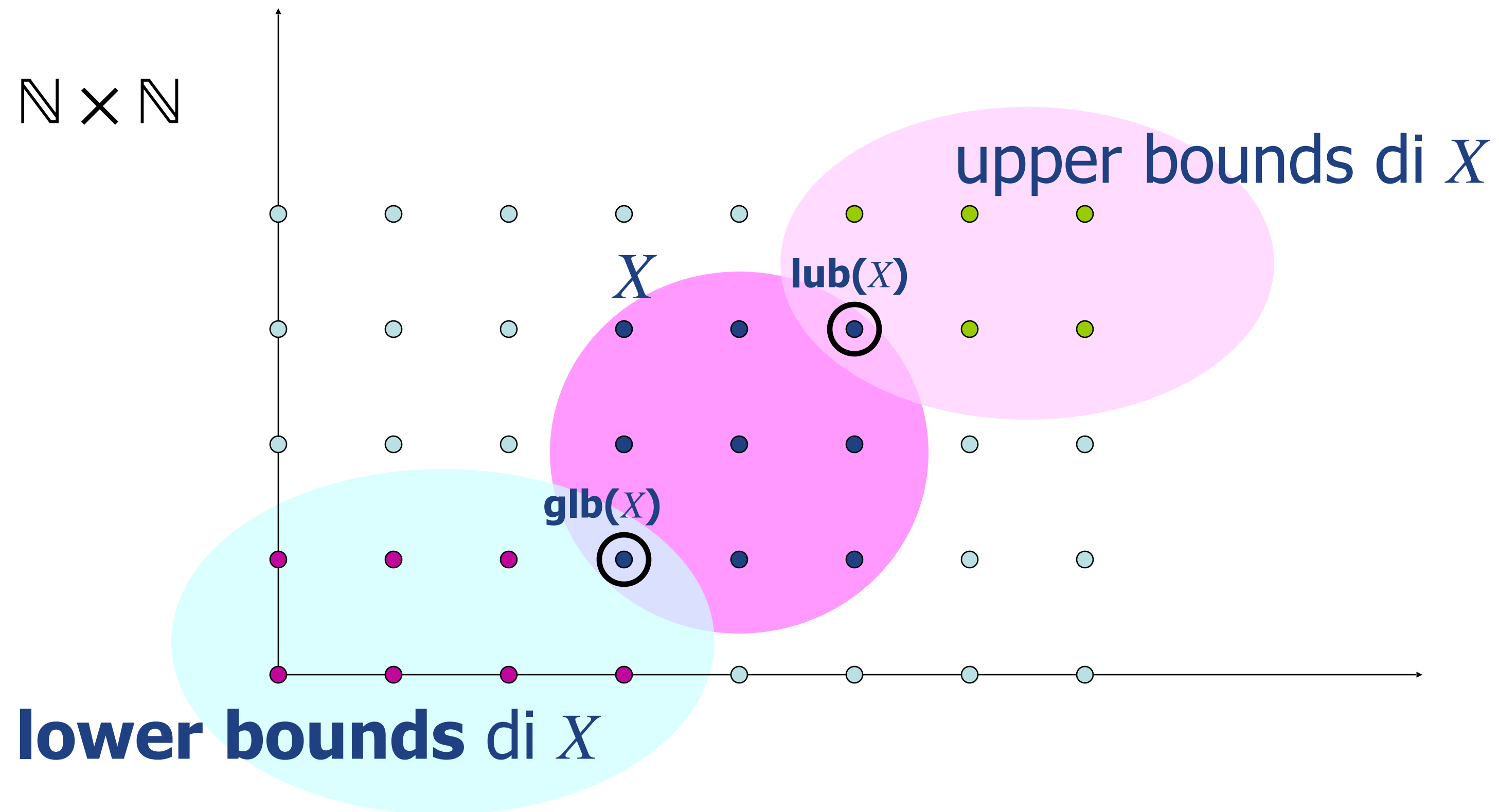
$$\alpha(S) = \mathbb{Z}_{<0} \quad \text{if } \emptyset \subset S \subseteq \{n \in \mathbb{Z} \mid n < 0\}$$

$$\alpha(S) = \mathbb{Z}_{>0} \quad \text{if } \emptyset \subset S \subseteq \{n \in \mathbb{Z} \mid n > 0\}$$

$$\alpha(S) = \mathbb{Z} \quad \text{otherwise}$$



lub and glb



$$(x_1, y_1) \leq_{\mathbb{N} \times \mathbb{N}} (x_2, y_2) \iff x_1 \leq_{\mathbb{N}} x_2 \wedge y_1 \leq_{\mathbb{N}} y_2$$

pointwise ordering

General definition

An **abstract interpretation** consists of:

- A concrete domain: (usually) a complete lattice C
- an abstract domain: (usually) a complete lattice A (smaller means "more precise")
- A monotone abstraction map $\alpha : C \rightarrow A$
- A monotone concretization map $\gamma : A \rightarrow C$
- The maps $\alpha : C \rightarrow A$ and $\gamma : A \rightarrow C$ form a **Galois insertion**
- **An abstract semantics $\llbracket \cdot \rrbracket^\#$ that is correct w.r.t. the concrete semantics $\llbracket \cdot \rrbracket$**

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\} = \top$$

$$S_2 = \{(1, n) \mid \exists m. (m, n) \in S_1\} = F_2(S_1)$$

$$S_3 = S_2 \cup S_6$$

$$S_4 = \{(m, n) \in S_3 \mid n > 0\} = F_4(S_3)$$

$$S_5 = \{(m \cdot n, n) \mid \exists m. (m, n) \in S_4\} = F_5(S_4)$$

$$S_6 = \{(m, n - 1) \mid \exists m. (m, n) \in S_5\} = F_6(S_5)$$

$$S_7 = \{(m, n) \in S_3 \mid n \leq 0\} = F_7(S_3)$$

each $F_i : \mathcal{P}(\Sigma) \rightarrow \mathcal{P}(\Sigma)$ is called a transfer function

Strongest invariant (not computable in general)

smallest solution of a system of equations

Kleene's fixpoint theorem again!

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \emptyset$$

$$S_2 = \emptyset$$

$$S_3 = \emptyset$$

$$S_4 = \emptyset$$

$$S_5 = \emptyset$$

$$S_6 = \emptyset$$

$$S_7 = \emptyset$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid \exists m. (m, n) \in S_1\}$$

$$S_3 = S_2 \cup S_6$$

$$S_4 = \{(m, n) \in S_3 \mid n > 0\}$$

$$S_5 = \{(m \cdot n, n) \mid \exists m. (m, n) \in S_4\}$$

$$S_6 = \{(m, n - 1) \mid \exists m. (m, n) \in S_5\}$$

$$S_7 = \{(m, n) \in S_3 \mid n \leq 0\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid \exists m. (m, n) \in \emptyset\} = \emptyset$$

$$S_3 = \emptyset \cup \emptyset = \emptyset$$

$$S_4 = \{(m, n) \in \emptyset \mid n > 0\} = \emptyset$$

$$S_5 = \{(m \cdot n, n) \mid \exists m. (m, n) \in \emptyset\} = \emptyset$$

$$S_6 = \{(m, n - 1) \mid \exists m. (m, n) \in \emptyset\} = \emptyset$$

$$S_7 = \{(m, n) \in \emptyset \mid n \leq 0\} = \emptyset$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \emptyset$$

$$S_3 = \emptyset$$

$$S_4 = \emptyset$$

$$S_5 = \emptyset$$

$$S_6 = \emptyset$$

$$S_7 = \emptyset$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid \exists m. (m, n) \in S_1\} \\ = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \emptyset$$

$$S_4 = \emptyset$$

$$S_5 = \emptyset$$

$$S_6 = \emptyset$$

$$S_7 = \emptyset$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = S_2 \cup S_6 \\ = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_4 = \emptyset$$

$$S_5 = \emptyset$$

$$S_6 = \emptyset$$

$$S_7 = \emptyset$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$\begin{aligned} S_4 &= \{(m, n) \in S_3 \mid n > 0\} \\ &= \{(1, n) \mid n > 0\} \end{aligned}$$

$$S_5 = \emptyset$$

$$S_6 = \emptyset$$

$$\begin{aligned} S_7 &= \{(m, n) \in S_3 \mid n \leq 0\} \\ &= \{(1, n) \mid n \leq 0\} \end{aligned}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_4 = \{(1, n) \mid n > 0\}$$

$$\begin{aligned} S_5 &= \{(m \cdot n, n) \mid \exists m. (m, n) \in S_4\} \\ &= \{(n, n) \mid n > 0\} \end{aligned}$$

$$S_6 = \emptyset$$

$$S_7 = \{(1, n) \mid n \leq 0\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_4 = \{(1, n) \mid n > 0\}$$

$$S_5 = \{(n, n) \mid n > 0\}$$

$$S_6 = \{(m, n - 1) \mid \exists m. (m, n) \in S_5\}$$

$$= \{(n, n - 1) \mid n > 0\}$$

$$S_7 = \{(1, n) \mid n \leq 0\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = S_2 \cup S_6$$

$$= \{(1, n) \mid n \in \mathbb{Z}\} \cup \{(n, n - 1) \mid n > 0\}$$

$$S_4 = \{(1, n) \mid n > 0\}$$

$$S_5 = \{(n, n) \mid n > 0\}$$

$$S_6 = \{(n, n - 1) \mid n > 0\}$$

$$S_7 = \{(1, n) \mid n \leq 0\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\} \cup \{(n, n - 1) \mid n > 0\}$$

$$S_4 = \{(m, n) \in S_3 \mid n > 0\}$$

$$= \{(1, n) \mid n > 0\} \cup \{(n, n - 1) \mid n > 1\}$$

$$S_5 = \{(n, n) \mid n > 0\}$$

$$S_6 = \{(n, n - 1) \mid n > 0\}$$

$$S_7 = \{(m, n) \in S_3 \mid n \leq 0\}$$

$$= \{(1, n) \mid n \leq 0\} \cup \{(n, n - 1) \mid n \leq 1\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\} \cup \{(n, n - 1) \mid n > 0\}$$

$$S_4 = \{(1, n) \mid n > 0\} \cup \{(n, n - 1) \mid n > 1\}$$

$$\begin{aligned} S_5 &= \{(m \cdot n, n) \mid \exists m. (m, n) \in S_4\} \\ &= \{(n, n) \mid n > 0\} \cup \{(n(n - 1), n - 1) \mid n > 1\} \end{aligned}$$

$$S_6 = \{(n, n - 1) \mid n > 0\}$$

$$S_7 = \{(1, n) \mid n \leq 0\} \cup \{(n, n - 1) \mid n \leq 1\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = \{(1, n) \mid n \in \mathbb{Z}\} \cup \{(n, n - 1) \mid n > 0\}$$

$$S_4 = \{(1, n) \mid n > 0\} \cup \{(n, n - 1) \mid n > 1\}$$

$$S_5 = \{(n, n) \mid n > 0\} \cup \{(n(n - 1), n - 1) \mid n > 1\}$$

$$S_6 = \{(m, n - 1) \mid \exists m. (m, n) \in S_5\}$$

$$= \{(n, n - 1) \mid n > 0\}$$

$$\cup \{(n(n - 1), n - 2) \mid n > 1\}$$

$$S_7 = \{(1, n) \mid n \leq 0\} \cup \{(n, n - 1) \mid n \leq 1\}$$

Analysis as fixpoint

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```

Concrete semantics $S_i \in \mathcal{P}(\Sigma)$ for $i \in \{1, 2, 3, 4, 6\}$

$$S_1 = \{(m, n) \mid m, n \in \mathbb{Z}\}$$

$$S_2 = \{(1, n) \mid n \in \mathbb{Z}\}$$

$$S_3 = S_2 \cup S_6$$

$$= \{(1, n) \mid n \in \mathbb{Z}\} \cup \{(n, n - 1) \mid n > 0\} \\ \cup \{(n(n - 1), n - 2) \mid n > 1\} \dots$$

$$S_4 = \{(1, n) \mid n > 0\} \cup \{(n, n - 1) \mid n > 1\}$$

$$S_5 = \{(n, n) \mid n > 0\} \cup \{(n(n - 1), n - 1) \mid n > 1\}$$

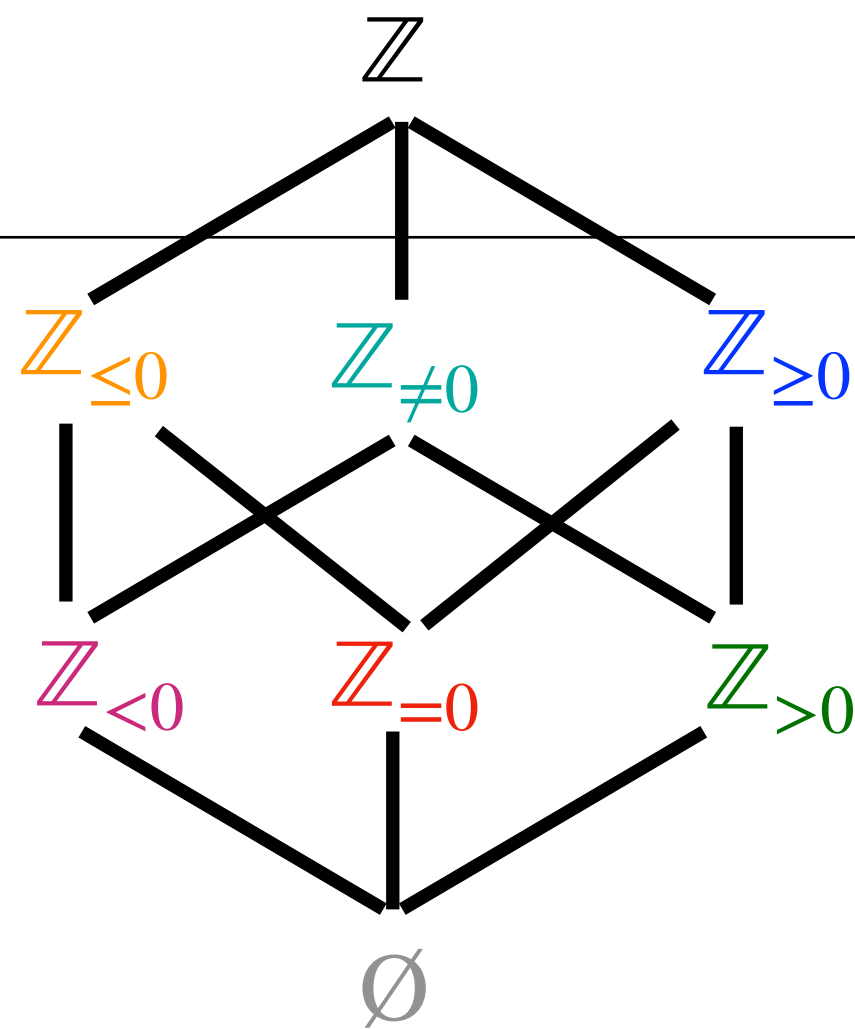
$$S_6 = \{(n, n - 1) \mid n > 0\}$$

$$\cup \{(n(n - 1), n - 2) \mid n > 1\}$$

$$S_7 = \{(1, n) \mid n \leq 0\} \cup \{(n, n - 1) \mid n \leq 1\}$$

Abstract interpretation

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)     m := m * n;  
(5)     n := n - 1  
(6) }  
(7) // end
```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \top$$

$$S_2^\# = F_2^\#(S_1^\#)$$

$$S_3^\# = S_2^\# \cup^\# S_6^\#$$

$$S_4^\# = F_4^\#(S_3^\#)$$

$$S_5^\# = F_5^\#(S_4^\#)$$

$$S_6^\# = F_6^\#(S_5^\#)$$

$$S_7^\# = F_7^\#(S_3^\#)$$

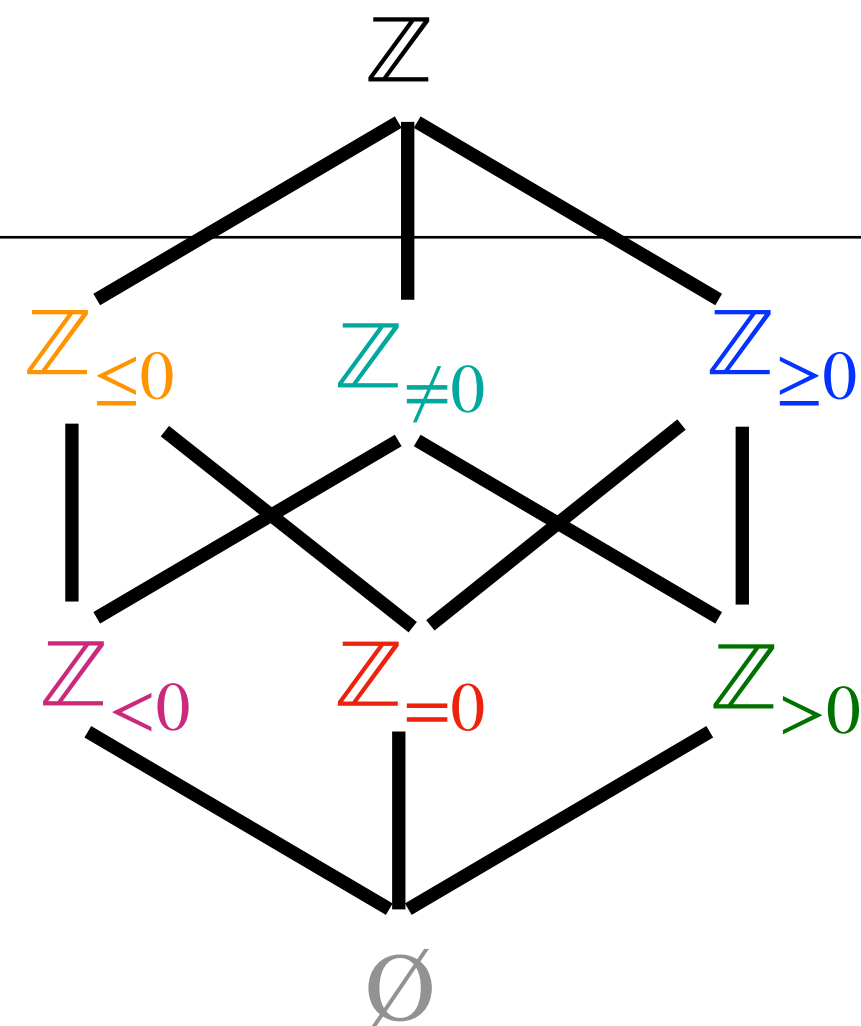
each $F_i^\#$ is called abstract transfer function

A is the set of properties of interest

$F_i^\# : A \rightarrow_{74} A$ is a sound over-approximation of F_i

Abstract interpretation

```
(1)  $m^\# := \mathbb{Z}_{>0};$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \emptyset$$

$$S_2^\# = \emptyset$$

$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

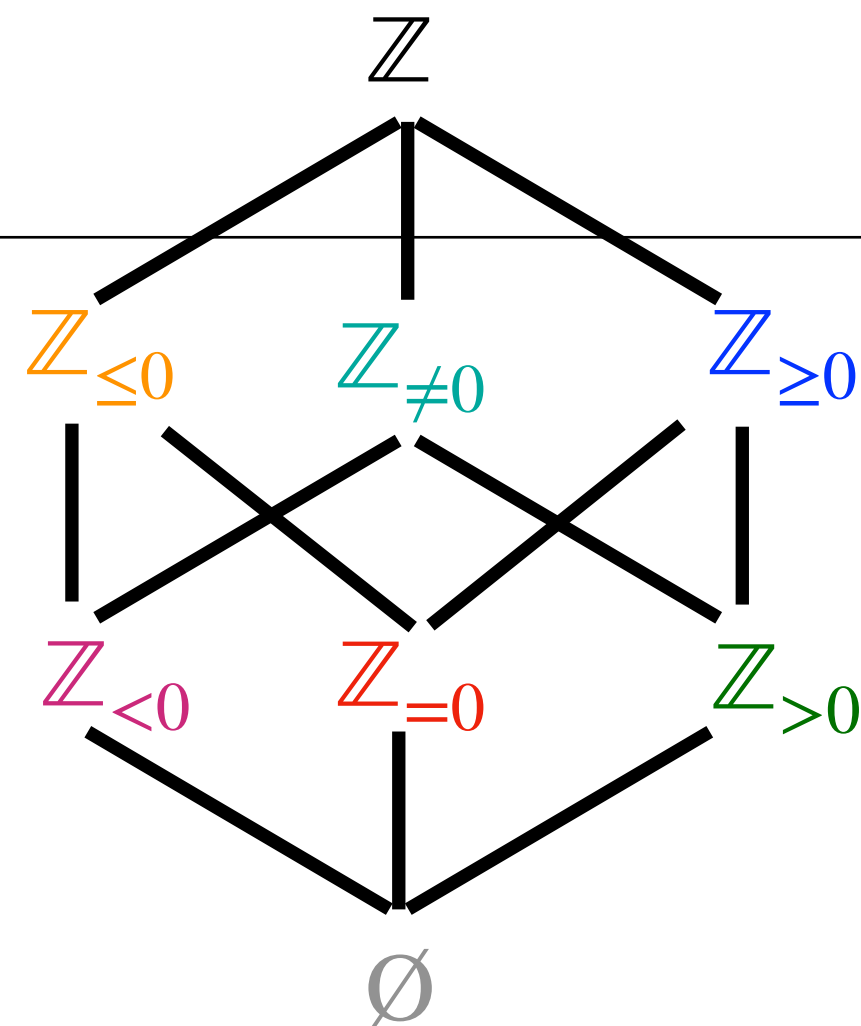
$$S_7^\# = \emptyset$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \emptyset$$

$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

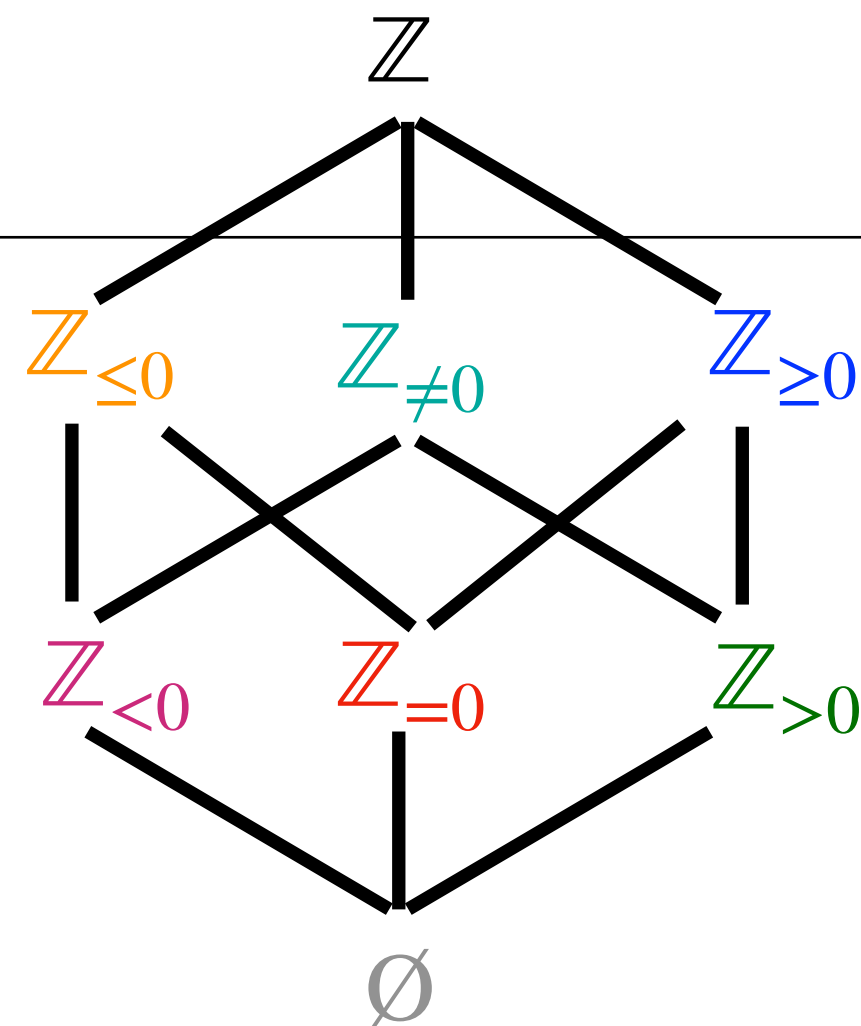
$$S_7^\# = \emptyset$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

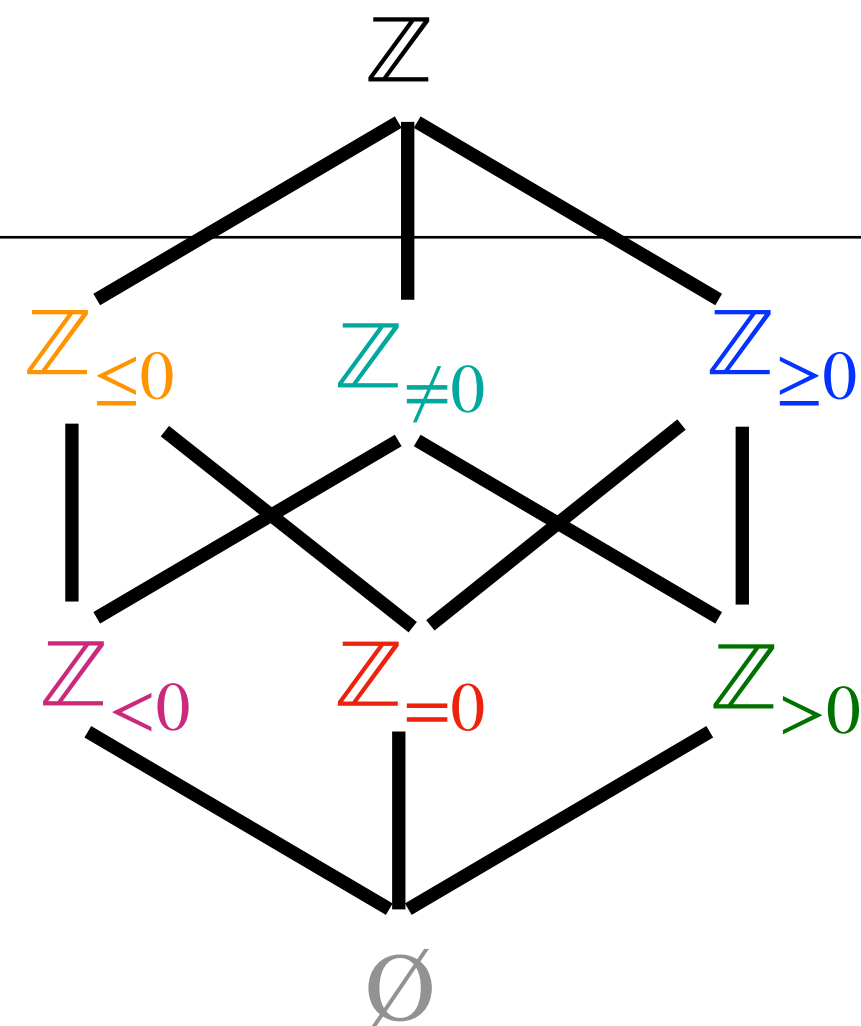
$$S_7^\# = \emptyset$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

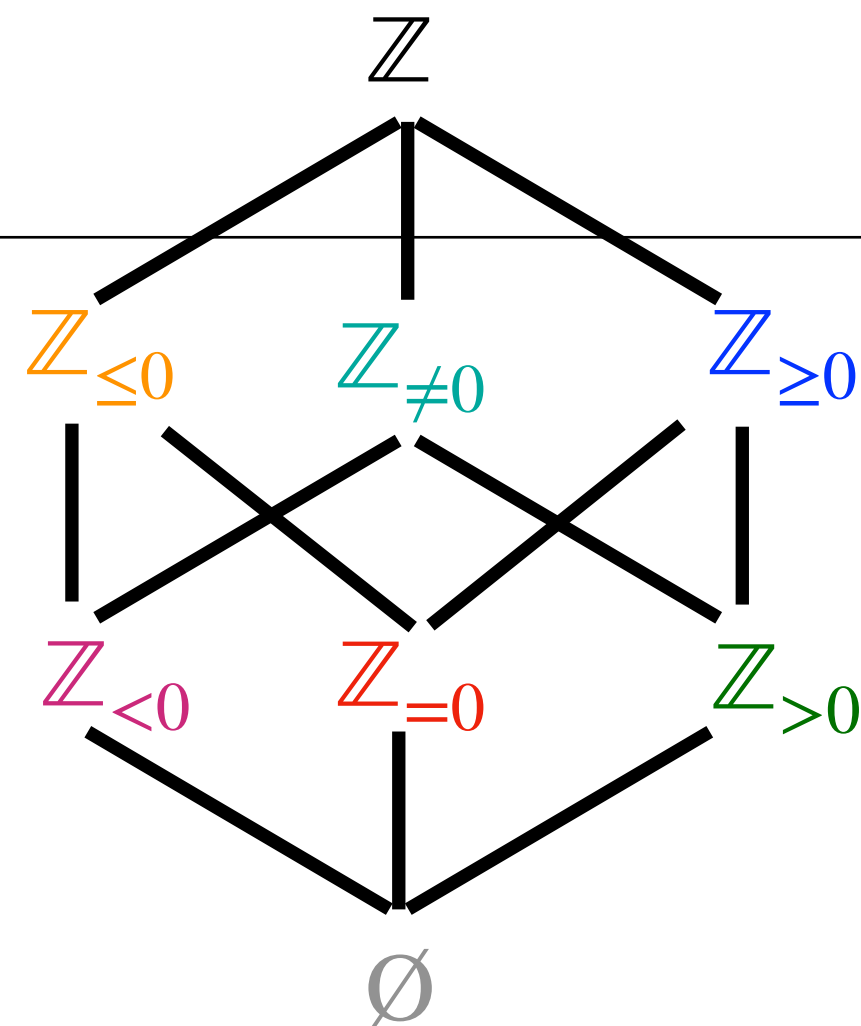
$$S_7^\# = \emptyset$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_4^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

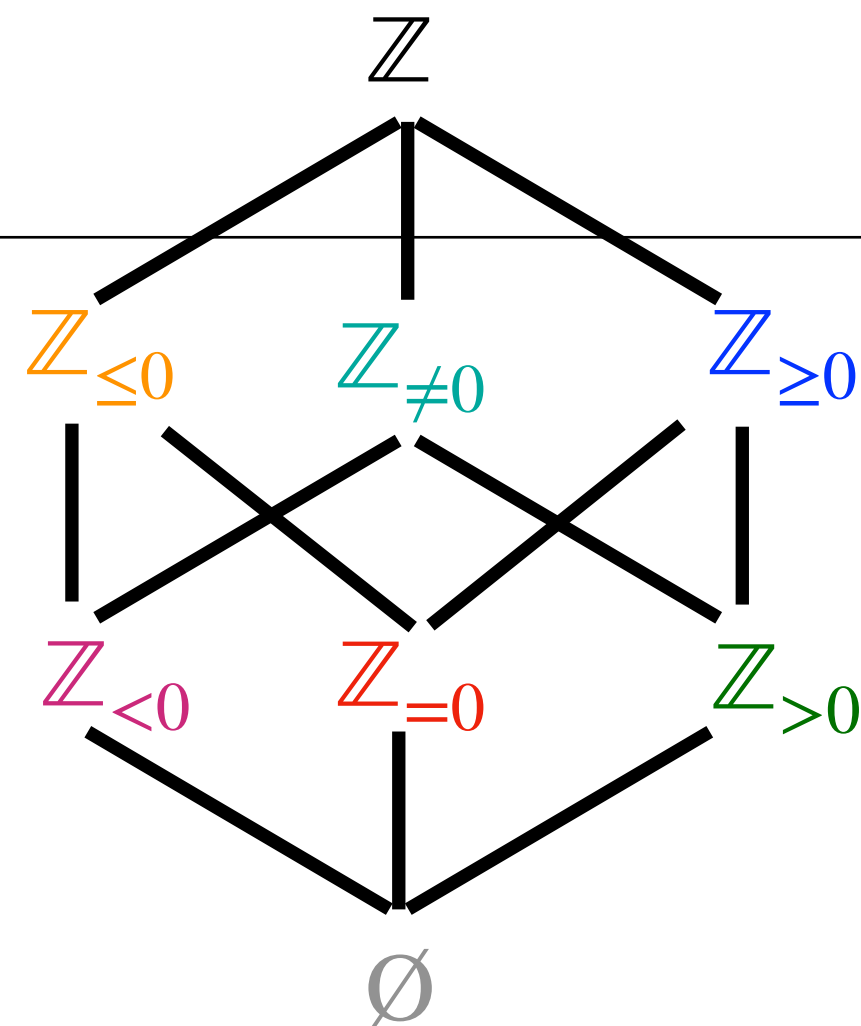
$$S_7^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_4^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_5^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_6^\# = \emptyset$$

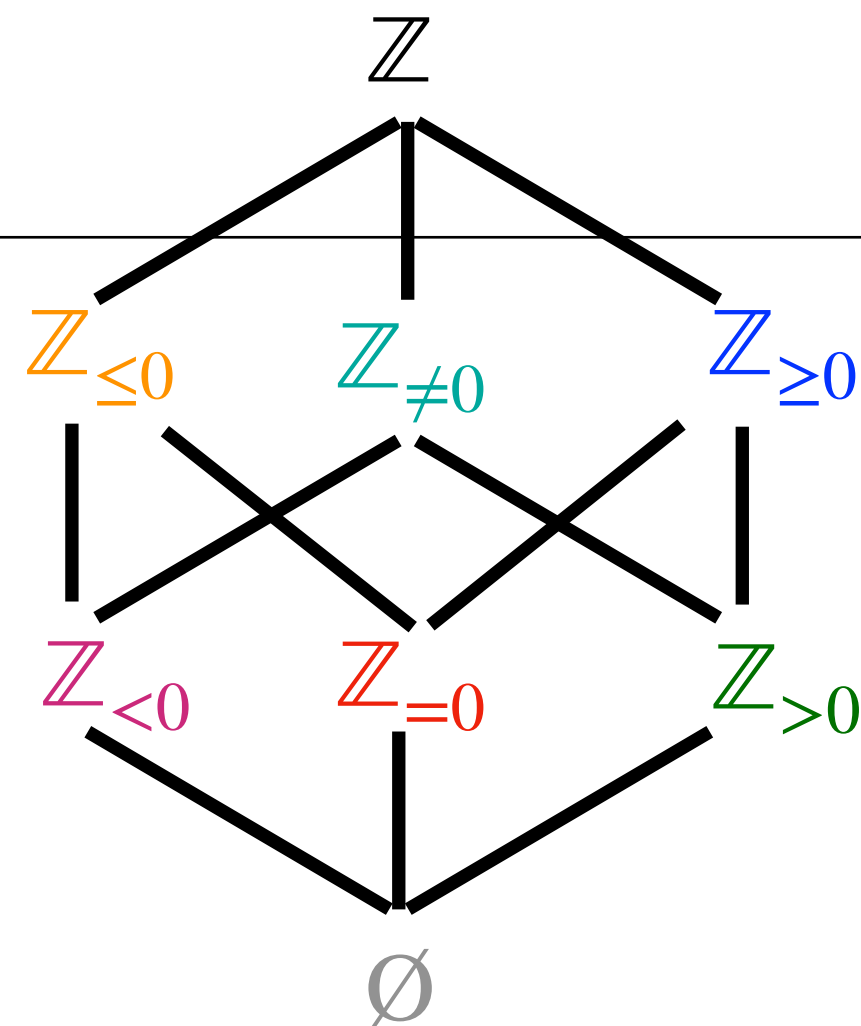
$$S_7^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_4^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_5^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_6^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

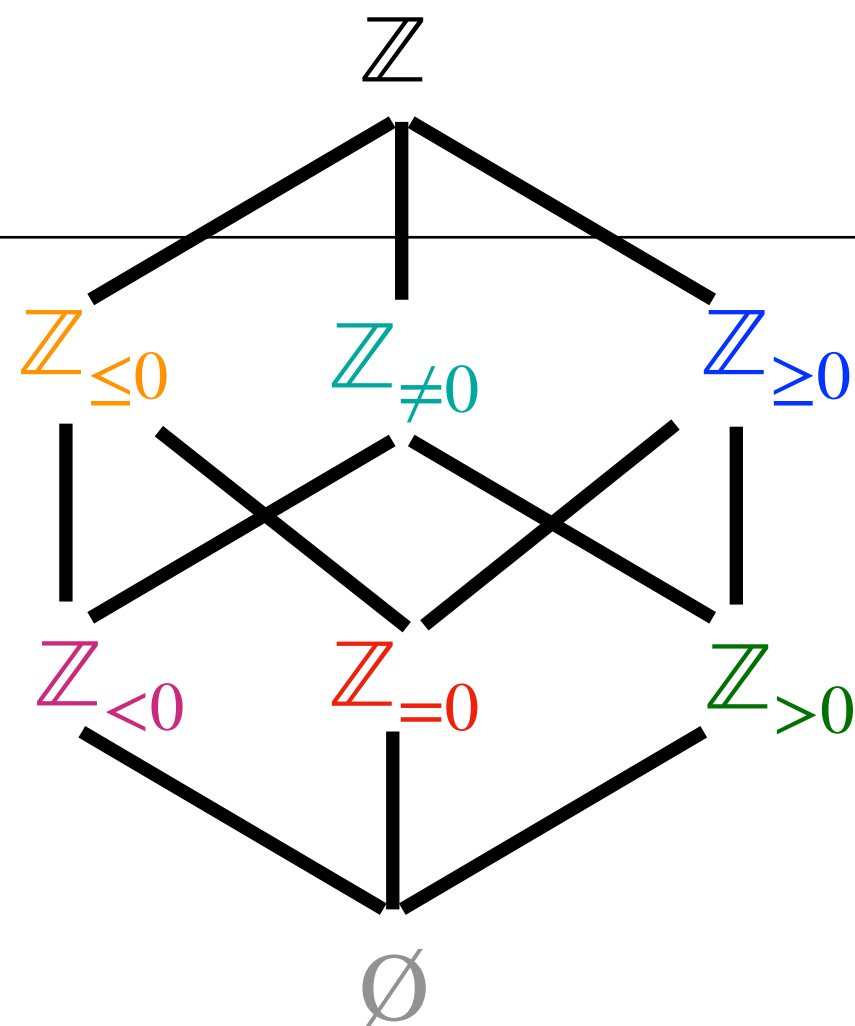
$$S_7^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

Abstract interpretation

```

(1)  $m^\# := \mathbb{Z}_{>0};$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Abstract semantics $S_i^\# \in A$ for $i \in \{1,2,3,4,6\}$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \mathbb{Z}_{>0} \times \mathbb{Z}$$

$$S_3^\# = (\mathbb{Z}_{>0} \times \mathbb{Z}) \cup^\# (\mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}) = (\mathbb{Z}_{>0} \times \mathbb{Z})$$

$$S_4^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_5^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$$

$$S_6^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

$$S_7^\# = \mathbb{Z}_{>0} \times \mathbb{Z}_{\geq 0}$$

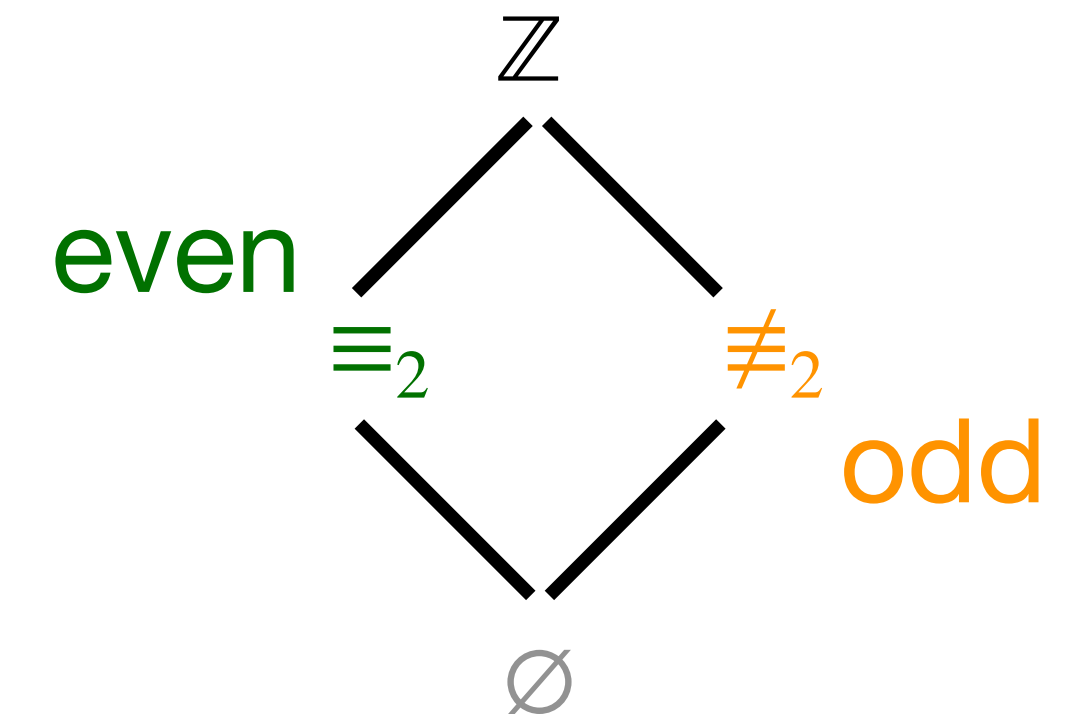
(ABSTRACT) FIXPOINT REACHED!

Exercises

Abstract operations

Given the "parity" domain $A = \{\emptyset, \equiv_2, \not\equiv_2, \mathbb{Z}\}$:

- define the (most precise) sound abstraction of product
- define the (most precise) sound abstraction of sum
- define the (most precise) sound abstraction of division



$\times^\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$		(commutative)		
$\equiv_2$				
$\not\equiv_2$				
$\mathbb{Z}$				

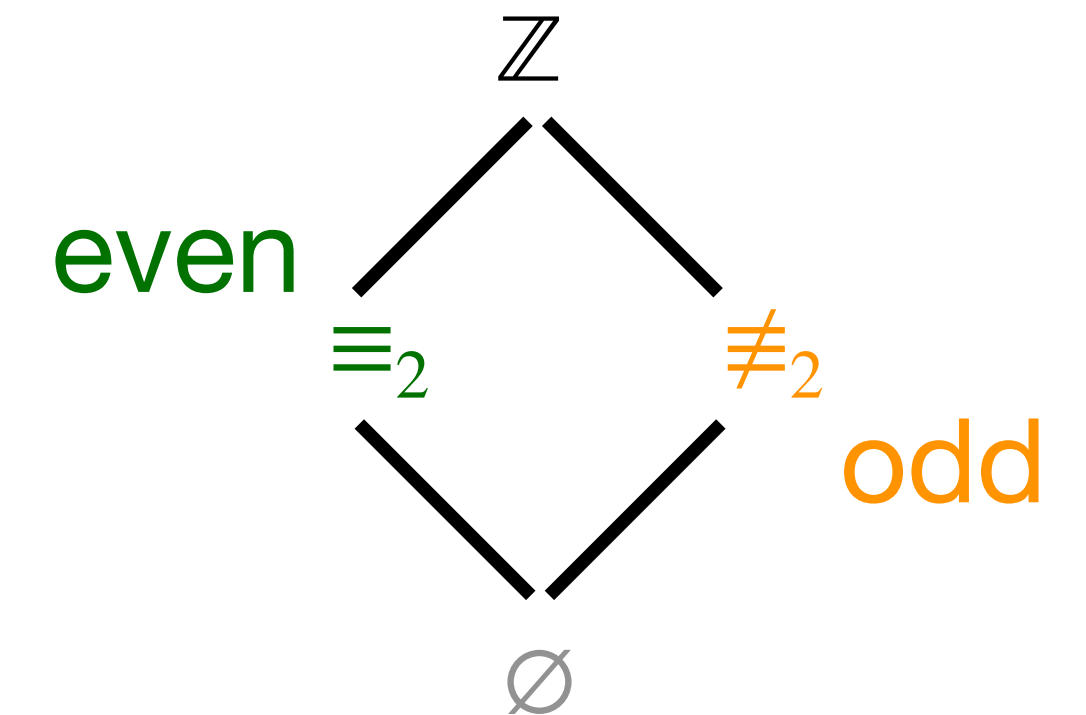
$+\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$		(commutative)		
$\equiv_2$				
$\not\equiv_2$				
$\mathbb{Z}$				

$/\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$		(commutative)		
$\equiv_2$				
$\not\equiv_2$				
$\mathbb{Z}$				

Abstract operations

Given the "parity" domain $A = \{\emptyset, \equiv_2, \not\equiv_2, \mathbb{Z}\}$:

- define the (most precise) sound abstraction of product
- define the (most precise) sound abstraction of sum
- define the (most precise) sound abstraction of division



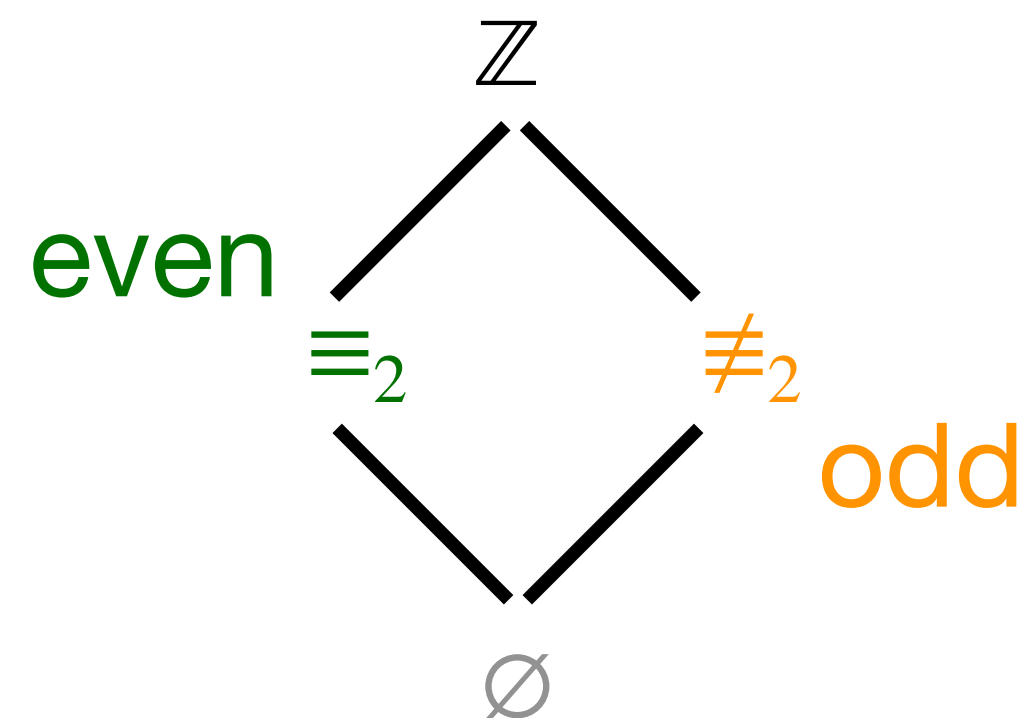
$\times^\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$	$\emptyset$	(commutative)		
$\equiv_2$	$\emptyset$	$\equiv_2$		
$\not\equiv_2$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	
$\mathbb{Z}$	$\emptyset$	$\equiv_2$	$\mathbb{Z}$	$\mathbb{Z}$

$+\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$	$\emptyset$	(commutative)		
$\equiv_2$	$\emptyset$	$\equiv_2$		
$\not\equiv_2$	$\emptyset$	$\not\equiv_2$	$\equiv_2$	
$\mathbb{Z}$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$

$/\#$	$\emptyset$	$\equiv_2$	$\not\equiv_2$	$\mathbb{Z}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
$\equiv_2$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
$\not\equiv_2$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
$\mathbb{Z}$	$\emptyset$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$

Abstract interpretation 1

```
(1) m := 1;  
(2) while (3) n > 0 do {  
(4)   m := m * n;  
(5)   n := n - 1  
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}$:

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = F_2^\#(S_1^\#)$$

$$S_3^\# = S_2^\# \cup^\# S_6^\#$$

$$S_4^\# = F_4^\#(S_3^\#)$$

$$S_5^\# = F_5^\#(S_4^\#)$$

$$S_6^\# = F_6^\#(S_5^\#)$$

$$S_7^\# = F_7^\#(S_3^\#)$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```

Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \emptyset$$

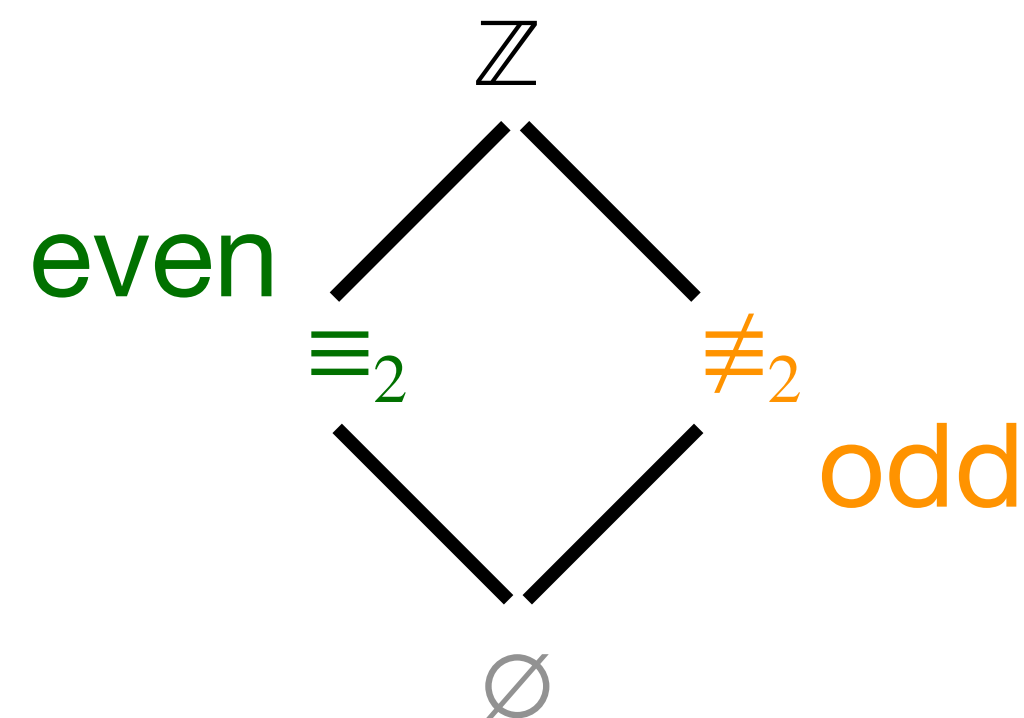
$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$



Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```

Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

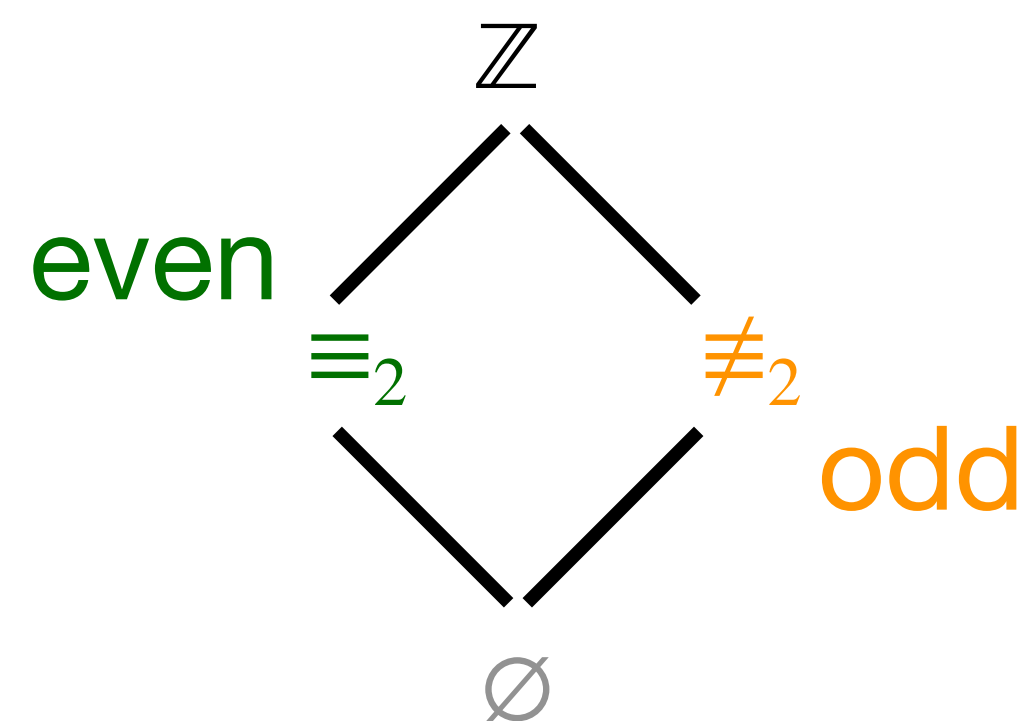
$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

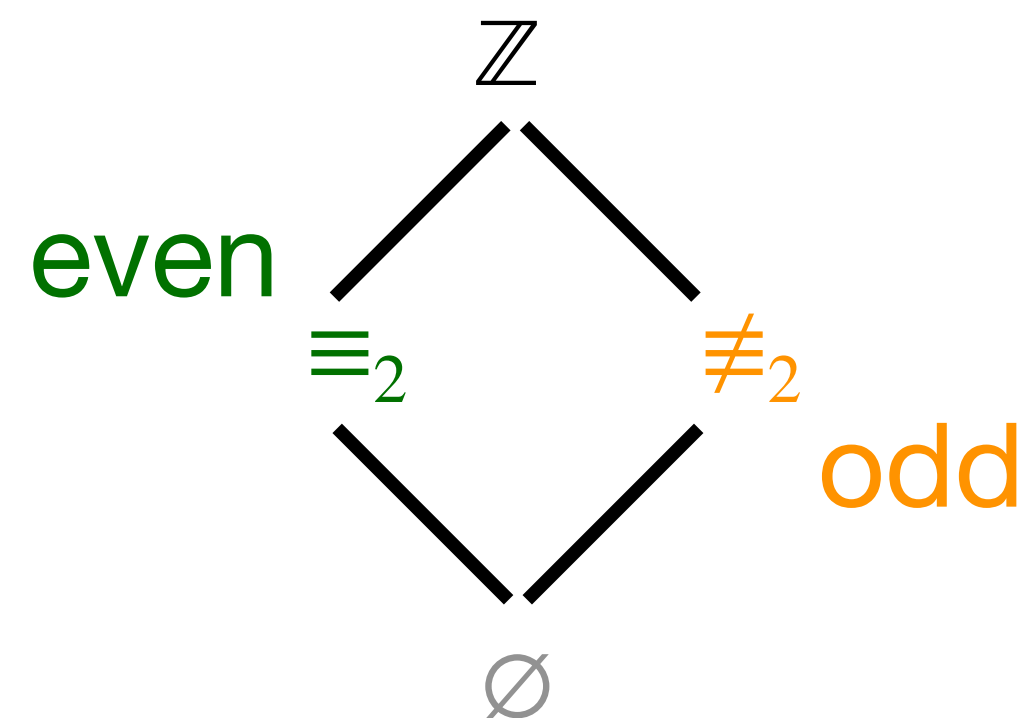
$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$



Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \neq_2 \times \mathbb{Z}$$

$$S_4^\# = \emptyset$$

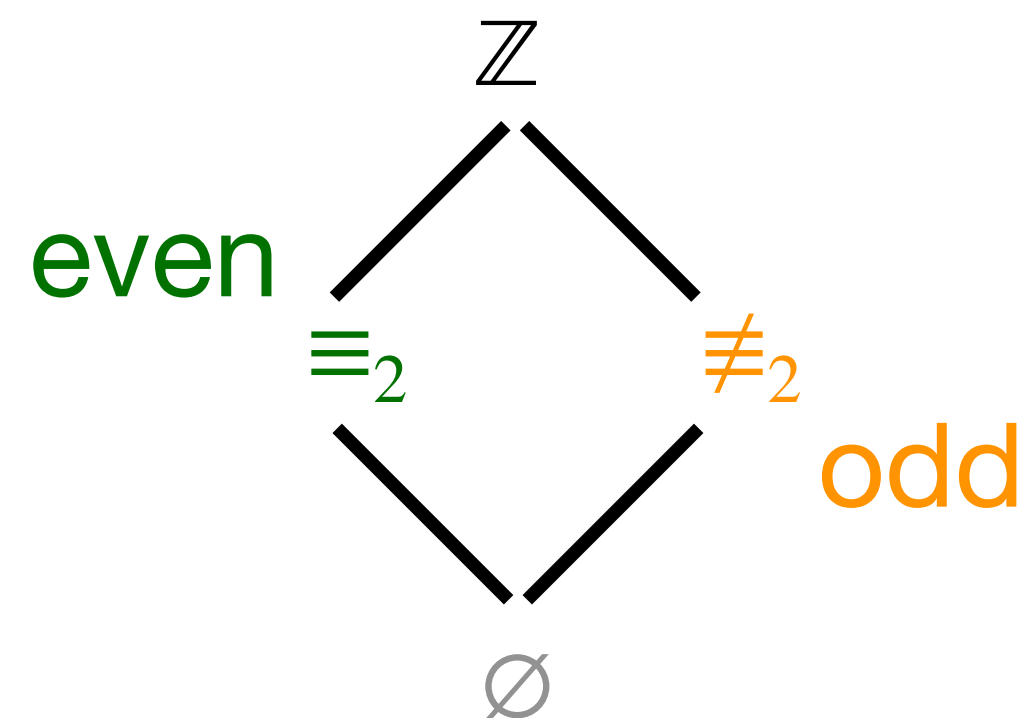
$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \neq_2 \times \mathbb{Z}$$

$$S_4^\# = \neq_2 \times \mathbb{Z}$$

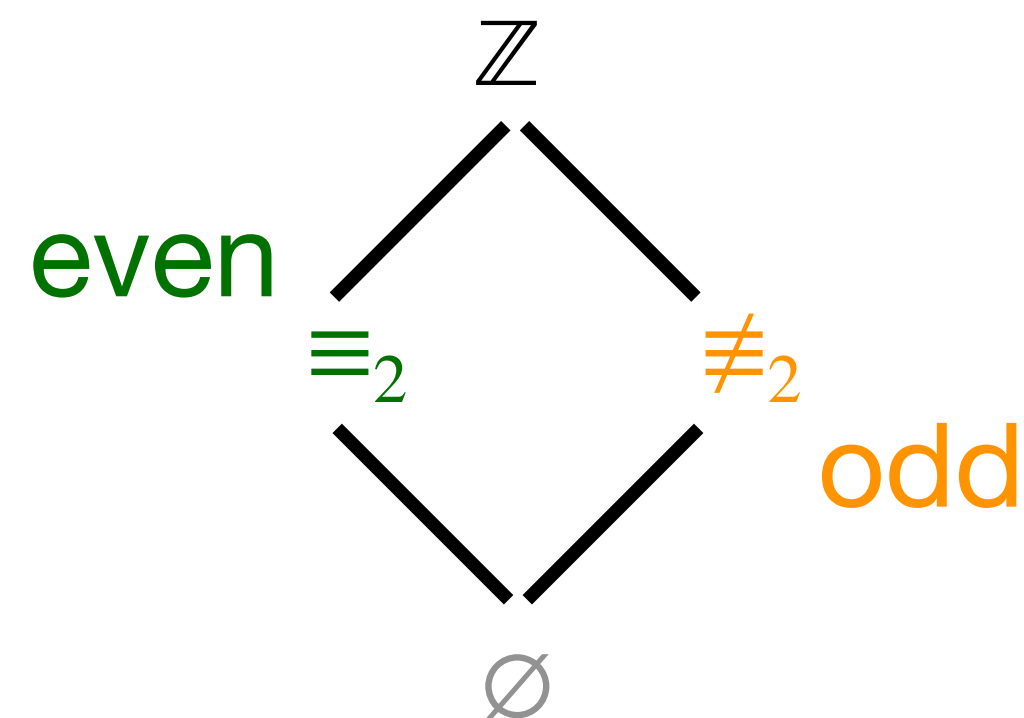
$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \neq_2 \times \mathbb{Z}$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \neq_2 \times \mathbb{Z}$$

$$S_4^\# = \neq_2 \times \mathbb{Z}$$

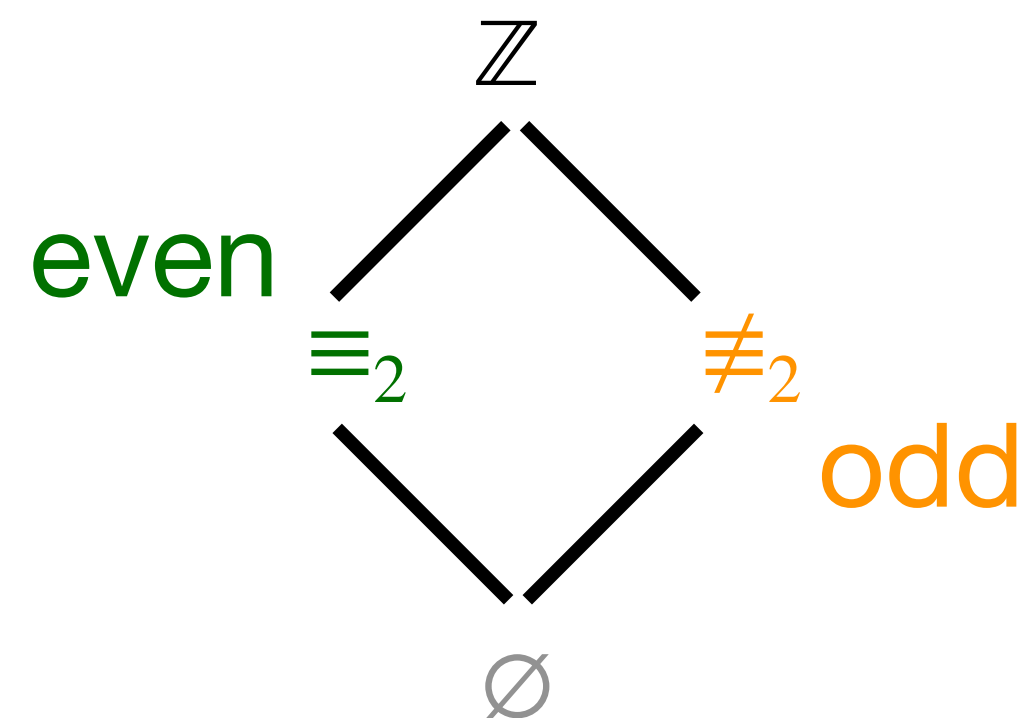
$$S_5^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \neq_2 \times \mathbb{Z}$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \neq_2 \times \mathbb{Z}$$

$$S_4^\# = \neq_2 \times \mathbb{Z}$$

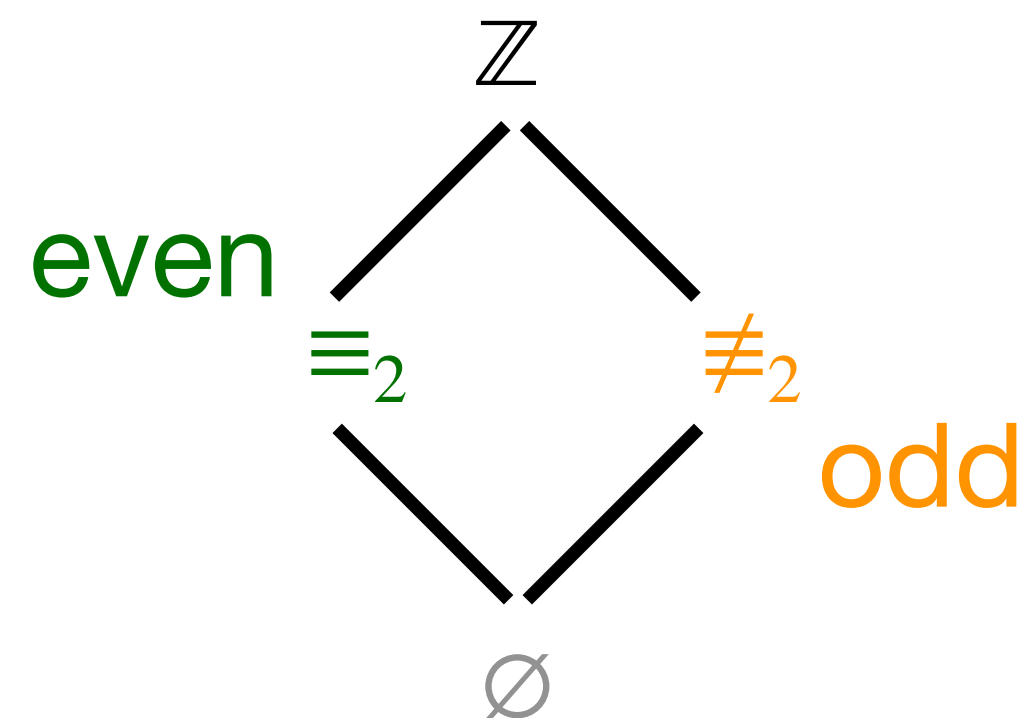
$$S_5^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_6^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_7^\# = \neq_2 \times \mathbb{Z}$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \neq_2 \times \mathbb{Z} \cup^\# \mathbb{Z} \times \mathbb{Z}$$

$$= \mathbb{Z} \times \mathbb{Z}$$

$$S_4^\# = \neq_2 \times \mathbb{Z}$$

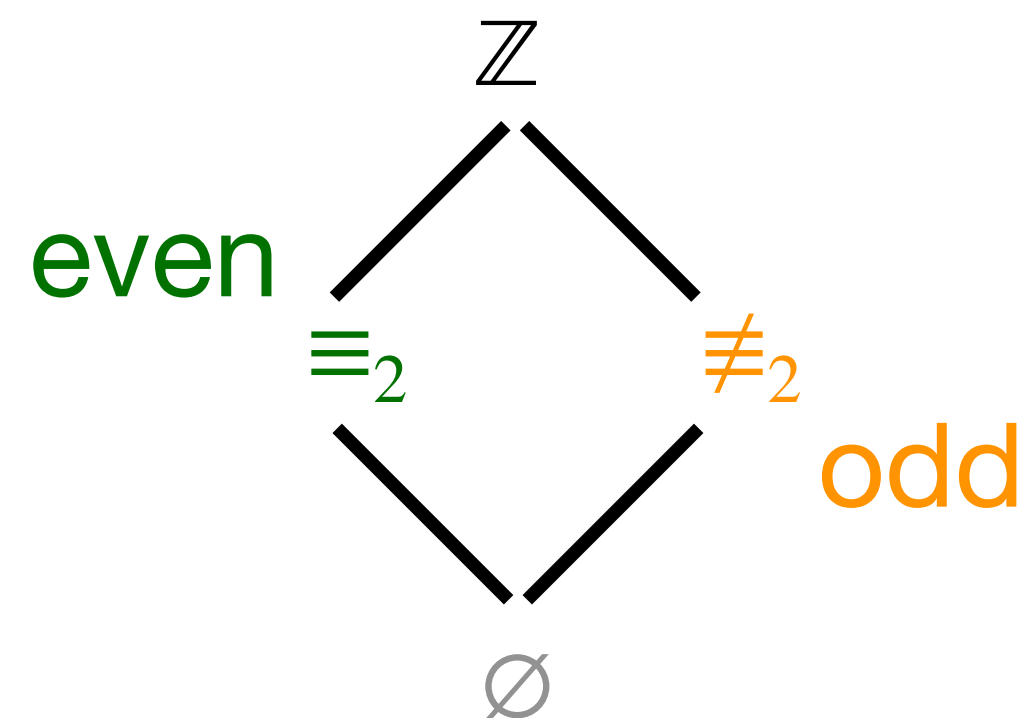
$$S_5^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_6^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_7^\# = \neq_2 \times \mathbb{Z}$$

Abstract interpretation 1

```
(1)  $m^\# := \neq_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \neq_2 \times \mathbb{Z}$$

$$S_3^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_4^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_5^\# = \mathbb{Z} \times \mathbb{Z}$$

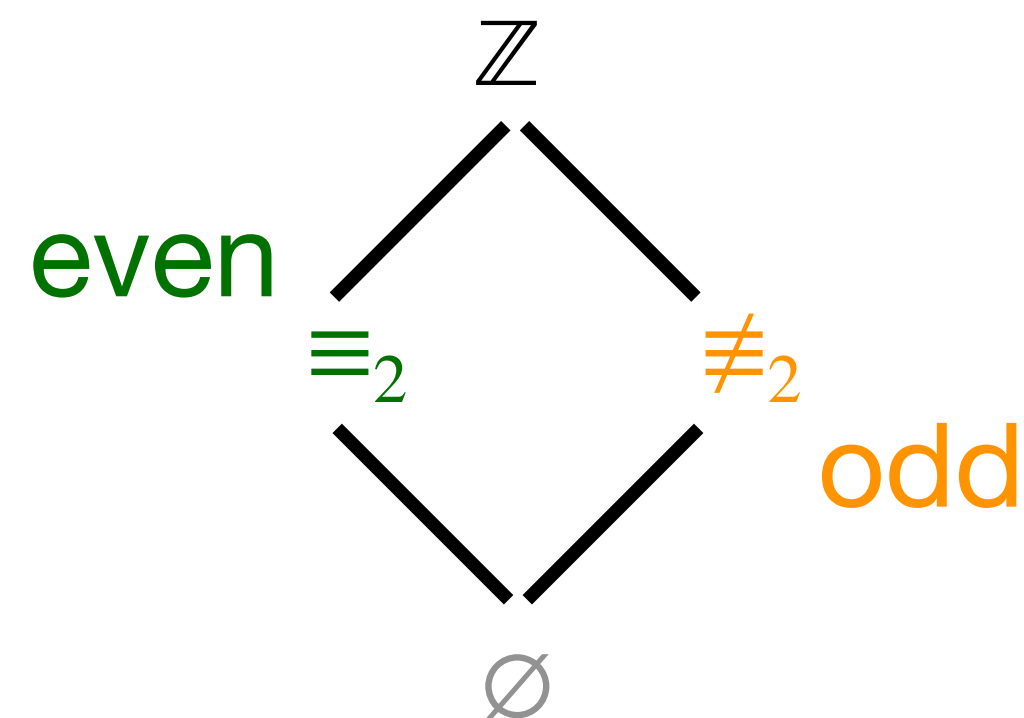
$$S_6^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_7^\# = \mathbb{Z} \times \mathbb{Z}$$

(ABSTRACT) FIXPOINT REACHED!

Abstract interpretation 2

```
(1) m := 2;  
(2) while (3) n > 0 do {  
(4)   m := m * n;  
(5)   n := n - 1  
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}$:

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = F_2^\#(S_1^\#)$$

$$S_3^\# = S_2^\# \cup^\# S_6^\#$$

$$S_4^\# = F_4^\#(S_3^\#)$$

$$S_5^\# = F_5^\#(S_4^\#)$$

$$S_6^\# = F_6^\#(S_5^\#)$$

$$S_7^\# = F_7^\#(S_3^\#)$$

Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```

Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \not\equiv_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \emptyset$$

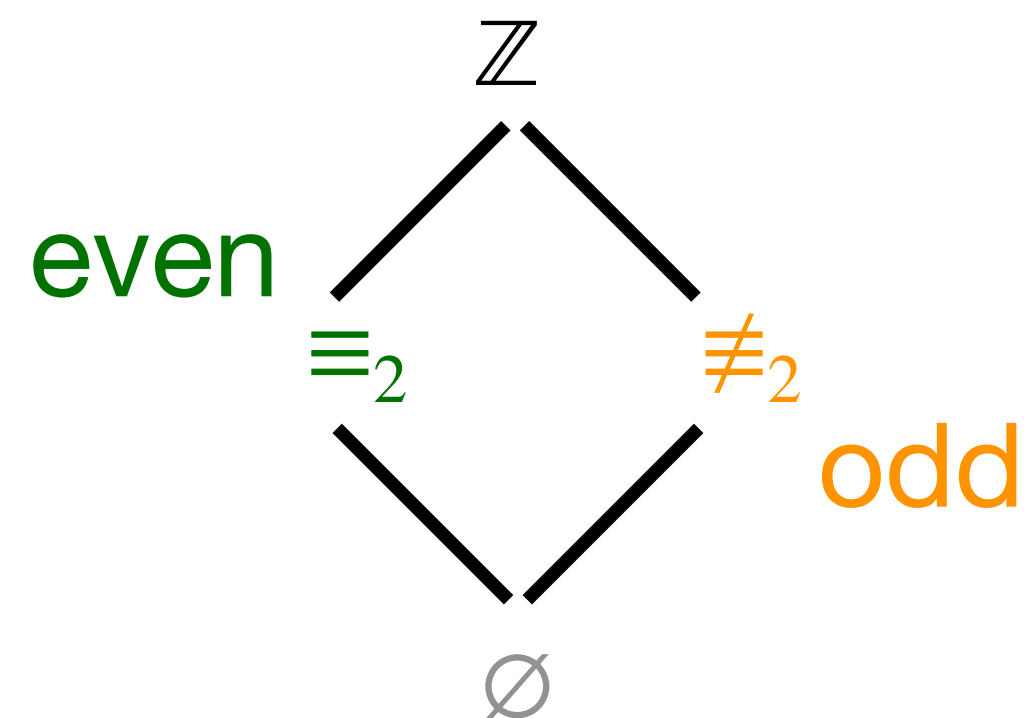
$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

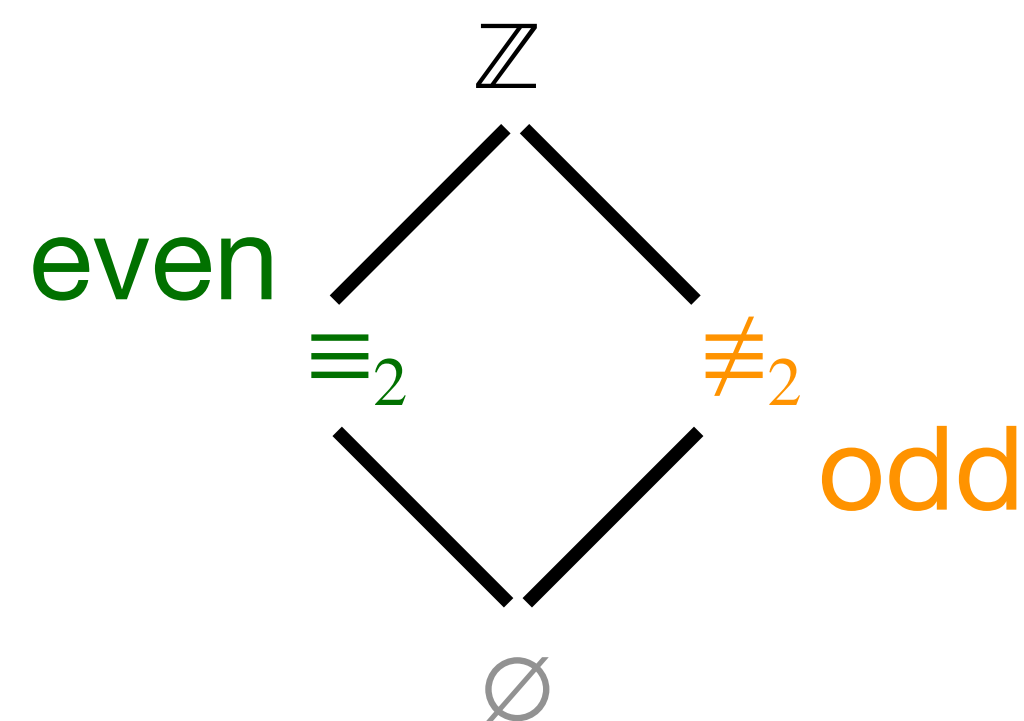
$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$



Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

$$S_3^\# = \emptyset$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$

Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```

Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \not\equiv_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

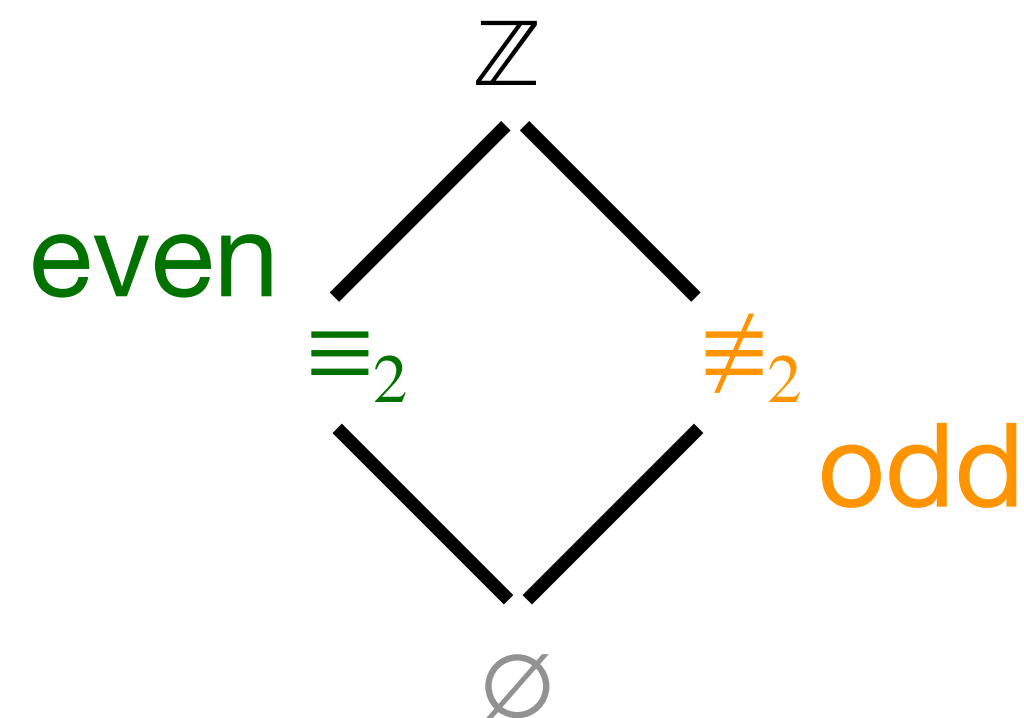
$$S_3^\# = \equiv_2 \times \mathbb{Z}$$

$$S_4^\# = \emptyset$$

$$S_5^\# = \emptyset$$

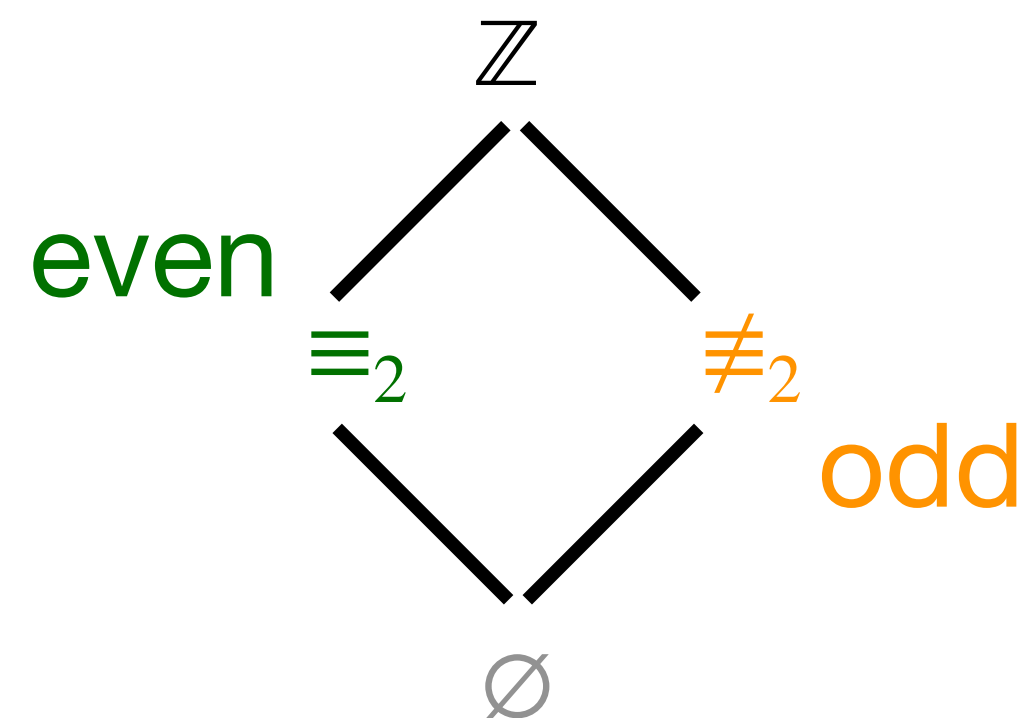
$$S_6^\# = \emptyset$$

$$S_7^\# = \emptyset$$



Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

$$S_3^\# = \equiv_2 \times \mathbb{Z}$$

$$S_4^\# = \equiv_2 \times \mathbb{Z}$$

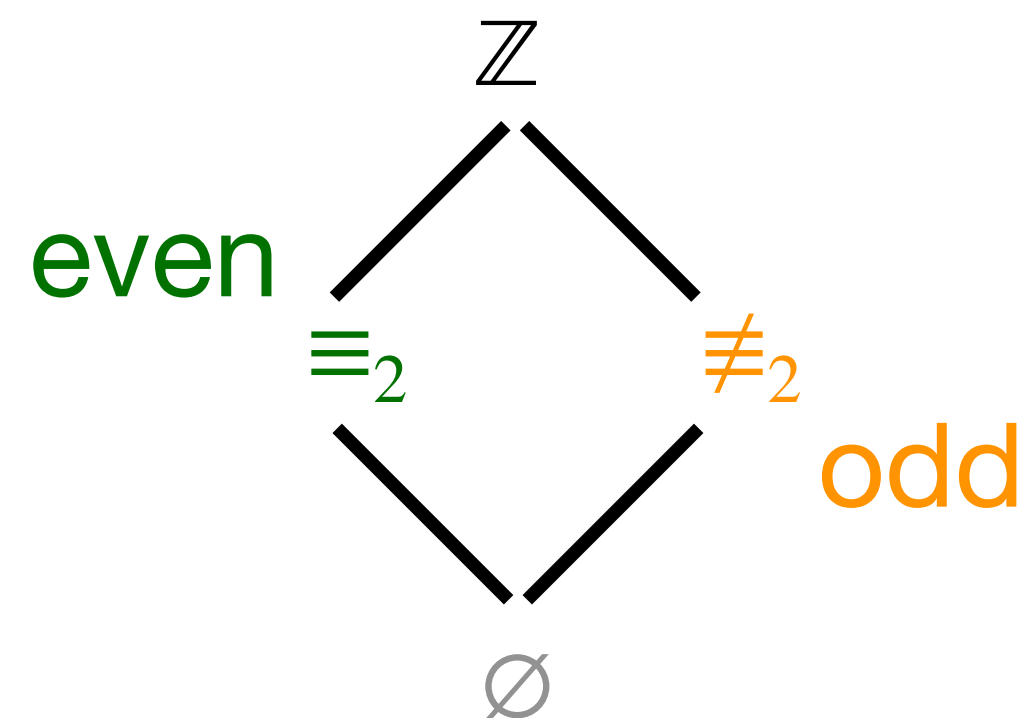
$$S_5^\# = \emptyset$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \equiv_2 \times \mathbb{Z}$$

Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

$$S_3^\# = \equiv_2 \times \mathbb{Z}$$

$$S_4^\# = \equiv_2 \times \mathbb{Z}$$

$$S_5^\# = \equiv_2 \times \mathbb{Z}$$

$$S_6^\# = \emptyset$$

$$S_7^\# = \equiv_2 \times \mathbb{Z}$$

Abstract interpretation 2

```
(1)  $m^\# := \equiv_2;$   
(2) while (3)  $n^\# > 0$  do {  
(4)    $m^\# := m^\# \times^\# n^\#;$   
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$   
(6) }  
(7) // end
```

Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

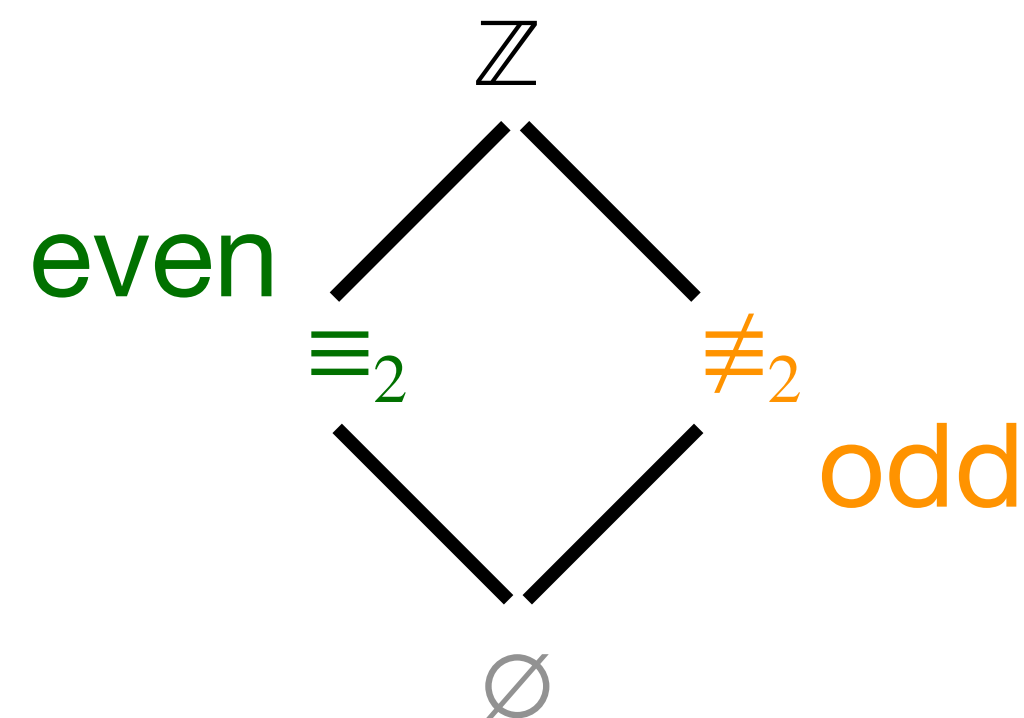
$$S_3^\# = \equiv_2 \times \mathbb{Z}$$

$$S_4^\# = \equiv_2 \times \mathbb{Z}$$

$$S_5^\# = \equiv_2 \times \mathbb{Z}$$

$$S_6^\# = \equiv_2 \times \mathbb{Z}$$

$$S_7^\# = \equiv_2 \times \mathbb{Z}$$

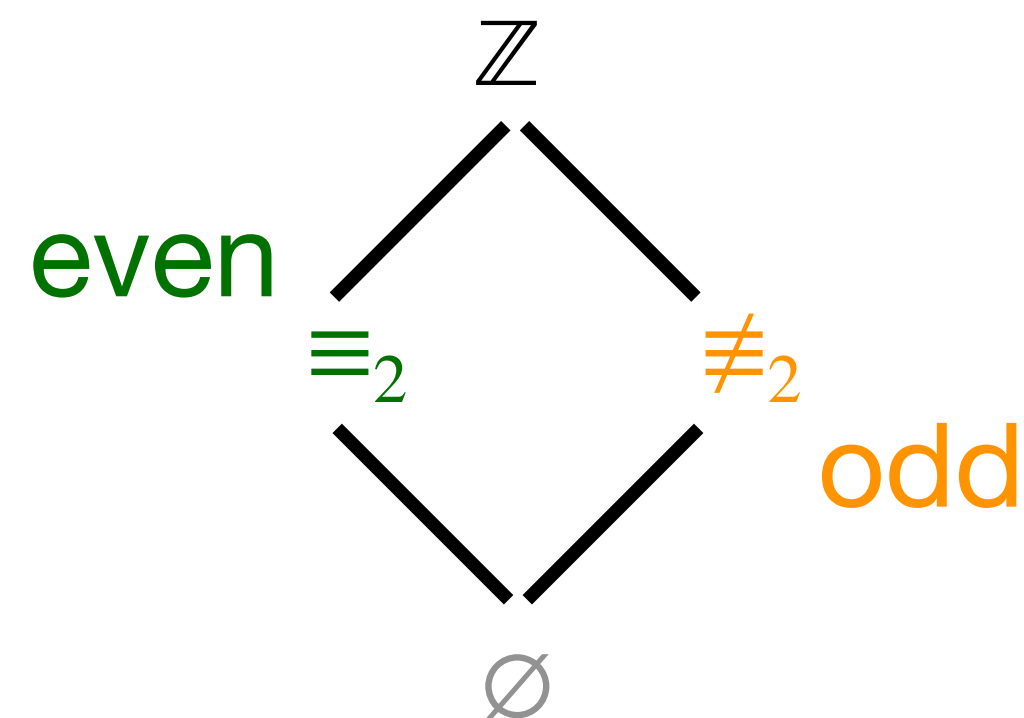


Abstract interpretation 2

```

(1)  $m^\# := \equiv_2;$ 
(2) while (3)  $n^\# > 0$  do {
(4)    $m^\# := m^\# \times^\# n^\#;$ 
(5)    $n^\# := n^\# ((\cdot) - 1)^\#$ 
(6) }
(7) // end

```



Evaluate the program in the abstract domain

$A = \{\emptyset, \equiv_2, \neq_2, \mathbb{Z}\}:$

$$S_1^\# = \mathbb{Z} \times \mathbb{Z}$$

$$S_2^\# = \equiv_2 \times \mathbb{Z}$$

$$S_3^\# = \equiv_2 \times \mathbb{Z} \cup^\# \equiv_2 \times \mathbb{Z} = \equiv_2 \times \mathbb{Z}$$

$$S_4^\# = \equiv_2 \times \mathbb{Z}$$

$$S_5^\# = \equiv_2 \times \mathbb{Z}$$

$$S_6^\# = \equiv_2 \times \mathbb{Z}$$

$$S_7^\# = \equiv_2 \times \mathbb{Z}$$

(ABSTRACT) FIXPOINT REACHED!